Coordination Chemistry- Bonding

Learning Objectives

- Draw a crystal field diagram for a given metal complex and use to identify the number of unpaired electrons
- Identify whether a given metal complex will be high or low spin
- Calculate the crystal field stabilization energy for a given metal complex
- Use ligand field theory to explain the spectrochemical series

- *A satisfactory theory of bonding in coordination compounds* must account for properties such as color and magnetism, as well as stereochemistry and bond strength.
- No single theory as yet does all this for us. Rather, several different approaches have been applied to transition metal complexes.
- There are essentially three bonding concepts that are used to describe the bonding in coordination compounds: **Valence Bond Theory (VBT), Crystal Field Theory (CFT),** and **Ligand Field Theory (LFT)**.

Coordination Chemistry- Bonding

- As an example, let us consider the following two transition metal complexes, $[Fe(NH_3)_6]^{2+}$ and $[Co(NH_3)_6]^{3+}$.
- The Fe^{2+} and the Co^{3+} centers both have a valence electron configuration of $3d^6$ and have the same ligands coordinated.
- Lewis, VSEPR, and valence bond theories predict that two isoelectronic molecules should have very similar properties, but in the case of $[Fe(NH_3)_6]^{2+}$ and $[Co(NH_3)_6]^{3+}$ that is not true.
- The light absorption properties (and colors) of the two are quite different. The iron complex absorbs very little visible light and therefore appears a very pale-yellow color and the cobalt complex strongly absorbs visible light and has a dark red-orange color.
- Also, the iron complex is paramagnetic (having unpaired electrons) and the cobalt complex is diamagnetic (having no unpaired electrons). Crystal field theory, and later ligand field theory were developed in part to explain the spectroscopic and magnetic properties of transition metal complexes.

Complex	**$[Fe(NH_3)_6]^{2+}$**	**$[Co(NH_3)_6]^{3+}$**
Valence electron configuration	$[Ar]\ 3d^6$	$[Ar]\ 3d^6$
Color	Pale-yellow	Red-orange
Magnetic properties	Paramagnetic	Diamagnetic

Crystal Field Theory

- Crystal field theory (CFT) describes the ***breaking of orbital degeneracy in transition metal complexes due to the presence of ligands***.
- CFT qualitatively describes the strength of the metal-ligand bonds. Based on the strength of the metal-ligand bonds, the energy of the system is altered. This may lead to a change in magnetic properties as well as color.
- The crystal field theory is actually **not a bonding theory** because it is based on **repulsive electrostatic interactions**.
- This theory was originally developed to explain color in ionic crystals. Later, it was found that it can also explain colors in molecular coordination compounds and is suitable to explain shapes and magnetism of complexes. However, because it is based on repulsive electrostatic interactions it cannot actually explain what holds the atoms in a molecule together. However, the crystal field theory is quite simple and convenient to use, and there is a lot of practicality to it.

❑ Crystal Field Theory (CFT)

- CFT makes some ***assumptions*** and ***simplifications:***
  - ***All ligands are modeled as negative point charges*** with ***no orbitals*** and the ***metal orbitals are modeled as clouds of negative charge***.
- This approach leads to the correct prediction that large cations of low charge, such as K^+ and Na^+, should form few coordination compounds.
- The situation is quite different for ***transition metal cations*** *that contain varying numbers of* ***d electrons in orbitals that are NOT spherically symmetric***.
- The ***shape and occupation of these d-orbitals*** then becomes important in an accurate description of the bond energy and properties of the transition metal compound.

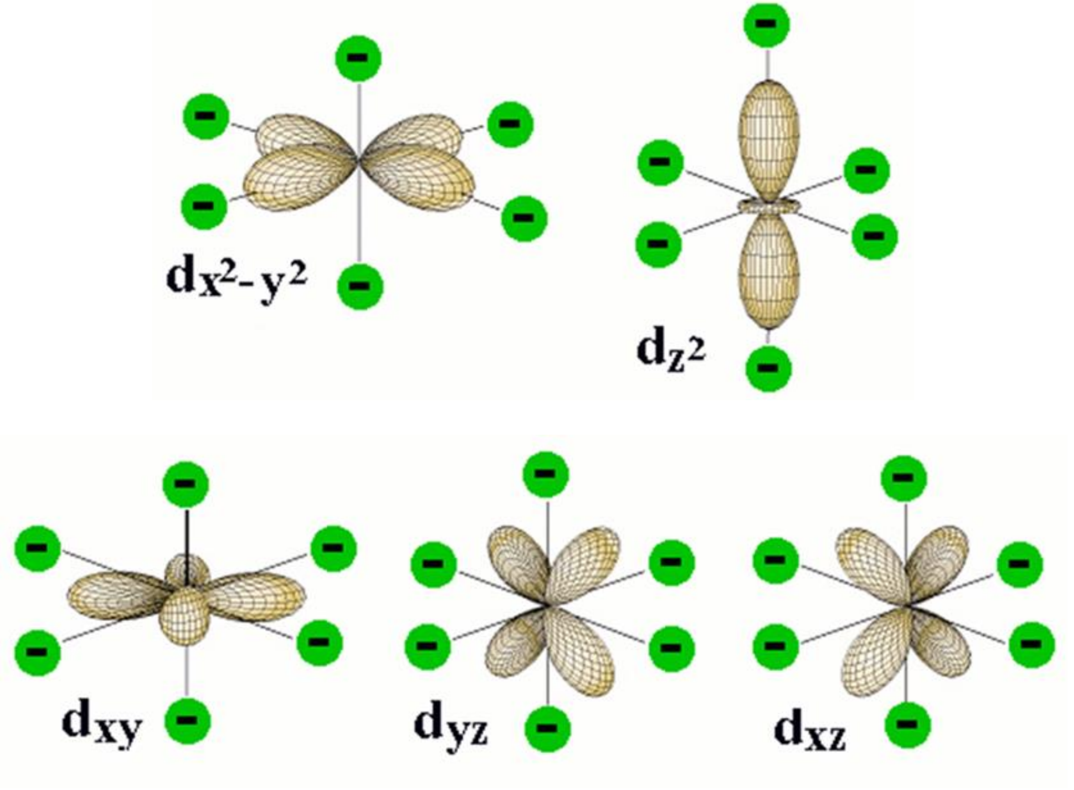


Figure 1: d-orbital orientations for a metal ion at the origin along with ligand point charges positions in the Cartesian coordinate for an *octahedral coordination*.
The green circles represent the coordinating electron-pairs of the ligands located at the six corners of the *octahedron* around the central atom. The two d orbitals at the top have regions of high electron density pointing directly toward the ligand orbitals.

Orientation of *d*-Orbitals

To understand CFT, one must first understand the ***shape*** and ***orientation*** of the d orbitals:

- d_{xy}: lobes lie in-between the x and the y axes.
- d_{xz}: lobes lie in-between the x and the z axes.
- d_{yz}: lobes lie in-between the y and the z axes.
- $d_{x^2-y^2}$: lobes lie on the x and y axes.
- d_{z2}: there are two lobes on the z axes and there is a donut shape ring that lies on the xy plane around the other two lobes.

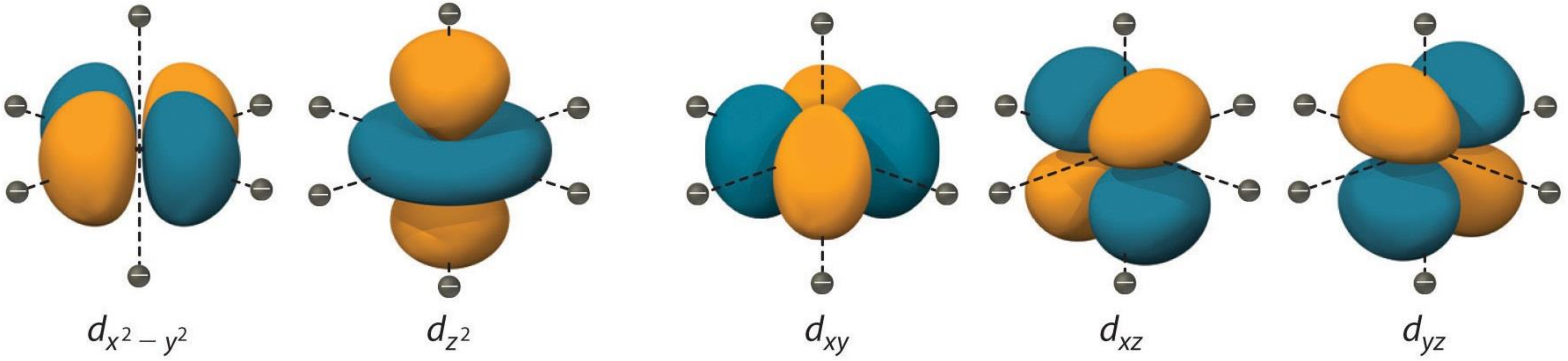
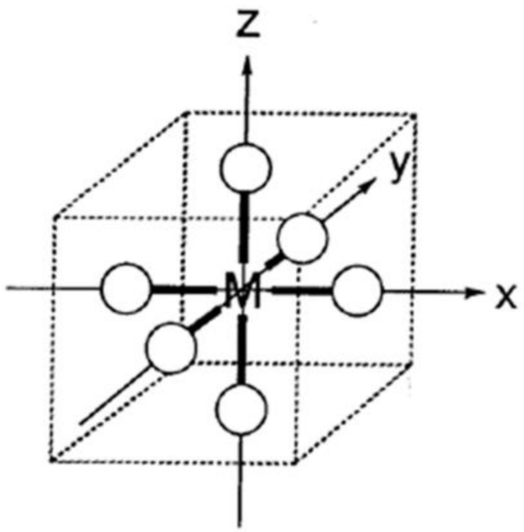
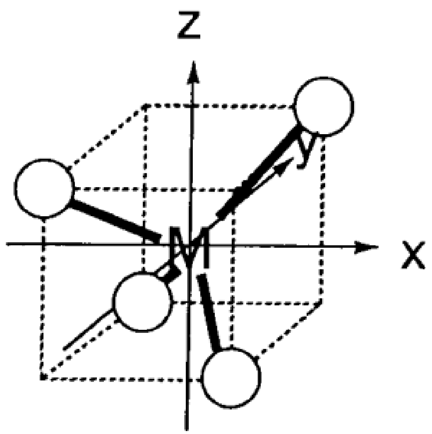


Figure 9.1.1: Spatial arrangement of ligands in the an octahedral ligand field with respect to the five d-orbitals.

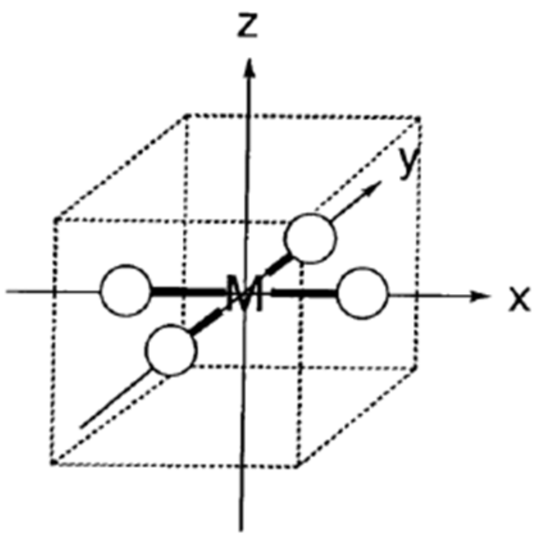
❑ Crystal Field Theory (CFT)



Octahedral coordination



Tetrahedral coordination



Square planar coordination

Figure 6.2.5: Ligand positions in the Cartesian coordinate with a metal ion at the origin.

- ➢ The interaction of a free metal ion in the gas phase with a sphere of negative charge causes the energy of the *d*-orbitals to increase as a result of the smaller value of *r*.

- ✓ All five of the *d* orbitals are **alike in energy** when in an **isolated atom or ion**, but they are **unlike in their spatial orientations**.

- ✓ One of them, is directed along the z axis, and another, has lobes along the x and y axes. The remaining three have lobes extending into regions between the perpendicular x, y, and z axes.

- ✓ According to the **crystal-field model**, ***when ligands approach the metal ion***, the ligand negative point charges destabilize the metal d-orbital based on the geometric structure of the molecule. Since ligands approach from different directions, ***not all d-orbitals interact equally***. These interactions, however, ***create a d-orbitals splitting due to the electrostatic environment***, *orbitals with* ***more interaction are higher in energy*** and orbitals with less interaction are lower in energy.

- ✓ Therefore, the energy of the *d*-orbitals depend on the ligand orientations around the central metal ion/atom.

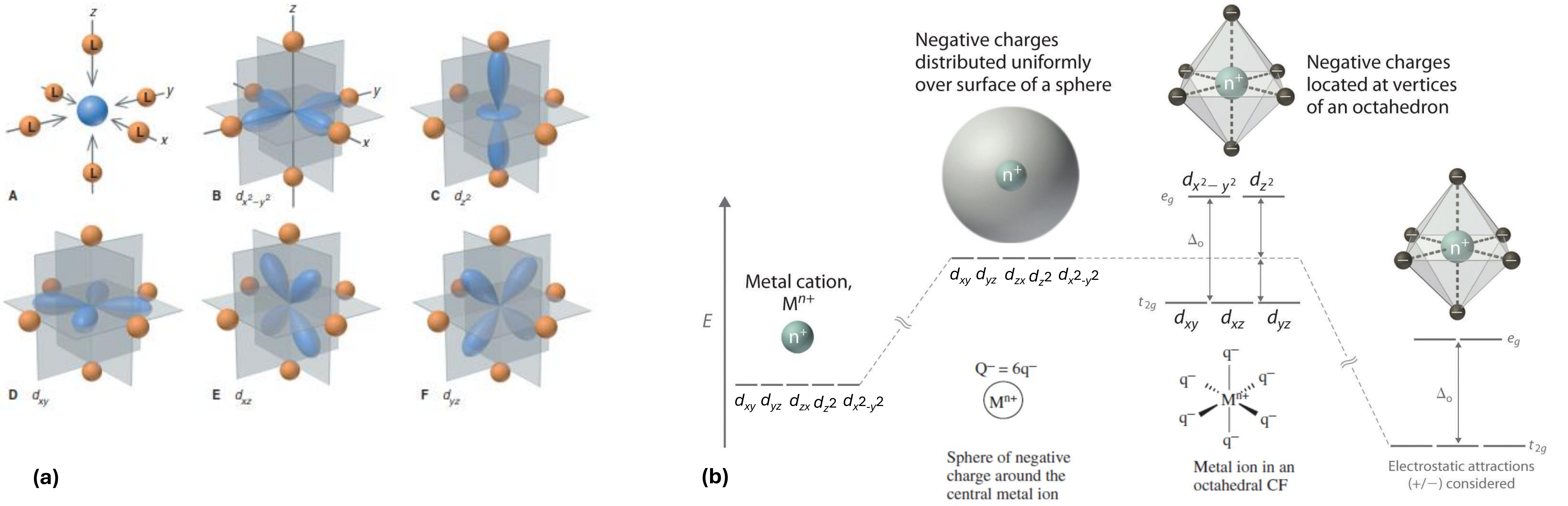
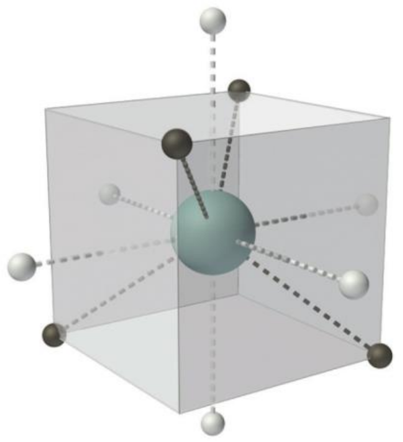
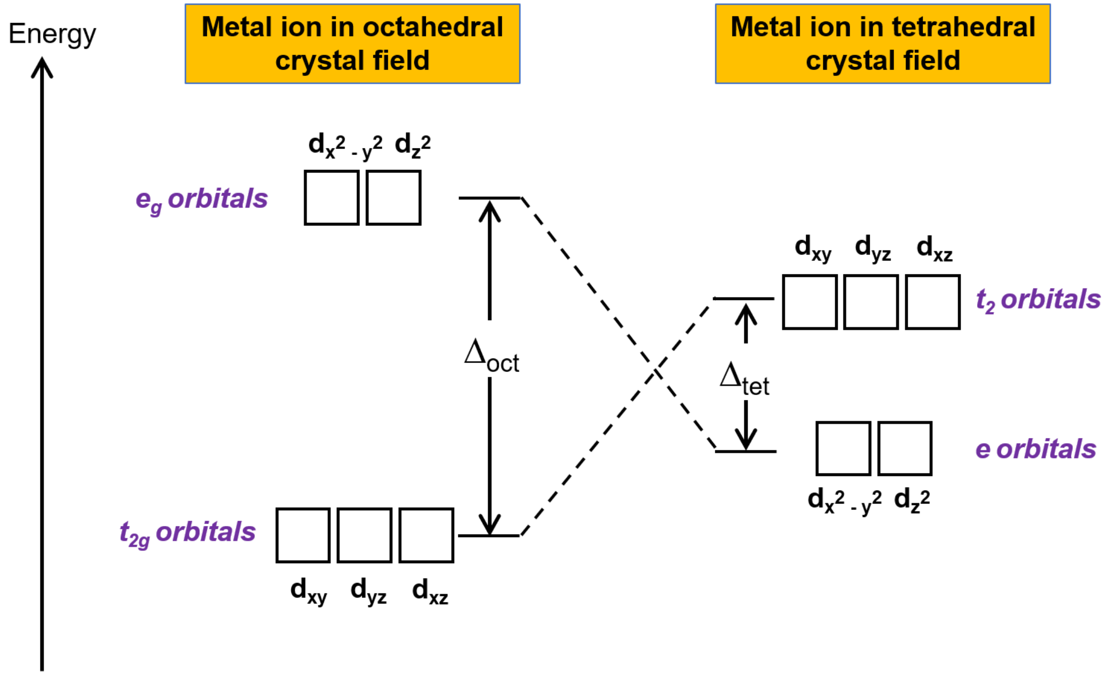


Figure 22.22. (a) The five d orbitals in an **octahedral field of ligands**. Orange balls: Ligands. A: Ligands approach along the x, y, and z axes. B and C: The two eg orbitals ($d_{x^2-y^2}$ and d_{z^2} orbitals point directly at some of the ligands. D to F: The d_{xy}, d_{xz}, and d_{yz} orbitals point between the ligands.

(b) In an isolated atom, all d-orbitals have the same energy. In an ***octahedral complex****, the* ***orbitals split into two sets***, with the d_z2 and $d_x2_{-y}2$ orbitals having higher energy than the other three. The two e_g orbitals point directly at the six negatively charged ligands, which increases their energy compared with a spherical distribution of negative charge. The repulsion is greater in the case of the d_z2 and $d_x2_{-y}2$ orbitals due to the direct interactions. In contrast, the three t_{2g} orbitals point between the negatively charged ligands, which decreases their energy compared with a spherical distribution of charge. Attractive electrostatic interactions between the negatively charged ligands and the positively charged metal ion (far right) cause all five d orbitals to decrease in energy but does not affect the splittings of the orbitals.

❑ Crystal Field Theory (CFT): d-orbitals splittings in octahedral vs. tetrahedral



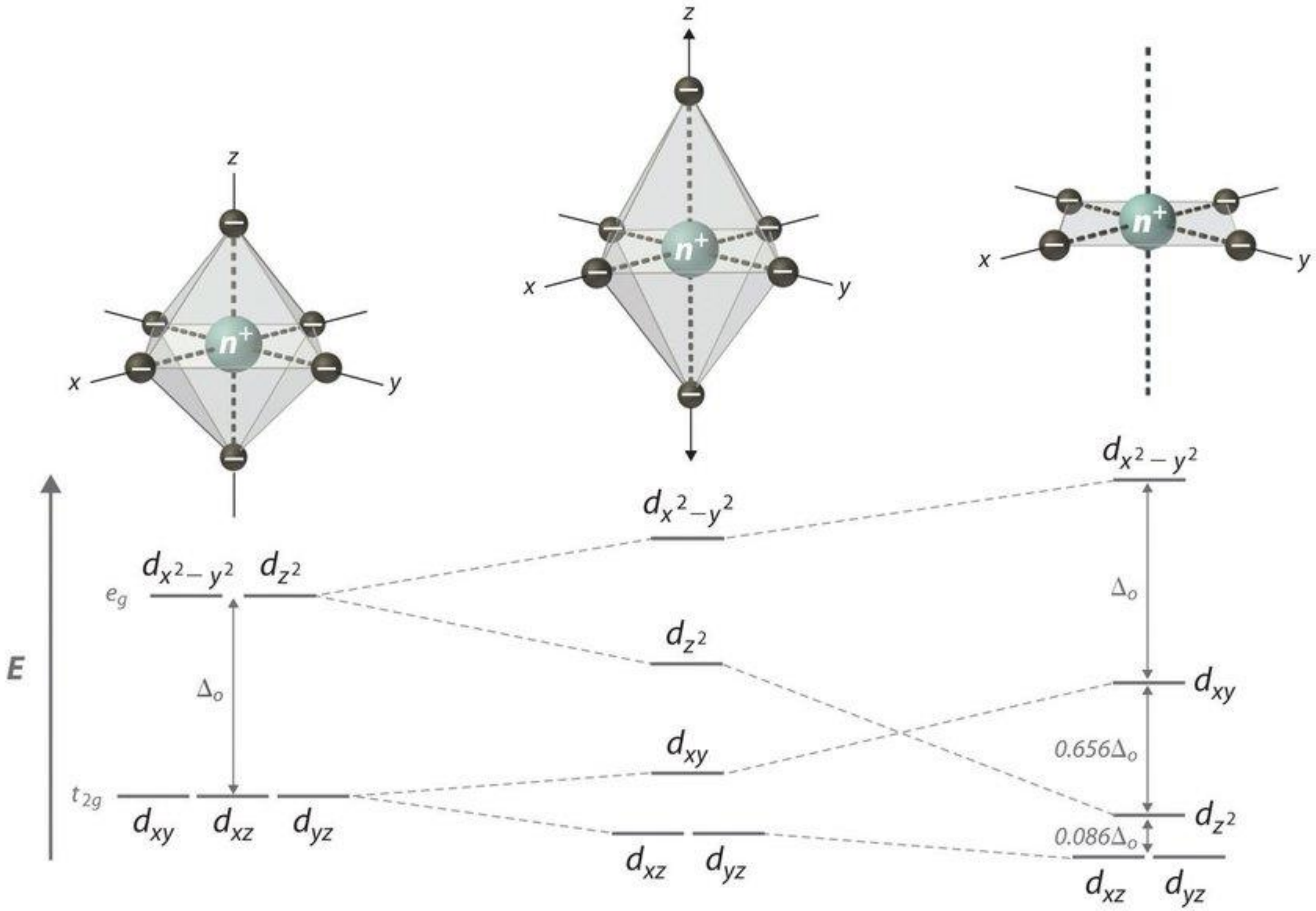


- In a tetrahedral complex (point group T_d), the four ligands are orientated at opposite corners of a cube rather than along the axes. Three d orbitals which lie between the axes, d_{xy}, d_{xz}, and d_{yz} (t_2 symmetry in tetrahedral), now interact more strongly with the ligands and are destabilized relative to the spherical crystal field. Conversely, $d_{x^2-y^2}$ and d_{z^2} (e symmetry in tetrahedral) now interact less with the ligands and are stabilized relative to the spherical crystal field. Thus, the ***splitting of the d orbitals in a tetrahedral crystal field is essentially the inverse of the octahedral*** crystal field.
- *Because there is **no center of inversion** in a tetrahedral complex, the orbital designation does not include the parity label g or u.*
- The **energy difference** between the e and the t_2 orbitals is **called tetrahedral splitting energy, Δ_t**.

❖ Square Planar Crystal Field

- In a complex with **four ligands**, the other possible arrangement of ligands is **square planar**.
- The easiest way to visualize the square planar geometry is to start with an octahedral complex and remove the two ligands that are along the z axis.
- The four remaining ligands are in the xy plane, along the x and y axes.
- From a crystal field splitting perspective, when the z ligands are removed the d orbitals split further.
- When the z ligands are removed the d_{z^2} orbital is stabilized because it no longer interacts with ligands on the z axis (in an octahedral crystal field both $d_{x^2-y^2}$ and d_{z^2} are destabilized).
- Conversely the $d_{x^2-y^2}$ orbital is further destabilized because all of the ligand negative charges are concentrated in the xy plane.
- Removing the z ligand has a similar effect on the d_{xy}, d_{yz}, and d_{zx} orbitals, although the magnitude of the stabilization and destabilization is less because those orbitals do not directly interact with the ligands. The d_{xy} orbital is destabilized because it also lies in the xy plane and the d_{yz}, and d_{zx} orbitals are stabilized because they have lost the (indirect) interaction with the ligands on the z axis.
- Overall, the $d_{x^2-y^2}$ orbital is significantly more destabilized than the other d orbitals, meaning the **square planar geometry is most favored in complexes with 8 electrons** where the $d_{x^2-y^2}$ is empty and the other more stabilized d orbitals are filled.

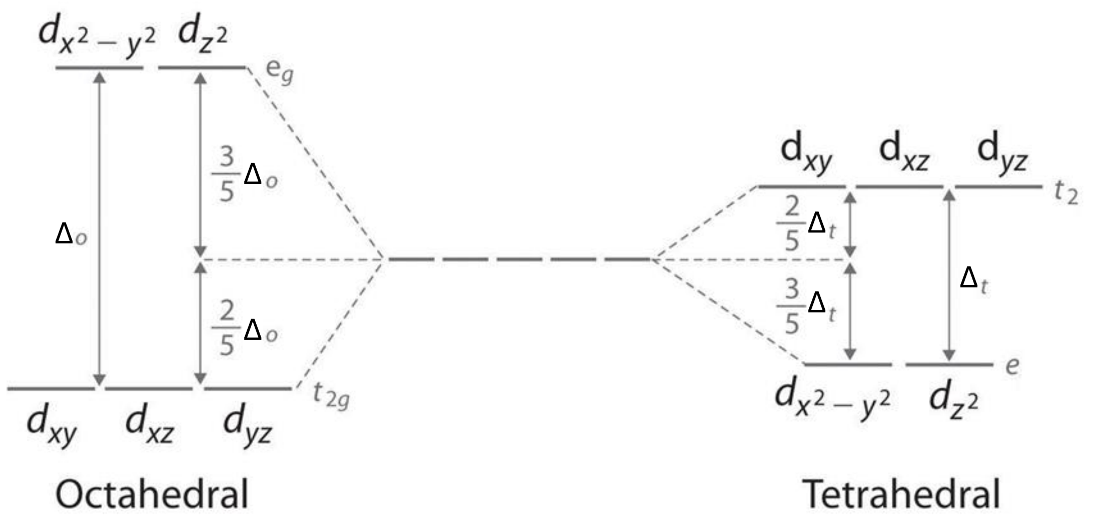
❑ **Crystal Field Theory (CFT): d-orbitals splittings in octahedral, elongated octahedral vs. square planar**



❖ CF splitting Energies

- The crystal field model explains that the ***properties of complexes result from the splitting of d-orbital energies***.

- The splitting of orbital energies is called the ***crystal field effect***, and the *energy difference* between e_g and t_{2g} orbitals is the **crystal field splitting energy, Δ_o** or 10 Dq.

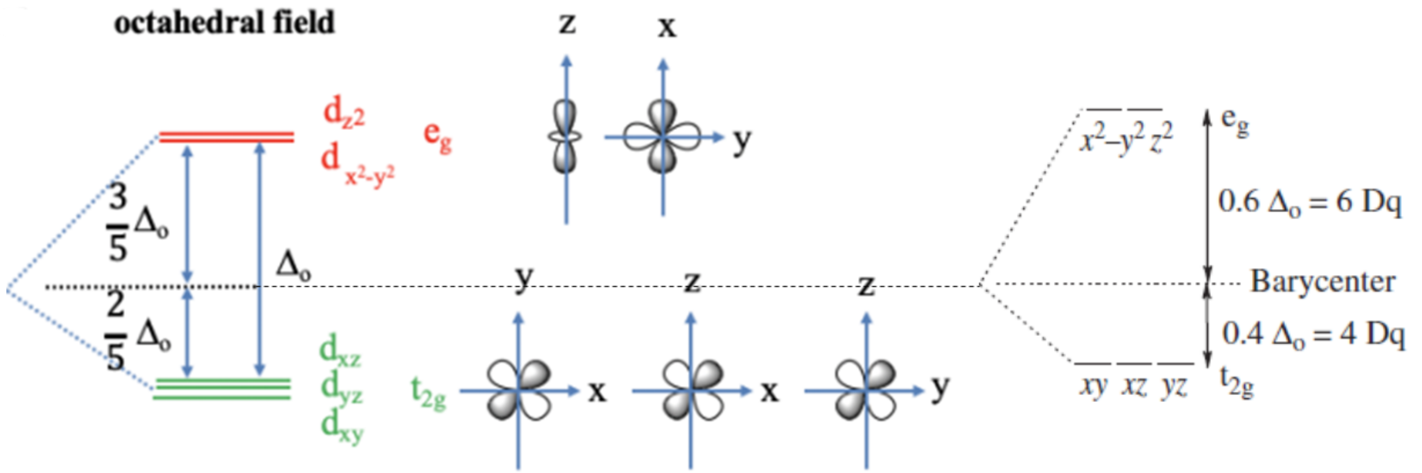


Definition: Barycenter

In crystal field theory the barycenter is the energy of the d orbitals in a spherical crystal field, where the negative ligand charges are evenly distributed around the metal ion in a sphere. The splitting of the d orbitals in other crystal fields will be symmetric around the barycenter (i.e. the total amount of stabilization will always be equal to the total amount of destabilization).

❖ Octahedral ***CF splitting Energies***:

FIGURE 16.6 CF splitting energy (Δ_o) arising from the octahedral coordination geometry.



- The energy of the t_{2g} orbitals is decreased by $(2/5)\Delta_o$ relative to the spherical crystal field (barycenter), and the energy of the e_g-orbitals is increased by $(3/5)\Delta_o$ relative to the spherical crystal field.

- Where do the factors 2/5 and 3/5 come from? They are due to the **law of the conservation of energy**. The overall energy reduction due to the energy decrease of the t_{2g}-orbitals must be equal to the overall energy increase due to the energy increase of the e_g orbitals: $\Sigma E(t_{2g}) = -\Sigma E(e_g)$. Because there are three t_{2g} orbitals but only two e_g orbitals this equation only holds when the energy of the e_g orbitals is increased by $(3/5)\Delta_o$ and the energy of the t_{2g} orbitals is decreased by $(2/5)\Delta_o$, or $3x\ (2/5)\Delta_o = 2\ x\ (3/5)\Delta_o$. Note that the energy of all orbitals will be greater in comparison to the case of no electrical field existing, but the energy is increased to a greater extent for the e -orbitals compared to the t orbitals.

❑ Strong-field and Weak-field ligands

➢ The **magnitude of Δ**, which can be *measured spectroscopically* (for d-d transitions), depends on **the identity** and **charge of the metal ion**, as well as **the type of ligand**.

• ***Strong-field ligands*** lead to a ***larger*** *splitting energy* (larger Δ).

• ***Weak-field ligands*** lead to a ***smaller*** *splitting energy* (**smaller Δ**).

✓ For instance, CN^- is a strong-field ligand and *H_2O is a weak-field ligand* (Figure 22.18). ***Note** the different orbital occupancies*; we discuss the reason for these differences shortly.

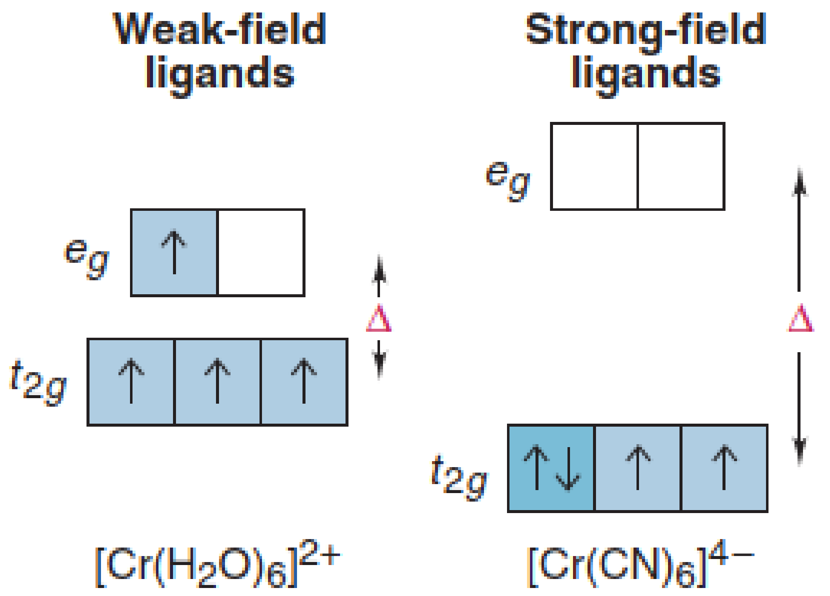


Figure 22.18 The effect of ligands and splitting energy orbital occupancy.

- By studying a range of different ligands, a ***spectrochemical series*** can be established which lists the ligands in the order of increasing Δ_o:

I^- < Br^- < S^{2-} < **S**CN$^-$ < Cl^- < N**O**$_2^-$ < N^{3-} < F^- < OH^- < $C_2O_4^{2-}$ (oxalate) < O^{2-} < H_2O < NCS$^-$ < H_3C—CN < py < NH_3 < ethylenediamine (en) < bpy (2,2'-bipyridyl) < phen (o-phenanthroline) < **N**O$_2^-$ < PPh_3 < CN^- < CO

← weak field – strong field →

- Given the same geometry of coordination and identical ligands, the crystal field splitting Δ_o for different *metals* increases in the following order (approximately:):

Mn^{2+} < Ni^{2+} < Co^{2+} < Fe^{2+} < V^{2+} < Fe^{3+} < Cr^{3+} < Co^{3+} < Mo^{3+} < Rh^{3+} < Ru^{3+} < Pd^{4+} < Ir^{3+} < Pt^{4+}

← weak field – strong field →

- The *value of Δ_O increases with an* ***increasing oxidation state*** *of the central metal ion* (compare the two entries for Fe and Co) and *increases* ***down a group*** (compare, for instance, the locations of Co, Rh, and Ir). The *variation with oxidation state reflects the smaller size of more highly charged ions and the consequently shorter metal-ligand distances and stronger interaction energies*. The *increase down a group reflects the larger size of the 4d and 5d orbitals compared with the compact 3d orbitals and the consequent stronger interactions with the ligands*.

The number of electrons in the *d* orbitals and the number of electrons can be removed in order to form the positive ion.

d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
3	4	5	6	7	8	9	10	11	12

Common Oxidation States of the First Series of Transition Metals

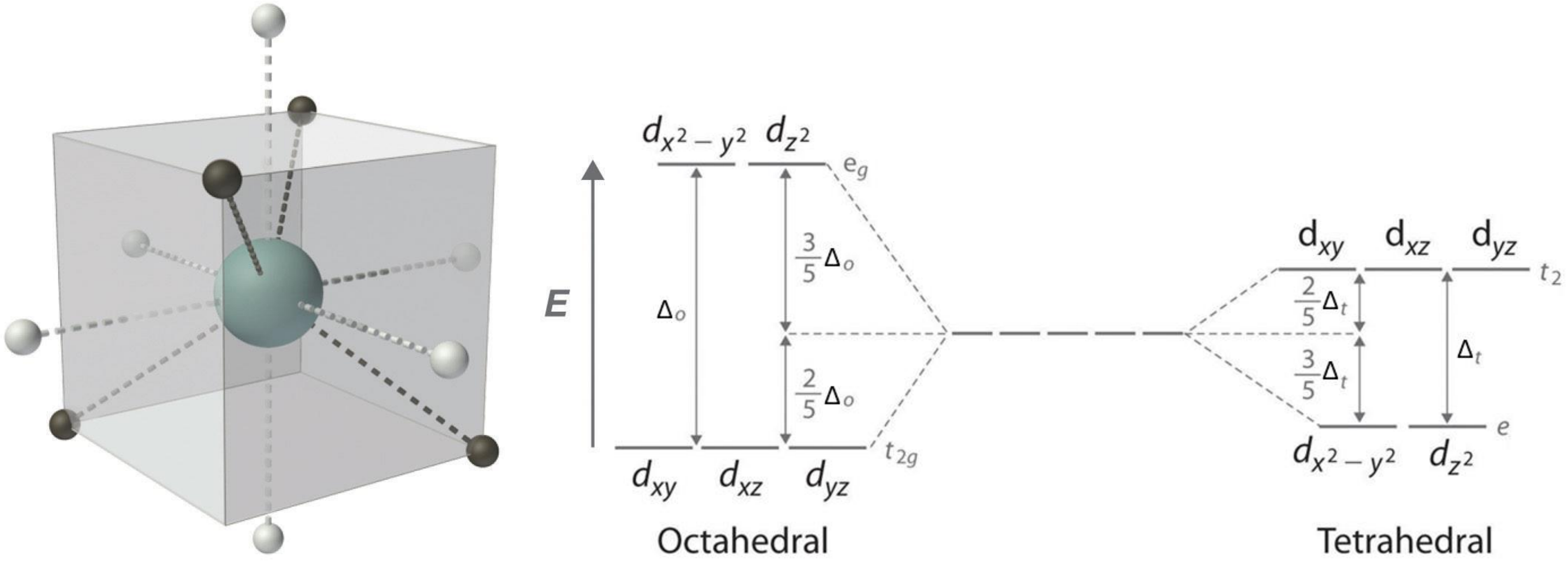
	Sc	**Ti**	**V**	**Cr**	**Mn**	**Fe**	**Co**	**Ni**	**Cu**	**Zn**
+1									d^{10}	
+2			d^3		d^5	d^6	d^7	d^8	d^9	d^{10}
+3	d^0			d^3		d^5	d^6			
+4		d^0			d^3					
+5			d^0							
+6				d^0						
+7					d^0					

❖ Tetrahedral Crystal Field Splitting Energy (Δ_t) :

Key points: In a tetrahedral complex, the e orbitals lie below the t_2 orbitals; Δ_t≈4 Δ_o/9 and only *the* ***high-spin case need be considered*.**

- The crystal field splitting energy in a tetrahedral crystal field is Δ_t. Because a tetrahedral complex has fewer ligands than an octahedral complex, and those ligands are not ideally orientated to overlap with the d orbitals Δ_t is less than Δ_o for the same metal and ligands: **Δ_t≈4 Δ_o/9.**

- Consequentially, **Δ_t is typically smaller** than **the spin pairing energy**, so **tetrahedral complexes are usually high spin**.
- The energy of the t_2 orbitals is increased by (2/5)Δ_t. The energy of the e orbitals is decreased by (3/5)Δ_t. This is again the due to the law of the conservation of energy. The total amount of decreased energy must equal the total amount of increased energy.



(a) (b)

Figure 9.1.6: (a) Tetrahedral ligand field surrounding a central transition metal (teal sphere). (b) Splitting of the degenerate d-orbitals (without a ligand field) due to an octahedral ligand field (left diagram) and the tetrahedral field (right diagram).

❖ Tetrahedral Crystal Field Splitting Energy (Δ_t)

- In a tetrahedral complex, the four ligands are orientated at opposite corners of a cube as shown in Figure 9.1.6a rather than along the axes. Three d orbitals which lie between the axes, d_{xy}, d_{xz}, and d_{yz} (t_2 symmetry in tetrahedral), now interact more strongly with the ligands and are destabilized relative to the spherical crystal field. Conversely, $d_{x^2-y^2}$ and d_{z^2} (e symmetry in tetrahedral) now interact less with the ligands and are stabilized relative to the spherical crystal field. Thus, the splitting of the d orbitals in a tetrahedral crystal field is essentially the inverse of the octahedral crystal field.
- The crystal field splitting energy in a tetrahedral crystal field is Δ_t. Because a tetrahedral complex has fewer ligands than an octahedral complex, and those ligands are not ideally orientated to overlap with the d orbitals Δ_t is less than Δ_o for the same metal and ligands:
- $\boldsymbol{\Delta_t \approx 4\,\Delta_o/9}$ (9.1.2)
- Consequentially, **Δ_t is typically smaller than the spin pairing energy**, so **tetrahedral complexes are usually high spin**.
- The **tetrahedral crystal field energy *is smaller than that of the octahedral field*** (**Δ_t is less than Δ_O**) because the octahedral field interacts more strongly with the d-orbitals compared to the tetrahedral field. One can calculate that it is actually just 4/9 of the octahedral field ($\Delta_t \approx {}^4/_9\,\Delta_O$).). This is mainly because the lobes of the t_2 orbitals to not point exactly toward the vertices of the tetrahedron, while the lobes of the e orbitals do point exactly toward the vertices of the octahedron. *The lack of direct interaction between orbitals and ligands in the tetrahedral configuration reduces the magnitude of the crystal field splitting* by a geometrical factor equal to 2/3, and the fact that there are only four ligands instead of six (4/6) reduces the ligand field by another 2/3. The total splitting in the tetrahedral case is thus approximately 2/3 x 2/3 = 4/9 of the *octahedral splitting*.

- ***The pairing energy** is invariably more **unfavorable than Δ_t**, and **normally only high-spin tetrahedral complexes** are **encountered**.* In theory, alternative electron configurations are possible for the cases d^3, d^4, d^5 and d^6 in the tetrahedral field but the gain in CFSE (crystal field stabilization energy) is reduced by the smaller size of the splitting, and the loss of exchange energy is never counterbalanced. Thus, all tetrahedral complexes are **high spin**.

❖ Tetrahedral vs Square Planar

- From VSEPR theory we know that sterically the most favorable way to arrange four ligands around a metal ion is in the tetrahedral arrangement. That puts the most possible space between the four ligands.
- Because of this, a four-coordinate metal complex will prefer a tetrahedral geometry UNLESS there is a larger crystal field stabilization energy (CFSE) for the square planar arrangement.
- As discussed above, this means square planar is the preferred geometry for most metal complexes with 8 d electrons, especially when the crystal field splitting (Δ) is large.
- While not every d^8 complex is square planar, almost all square planar complexes are d^8 (there are examples of d^7 and d^9 square planar complexes but they are less common).

❑ Filling of Electrons in Orbitals: Δ vs. Spin Pairing Energy (P) and Low-Spin and High-Spin Complexes

❖ ***Keypoint:*** ***If the electron configurations for any d-electron count is different depending on Δ, the configuration with more paired electrons is called low spin while the one with more unpaired electrons is called high spin.***

- As illustrated in Fig. 11.11, electrons are filled into the orbitals according to the principles we have discussed earlier. One electron is added to each of the degenerate orbitals until each orbital has one electron with the same spin. In an octahedral complex, this works straightforwardly up to the third electron. *The **fourth electron** could either be placed as a paired electron in the lower set of d orbitals (**low-spin complex**) or as an unpaired electron in the energetically higher set of d orbitals (**high-spin complex**).*

- Whether the first or the second scenario happens **depends on the magnitude of crystal field splitting energy (Δ_o) and the total pairing energy (P_{tot})**. If the ***Δ_o is larger than the spin pairing energy***, then a **low spin** complex is favored, if Δ_o is smaller than the spin-pairing energy, then the high spin complex is favored.

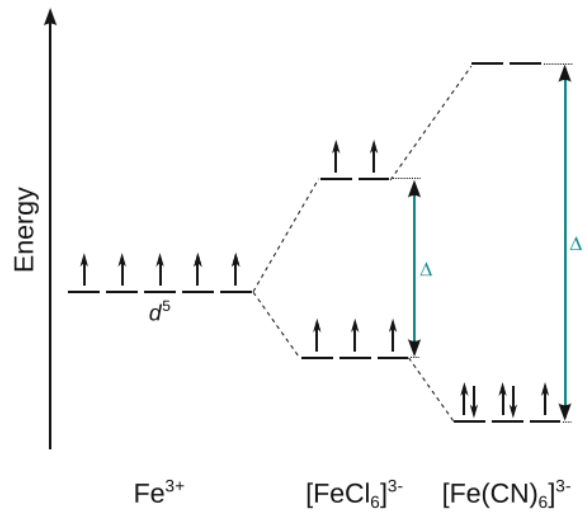


Fig. 11.11 Comparison of crystal field splitting for two Fe^{3+} complexes.

Definition: Spin Pairing Energy (P)

The spin pairing energy is the energy required to pair two electrons in an orbital. It is based on the Coulombic repulsion between the two negative charges. Larger, more diffuse orbitals will have lower spin pairing energy because the two electrons can be further apart.

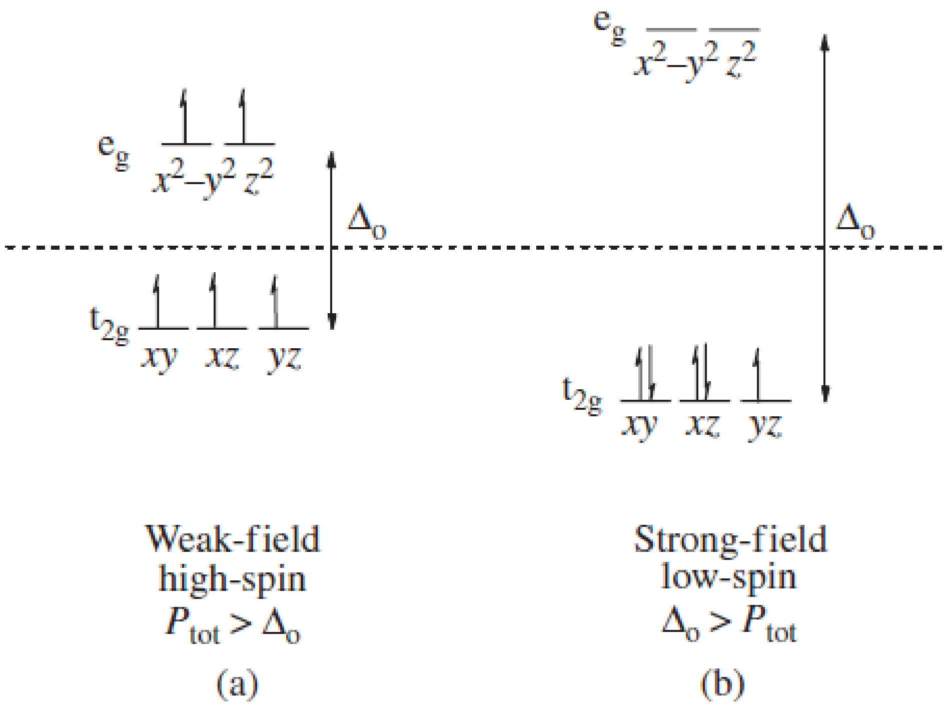


FIGURE 16.8

The relative energies of the five *d*-orbitals (a) in a weak CF and (b) in a strong CF. The dashed line represents the barycenter.

- As the valence *d*-electrons fill the empty orbitals in Figure according to the Aufbau principle, they will first populate the lower-lying t_{2g} set, going in with their spins paired according to Hund's rule.
- *Because the degenerate orbitals are indistinguishable, there is only one way to write the electron configurations for d^1 – d^3.*
- For the d^4 –d^7 electron configurations, however, there are two possible ways for the valence electrons in a metal to occupy the t_{2g} and e_g sets of orbitals.

✓ If $P_{tot} > \Delta_o$(where P_{tot} is the total spin pairing energy), then it will be easier for the electrons to *fill the upper e_g level before they pair up*, leading to the ***weak-field,*** *high-spin (HS) configuration* shown in Figure 16.8(a).

✓ If the magnitude of $\Delta_o > P_{tot}$, *then the electrons will pair up first in a* ***strong-field,*** *low-spin (LS) configuration*, as shown in Figure here (16.18 (b)).

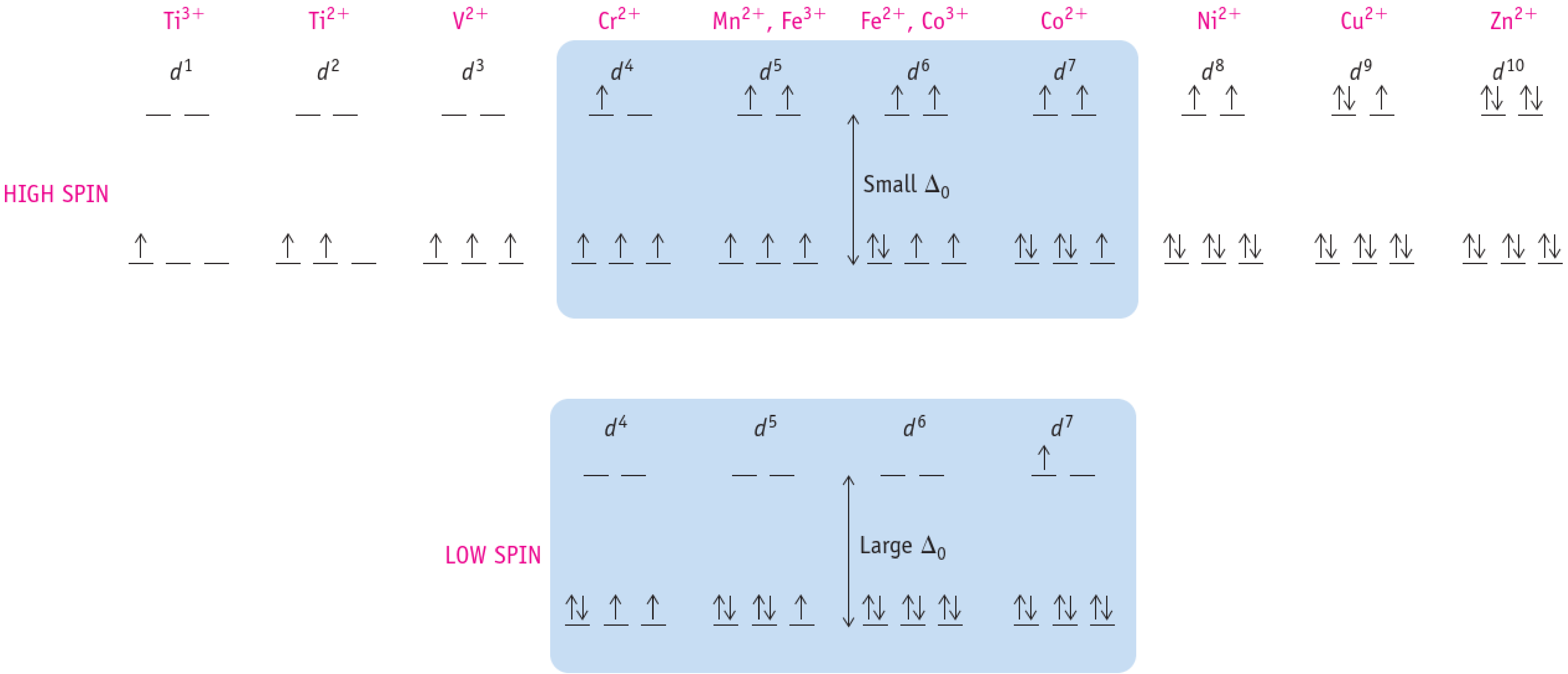


Figure 22.24 High- and low-spin octahedral complexes. d-Orbital occupancy for octahedral complexes of metal ions. Only the d^4 through d^7 cases have both high- and low-spin configurations.

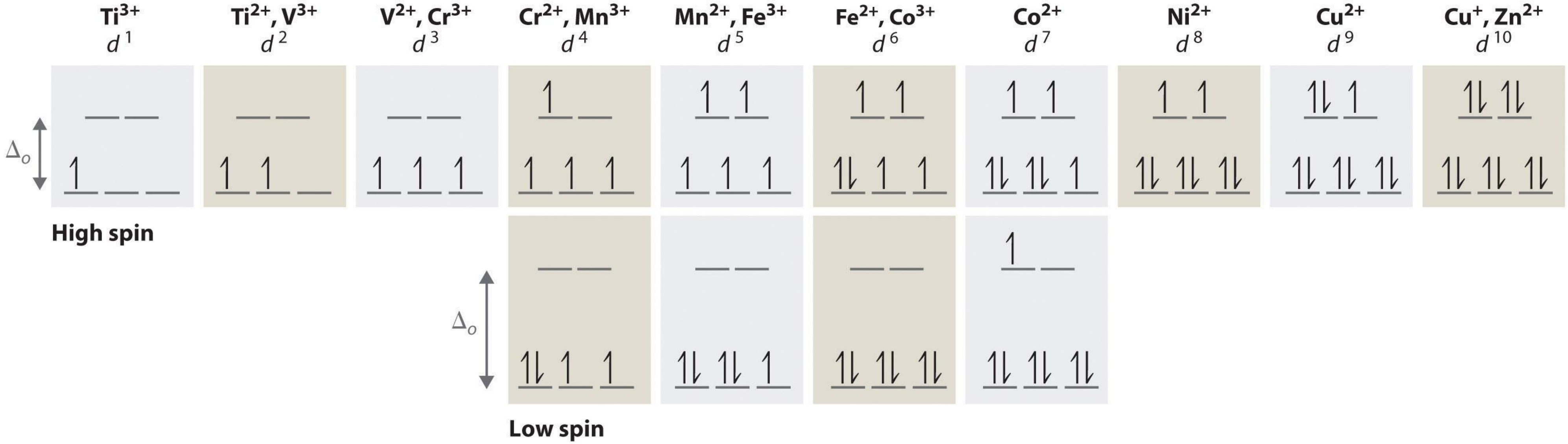


Figure 8.5.2: The Possible Electron Configurations for Octahedral d^n Transition-Metal Complexes (n = 1–10). Two different configurations are possible for octahedral complexes of metals with d^4, d^5, d^6, and d^7 configurations; the magnitude of Δ_o determines which configuration is observed.

- Therefore, *high-spin complexes are typically found with metals/ligands at the weak-field end of the spectrochemical series* (such as e.g. $[FeCl_6]^{3-}$); vice versa, low-spin complexes are expected in complexes that comprise metals and ligands at the high-field end of the spectrochemical series (such as e.g. $[Fe(CN)_6]^{3-}$).
- In the electron configurations $d^1 – d^3$ all electrons are unpaired in the t_{2g} orbitals. Therefore, there is no electron that could be moved from t_{2g} to an e_g orbital.
- Importantly, *with octahedral geometry, the options of forming either low-spin or high-spin complexes only exist for **systems with 4–7 electrons** in the d orbitals* (d^4, d^5, d^6, d^7 complexes). For d^1, d^2 d^3 and d^8, d^9, d^{10} complexes, there is only one possible electron configuration.
- For the configurations $d^8 – d^{10}$ all t_{2g} orbitals are necessarily full. Therefore, there is no possibility to move an electron from an e_g to a t_{2g} orbital regardless the crystal field strength.
- Since the formation of either a high-spin or a low-spin complex affects the pairing of electrons, the crystal field splitting has repercussions in the magnetic properties of a complex.

❑ MAGNETIC MEASUREMENTS

Key points: ***Magnetic measurements are used to determine the number of unpaired spins in a complex*** *and hence to identify its ground-state configuration. A spin-only calculation may fail for low-spin d^5 and for high-spin $3d^6$ and $3d^7$ complexes.*

- The experimental distinction between high-spin and low-spin octahedral complexes is based on the determination of their magnetic properties. Compounds are classified as diamagnetic if they are repelled by a magnetic field and paramagnetic if they are attracted by a magnetic field. The two classes are distinguished experimentally by magnetometry.

- The *magnitude of the paramagnetism of a complex is commonly reported in terms of the magnetic dipole moment* it possesses: the higher the magnetic dipole moment of the complex, the greater the paramagnetism of the sample.

- In a free atom or ion, ***both the orbital and the spin angular momenta*** give rise to a magnetic moment and contribute to the paramagnetism. The following formula for the magnetic moment (μ) of a molecule in Bohr magnetons is:

$$\mu_{\text{total}} = -[\{L(L+1)\} + 4\{S(S+1)\}]^{1/2}\,\mu_B$$

- When *the atom or ion is part of a complex, any* ***orbital angular momentum*** *is* ***normally quenched***, or ***suppressed***, as a result of the interactions of the electrons with their nonspherical environment. However, if any electrons are unpaired, *the net electron spin angular momentum survives* and gives rise to ***spin-only paramagnetism***, which is characteristic of many d-metal complexes.

- The spin-only magnetic moment, μ, of a complex with total spin quantum number S is: $\mu_{\text{spin-only}} = 2\{S(S+1)\}^{1/2}\,\mu_B$ where μ_B is the Bohr magneton, $\mu_B = e\hbar/2m_e$ with the value 9.274×10^{-24} J T^{-1}.

- Because ***S = N/2***, where ***N is the number of unpaired electrons***, each with spin $s = 1/2$, $\mu_{\text{spin-only}} = \{N(N+2)\}^{1/2}\,\mu_B$.

- For example, magnetic measurements on a d^6 complex easily distinguish between a high-spin $t_{2g}^4 e_g^2$ (N = 4, S = 2, μ = 4.90μ_B) configuration and a low-spin t_{2g}^6 (N = 0, S = 0, μ = 0) configuration.

✓ The high-spin complex $[Fe(H_2O)_6]^{2+}$ has four unpaired electrons and is paramagnetic (*attracted by a magnet*), whereas the low-spin $[Fe(CN)_6]^{4-}$ complex has no unpaired electrons and is diamagnetic (*repelled by a magnet*).

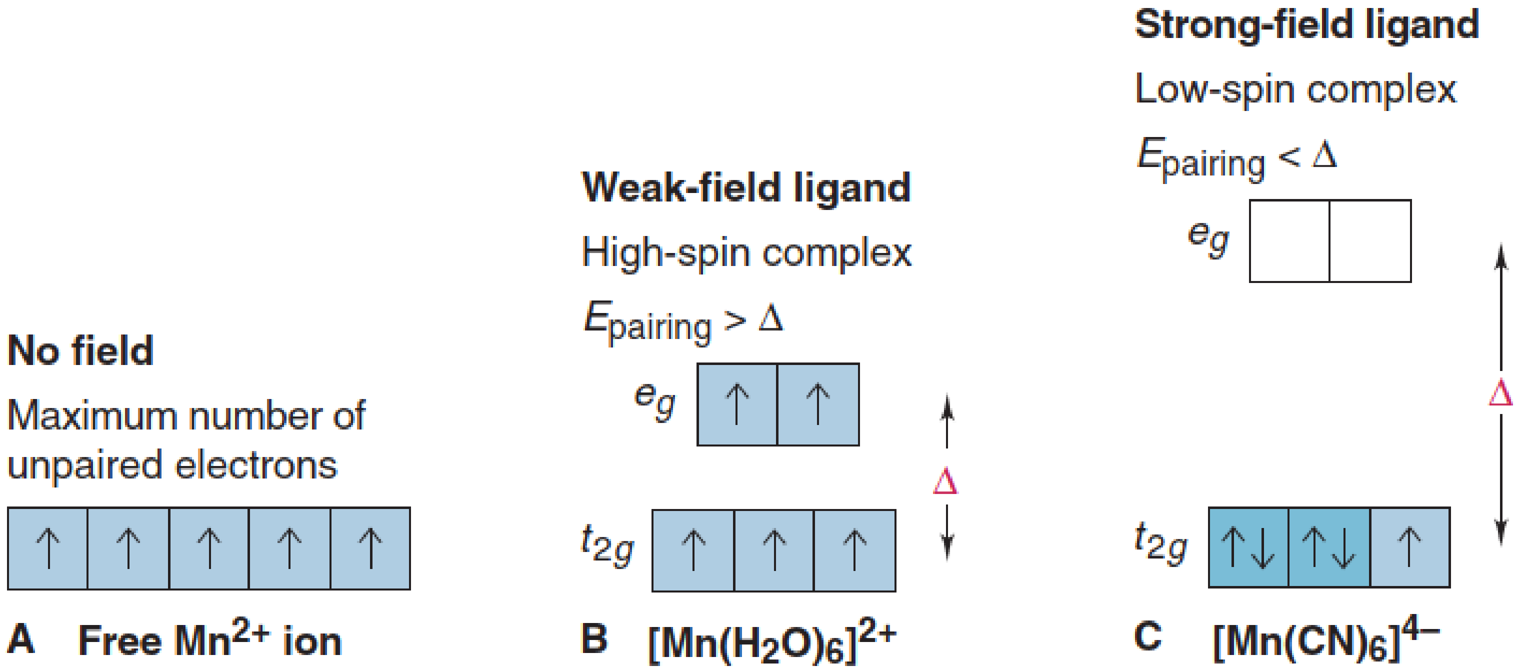


Figure 22.22 High-spin and low-spin octahedral complex ions of Mn^{2+}.
A: Free Mn^{2+} has *five unpaired electrons*.
B: Bonded to a weak-field ligand, Mn^{2+} still has five unpaired electrons (high-spin complex).
C: Bonded to a strong-field ligand, Mn^{2+} has one unpaired electron (low-spin complex).

❖ **Test:** *The magnetic moment of the complex* $[Mn(NCS)_6]^{4-}$ is 6.06 μ_B. *What is its d-electron configuration*?

- The spin-only magnetic moments for some electron configurations are listed in Table 20.3 and compared there with experimental values for a number of 3d complexes.
- For most 3d complexes (and some 4d complexes), experimental values lie reasonably close to spin-only predictions, so it becomes possible to identify correctly the number of unpaired electrons and assign the ground-state configuration.
- For instance, $[Fe(OH_2)_6]^{3-}$ is paramagnetic with a magnetic moment of $5.9\mu_B$. As shown in Table 20.3, this value is consistent with there being five unpaired electrons (N = 5 and S =5/2), which implies a **high-spin $t_{2g}^3e_g^2$ configuration**.

Table 20.3 Calculated spin-only magnetic moments

Ion	Electron configuration	S	μ/μ_B Calculated	Experimental
Ti^{3+}	t_{2g}^1	$\frac{1}{2}$	1.73	1.7–1.8
V^{3+}	t_{2g}^2	1	2.83	2.7–2.9
Cr^{3+}	t_{2g}^3	$\frac{3}{2}$	3.87	3.8
Mn^{3+}	$t_{2g}^3e_g^1$	2	4.90	4.8–4.9
Fe^{3+}	$t_{2g}^3e_g^2$	$\frac{5}{2}$	5.92	5.9

- The interpretation of magnetic measurements is sometimes less straightforward than the above examples might suggest.

- ✓ For example, the potassium salt of $[Fe(CN)_6]^{3-}$ has $\mu = 2.3\mu_B$, which is between the spin-only values for one and two unpaired electrons ($1.7\mu_B$ and $2.8\mu_B$, respectively). In this case, the ***spin-only assumption has failed*** because the orbital contribution to the magnetic moment is substantial.

Table 7.3 Calculated magnetic moments, in Bohr magnetons, of the ground states of some free transition-metal ions and the experimental magnetic moments for the octahedrally co-ordinated ions in their high-spin configurations

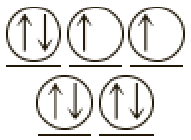
Ion	S	L	n	n′	μ_S	μ_{total}	μ_{exp}
Ti^{3+}	$\frac{1}{2}$	2	1	1	1.73	3.00	1.7 – 1.8†
V^{4+}	$\frac{1}{2}$	2	1	1	1.73	3.00	1.7 – 1.8†
V^{3+}	1	3	2	2	2.83	4.47	2.6 – 2.8†
Cr^{3+}	$\frac{3}{2}$	3	3	3	3.87	5.20	3.7 – 3.9
Cr^{2+}	2	2	4	4	4.90	5.48	4.7 – 4.9
Mn^{3+}	2	2	4	4	4.90	5.48	4.9 – 5.0
Mn^{2+}	$\frac{5}{2}$	0	5	5	5.92	5.92	5.6 – 6.1
Fe^{3+}	$\frac{5}{2}$	0	5	5	5.92	5.92	5.7 – 6.0
Fe^{2+}	2	2	6	4	4.90	5.48	5.1 – 5.5†
Co^{3+}	2	2	6	4	4.90	5.48	ca. 5.4†
Co^{2+}	$\frac{3}{2}$	3	7	3	3.87	5.20	4.1 – 5.2†
Ni^{2+}	1	3	8	2	2.83	4.47	2.8 – 3.5
Cu^{2+}	$\frac{1}{2}$	2	9	1	1.73	3.00	1.7 – 2.2

†The configurations marked † are those for which there can be an orbital contribution to the magnetic moment, see text. n = number of d electrons, n′ = number unpaired.

❑ Describe the d-electron distributions of the complexes $[Ni(NH_3)_4]^{2+}$ and $[Ni(CN)_4]^{2-}$, according to crystal field theory. The tetraamminenickel(II) ion is *paramagnetic*, and the tetracyanonickelate(II) ion is *diamagnetic*.

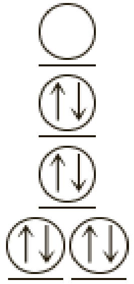
….The complex ions given in the problem are 4 C.N.: ***either tetrahedral or square planar***.

- **The tetrahedral field to give high-spin complexes and the square planar field to give low-spin complexes**. Therefore, the geometry of the $[Ni(NH_3)_4]^{2+}$ ion, which is paramagnetic (8 electrons are **NOT** paired), is probably tetrahedral. The distribution of d electrons in the Ni^{2+} ion (configuration d^8) is:



$[Ni(NH_3)_4]^{2+}$(tetrahedral)

The geometry of $[Ni(CN)_4]^{2-}$, which is diamagnetic (all 8 electrons are **paired**), is probably square planar; the distribution of d electrons



$[Ni(CN)_4]^{2-}$(square planar)

Q. Predict the number of unpaired spins in the $[Cr(en)_3]^{2+}$ ion. Calculate magnetic moment.

Strategy The magnetic properties of a complex ion depend on the ***strength of the ligands***. Strong-field ligands cause a high degree of splitting among the d orbital energy levels, resulting in low-spin complexes. Weak-field ligands, which cause a small degree of splitting among the d orbital energy levels, result in high-spin complexes.

Solution

The electron configuration of Cr^{2+} is $[Ar]3d^4$. Because ***en*** *is a* ***strong-field ligand***, we expect $[Cr(en)_3]^{2+}$ to be a low-spin complex. All four electrons will be placed in the lower-energy d orbitals (d_{xy}^2, d_{yz}^1, and d_{xz}^1) and there will be *two* ***unpaired spins*** *and hence paramagnetic complex*.

- **But the *complex ion is diamagnetic*. Use *ideas from the crystal field theory to speculate* on its probable structure.**

❖ **To determine whether a complex is high spin or low spin:**

- ❑ Look at the **transition metal**.
- ✓ ***First row metals will want to be high spin*** unless the ligand(s) force(s) it to be low spin.
- ✓ ***Second row metals want to be low spin*** unless the ligand(s) forces it high spin.
- ✓ ***Third row metals will be low spin***.
- ❑ Look at the **ligand(s)** as they will be the ultimate determining factor.
- ✓ If the ligand matches the transition metal in terms of high spin/low spin, then the complex's spin will be whatever is "agreed" upon.
- ✓ *If they differ*, ***follow the ligand's spin type***. If the ligand is classified *neither as high nor low spin*, follow the transition metal's spin type. Ligand spin types are enumerated below.
- ✓ If there are *multiple ligands with differing spin types*, *go with whichever spin type is most abundant* in the complex.
- ➢ The ligands below are ranked ***from low spin*** (greatest energy difference between t_{2g} and e_g) to ***high spin*** (lowest energy difference):

$$\mathbf{CO > CN^- > PR_3 > ethylene\ diamine\ (en) > NH_3 > H_2O > F^- > Cl^- > Br^- > I^-}$$

To illustrate this concept, let's take the following complexes: $[Ni(NH_3)_6]^{2+}$ and $[Ni(CN)_4]^{2-}$.
The nickel complexes all have the same oxidation state on the metal (2+), and thus the same d-electron count. In the first complex, nickel wants to be high spin, while ammonia prefers neither high nor low spin. Therefore, the complex will be high spin and emit blue light, which is an absorbance of orange—weak energy—light. For the second complex, nickel again wants to be high spin, but cyanide prefers low spin. As a result, the complex becomes low spin and will emit yellow light, which is an absorbance of violet—strong energy—light.

❖ Crystal Field Stabilization Energy (CFSE)

➢ A consequence of crystal field theory is that the distribution of electrons in the d orbitals may lead to net stabilization (decrease in energy) of some complexes depending on the specific ligand geometry and metal d-electron configurations. It is a simple matter to calculate this stabilization since all that is needed is the electron configuration and knowledge of the splitting patterns.

Definition: Crystal Field Stabilization Energy (CFSE)
The crystal field stabilization energy is defined as the energy of the electron configuration in the crystal field minus the energy of the electronic configuration in the spherical field.

➢ $\boldsymbol{CFSE = \Delta E = E_{\text{crystal field}} - E_{\text{spherical field}}}$

➢ The CSFE will depend on multiple factors including:

- Geometry (crystal field splitting pattern)
- Number of d-electrons
- Spin Pairing Energy (the contribution of spin pairing energy is often negligible in comparison to other contributions and is omitted at times)

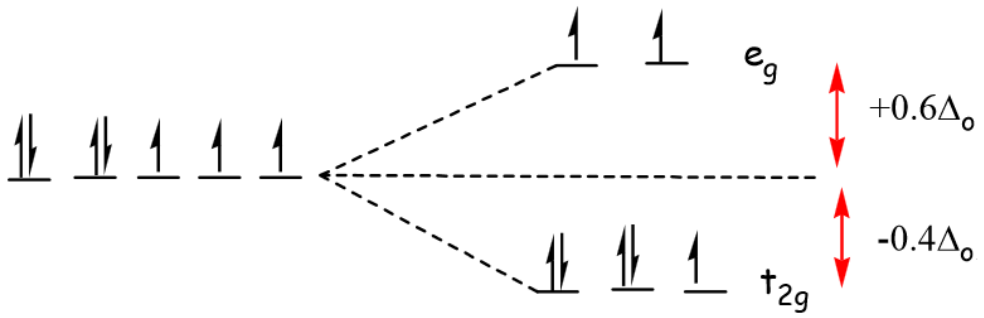
➢ For an octahedral complex, each of the more stable **t_{2g} orbitals are stabilized by −0.4Δo** relative to spherical field whereas the higher energy **e_g orbitals are destabilized by $+0.6\Delta_o$**. For a complex the stabilization or destabilization of each electron is summed. Any electrons that are paired within a single orbital, also contribute **spin pairing energy (+P)**. The overall CFSE for a complex is expressed as a multiple of the crystal field splitting parameter Δ_o.

Example 9.2.1: CFSE for a high Spin d^7 complex:

What is the CFSE for a high spin d^7 octahedral complex?

Solution

The splitting pattern and electron configuration for both spherical and octahedral ligand fields are compared below.



The ***energy of the spherical*** field is: $E_{spherical\ field}=7\times0+2P=2P$.

The ***energy of the octahedral*** crystal field $E_{oct\ field}$ is: $E_{oct\ field} = \{5\times(-0.4\Delta_o)\} + (2\times0.6\Delta_o) + 2P = -0.8\Delta_o + 2P$.

So, the **CFSE is = $E_{oct\ field} - E_{spherical\ field} = (-0.8\Delta_o + 2P) - 2P = -0.8\Delta_o$**

➢ Notice that the spin pairing energy cancels out in this case (and will when calculating the CFSE of high spin complexes) since the number of paired electrons in the octahedral field is the same as that in spherical field of the free metal ion.

Table 9.2.1: Crystal field stabilization energies (CFSE) for high and low spin octahedral complexes

Total d-electrons	Spherical Field	Octahedral Field				Crystal Field Stabilization Energy	
		High Spin		Low Spin			
	$E_{\text{spherical field}}$	Configuration	$E_{\text{crystal field}}$	Configuration	$E_{\text{crystal field}}$	High Spin	Low Spin
d^0	0	$t_{2g}^0\ e_g^0$	0	$t_{2g}{}^0 e_g{}^0$	0	0	0
d^1	0	$t_{2g}^1\ e_g^0$	-2/5 Δ_o	$t_{2g}{}^1 e_g{}^0$	-2/5 Δ_o	-2/5 Δ_o	-2/5 Δ_o
d^2	0	$t_{2g}^2\ e_g^0$	-4/5 Δ_o	$t_{2g}{}^2 e_g{}^0$	-4/5 Δ_o	-4/5 Δ_o	-4/5 Δ_o
d^3	0	$t_{2g}^3\ e_g^0$	-6/5 Δ_o	$t_{2g}{}^3 e_g{}^0$	-6/5 Δ_o	-6/5 Δ_o	-6/5 Δ_o
d^4	0	$t_{2g}^3\ e_g^1$	-3/5 Δ_o	$t_{2g}{}^4 e_g{}^0$	-8/5 Δ_o + P	-3/5 Δ_o	-8/5 Δ_o + P
d^5	0	$t_{2g}^3\ e_g^2$	0 Δ_o	$t_{2g}{}^5 e_g{}^0$	-10/5 Δ_o + 2P	0 Δ_o	-10/5 Δ_o + 2P
d^6	P	$t_{2g}^4\ e_g^2$	-2/5 Δ_o + P	$t_{2g}{}^6 e_g{}^0$	-12/5 Δ_o + 3P	-2/5 Δ_o	-12/5 Δ_o + P
d^7	2P	$t_{2g}^5\ e_g^2$	-4/5 Δ_o + 2P	$t_{2g}{}^6 e_g{}^1$	-9/5 Δ_o + 3P	-4/5 Δ_o	-9/5 Δ_o + P
d^8	3P	$t_{2g}^6\ e_g^2$	-6/5 Δ_o + 3P	$t_{2g}{}^6 e_g{}^2$	-6/5 Δ_o + 3P	-6/5 Δ_o	-6/5 Δ_o
d^9	4P	$t_{2g}^6\ e_g^3$	-3/5 Δ_o + 4P	$t_{2g}{}^6 e_g{}^3$	-3/5 Δ_o + 4P	-3/5 Δ_o	-3/5 Δ_o
d^{10}	5P	$t_{2g}^6\ e_g^4$	0 Δ_o + 5P	$t_{2g}{}^6 e_g{}^4$	0 Δ_o + 5P	0	0

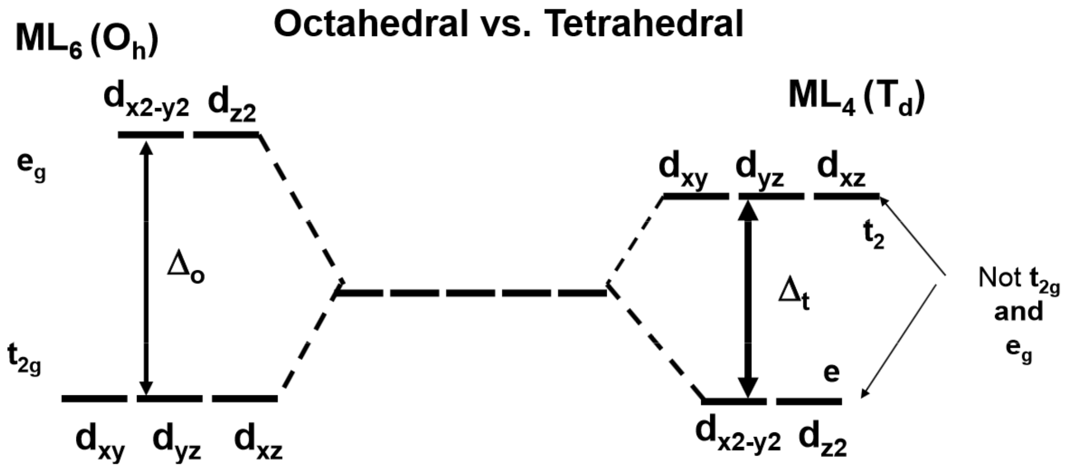
P is the spin pairing energy and represents the energy required to pair up electrons within the same orbital. For a given metal ion P (pairing energy) is **constant**, it **does not vary** with ligand and oxidation state of the metal ion).

Octahedral Preference

- Similar CFSE values can be constructed for non-octahedral ligand field geometries once the knowledge of the d-orbital splitting is known and the electron configuration within those orbitals known, e.g., the tetrahedral complexes in Table 9.2.2. These energies geometries can then be contrasted to the octahedral CFSE to calculate a thermodynamic preference for a metal-ligand combination to favor the octahedral geometry. This is quantified via an octahedral site preference energy defined below.

Definition: Octahedral Site Preference Energies (OSPE)

- The octahedral site preference energy (OSPE) is defined as the difference of CFSE energies for a non-octahedral complex and the octahedral complex. For comparing the preference of forming an octahedral ligand field vs. a tetrahedral ligand field, the OSPE is thus:
- OSPE = CFSE(oct) − CFSE(tet) (9.2.2)
- The OSPE quantifies the preference of a complex to exhibit an octahedral geometry vs. another geometry. The OSPE for octahedral vs tetrahedral geometries is illustrated below.



- Note: the conversion between Δo and Δt used for these calculations is:
- $\Delta t \approx 4\Delta_o/9$ (9.2.3)

which is applicable for comparing octahedral and tetrahedral complexes that involve same ligands only.

Table 9.2.2: Octahedral Site Preference Energies (OSPE)

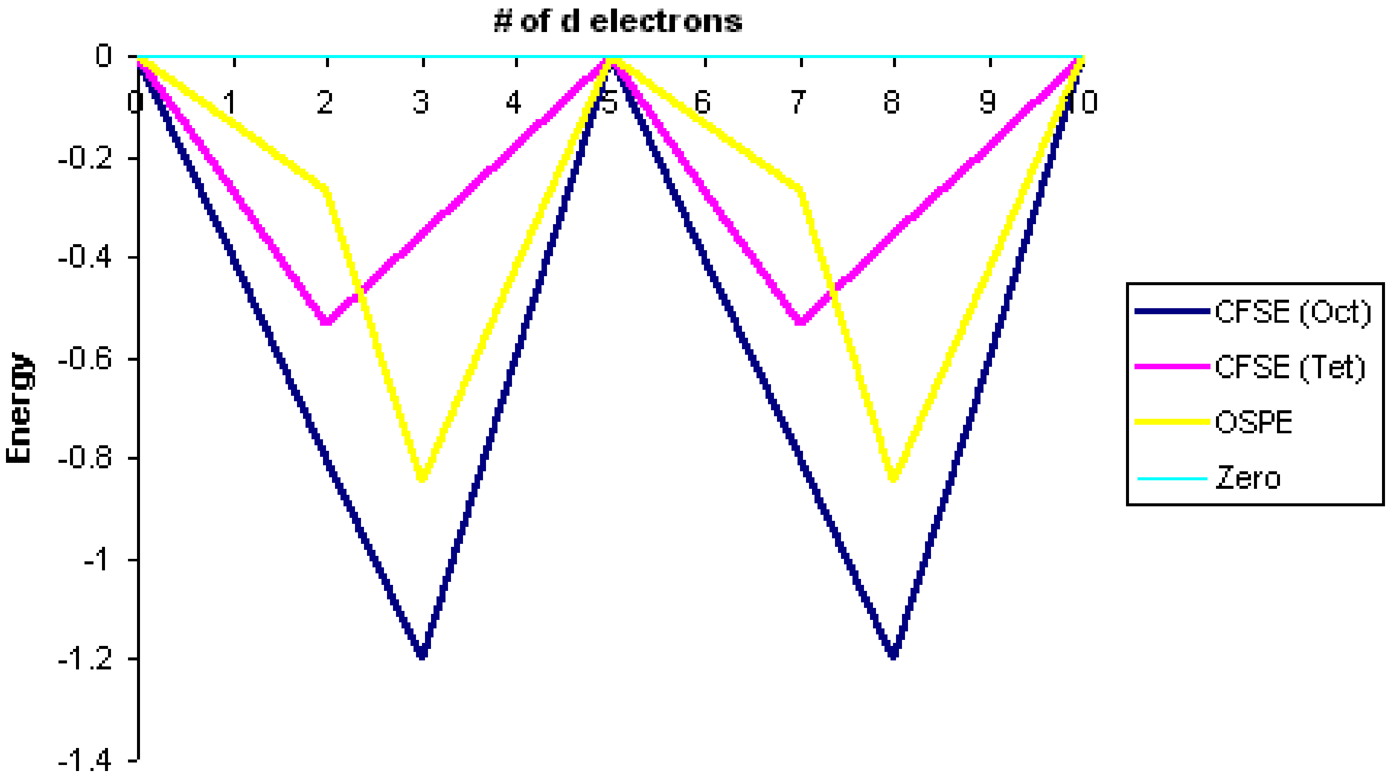
Total d-electrons	CFSE(Octahedral)		CFSE(Tetrahedral)		OSPE (for high spin complexes)**
	High Spin	Low Spin	Configuration	**Always High Spin***	
d^0	0 Δ_o	0 Δ_o	e^0	0 Δ_t	0 Δ_o
d^1	-2/5 Δ_o	-2/5 Δ_o	e^1	-3/5 Δ_t	-6/45 Δ_o
d^2	-4/5 Δ_o	-4/5 Δ_o	e^2	-6/5 Δ_t	-12/45 Δ_o
d^3	-6/5 Δ_o	-6/5 Δ_o	$e^2t_2^1$	-4/5 Δ_t	-38/45 Δ_o
d^4	-3/5 Δ_o	-8/5 Δ_o + P	$e^2t_2^2$	-2/5 Δ_t	-19/45 Δ_o
d^5	0 Δ_o	-10/5 Δ_o + 2P	$e^2t_2^3$	0 Δ_t	0 Δ_o
d^6	-2/5 Δ_o	-12/5 Δ_o + P	$e^3t_2^3$	-3/5 Δ_t	-6/45 Δ_o
d^7	-4/5 Δ_o	-9/5 Δ_o + P	$e^4t_2^3$	-6/5 Δ_t	-12/45 Δ_o
d^8	-6/5 Δ_o	-6/5 Δ_o	$e^4t_2^4$	-4/5 Δ_t	-38/45 Δ_o
d^9	-3/5 Δ_o	-3/5 Δ_o	$e^4t_2^5$	-2/5 Δ_t	-19/45 Δ_o
d^{10}	0	0	$e^4t_2^6$	0 Δ_t	0 Δ_o

P is the spin pairing energy and represents the energy required to pair up electrons within the same orbital.

Note:
Tetrahedral complexes are always high spin since the splitting is appreciably smaller than P (Equation 9.2.3).

After conversion with Equation 9.2.3. The data in Tables 9.2.1 and 9.2.2 are represented graphically by the curves in Figure 9.2.1 below for the high spin complexes only. The low spin complexes require knowledge of P to graph, which will vary depending on the identity of the metal ion.

Figure 9.2.1: Crystal Field Stabilization Energies for both high spin octahedral fields (CFSEoct) and tetrahedral fields (CFSEtet). Octahedral Site Preference Energies (OSPE) are in yellow. This is for high spin complexes only.



From a simple inspection of Figure 9.2.1, the following observations can be made:

- The OSPE is small in d^1, d^2, d^5, d^6, d^7 high spin complexes and other factors influence the stability of the complexes including steric factors.
- The **OSPE is large in d^3 and d^8 high spin complexes** which strongly favor octahedral geometries

- It is apparent that, in terms of CFSE (LFSE), ***octahedral geometries are strongly preferred over tetrahedral for d^3 and d^8 complexes***: chromium(III) (d^3) and nickel(II) (d^8) do indeed show an exceptional preference for octahedral geometries.

- *Similarly, d^4 and d^9 configurations show a preference for octahedral complexes (for example Mn(III) and Cu(II); note that the Jahn–Teller effect enhances this preference),*

- ✓ whereas *tetrahedral complexes* of d^1, d^2, d^6, and d^7 ions *will **not be too disfavored***; thus V(II) (d^2) and Co(II) (d^7) form tetrahedral complexes ($[MX_4]^{2-}$) with chloride, bromide, and iodide ligands.

❖ CFSE: Stabilization of Oxidation States

The standard electrode potential for the reduction of Co(III) to Co(II) is

Co^{3+}(**aq**) + e^- → Co^{2+}(**aq**) E^0 = +1.82 V (*Compare CFSE of* $[Co(H_2O)_6]^{3+}$ vs. $[Co(H_2O)_6]^{2+}$)

This large positive value suggests that Co^{3+}(aq) is a strong oxidizing agent, strong enough to oxidize water to O_2 (g)

4 Co^{3+}(aq) + 2 H_2O(l) → Co^{2+}(aq) + O_2(g) + 4 H^+(aq) E^0 = +0.58 V

Yet one of the complex ions, $[Co(NH_3)_6]^{3+}$, is stable in water solution, even though it contains cobalt in the +3 oxidation state. **$[Co(NH_3)_6]^{2+}$ is easily oxidized to Co(III) complex**. *On the other hand*, $[Co(H_2O)_6]^{2+}$ is a stable complex.

This phenomenon can be explained as follows: Co^{2+} is a d^7 electron system whereas Co^{3+} is a d^6 one.

- ➢ ***H_2O is a weak field ligand,*** *and* ***we expect a high spin (HS) complex*** with it. The HS d-electron configuration for d^7 and d^6 are, respectively: $t_{2g}^5e_g^2$ and $t_{2g}^4e_g^2$. The crystal field stabilization energy (CFSE) is higher for $t_{2g}^5e_g^2$ configuration, i.e., d^7, and hence Co^{2+} is preferred for the ligand, H_2O.
- ➢ On the other hand, **NH_3** is a moderately ***strong field ligand, and we expect a low spin (LS) complex*** with it. The LS d-electron configurations for d^7 and d^6 are: $t_{2g}^6e_g^1$ and t_{2g}^6 respectively. The crystal field stabilization energy (CFSE) is higher for the t_{2g}^6 configuration, i.e., d^6 and hence Co^{3+} is preferred for the ligand, NH_3.

- **The ability of a strong electron-pair donor** (strong Lewis base) to stabilize high oxidation states in the way that NH_3 does in Co(III) complexes and O^{2-} in Mn(VII) complexes (such as MnO_4^-) **affords a means of attaining certain oxidation states that might otherwise be difficult or impossible to attain**.

❑ **Tetragonally Distorted Octahedral** and **Square Planar** Crystal Fields

- One nice feature of crystal field theory is that can readily explain ***distortions such as the tetragonal distortion*** of octahedral complexes (Fig. 7.1.11).

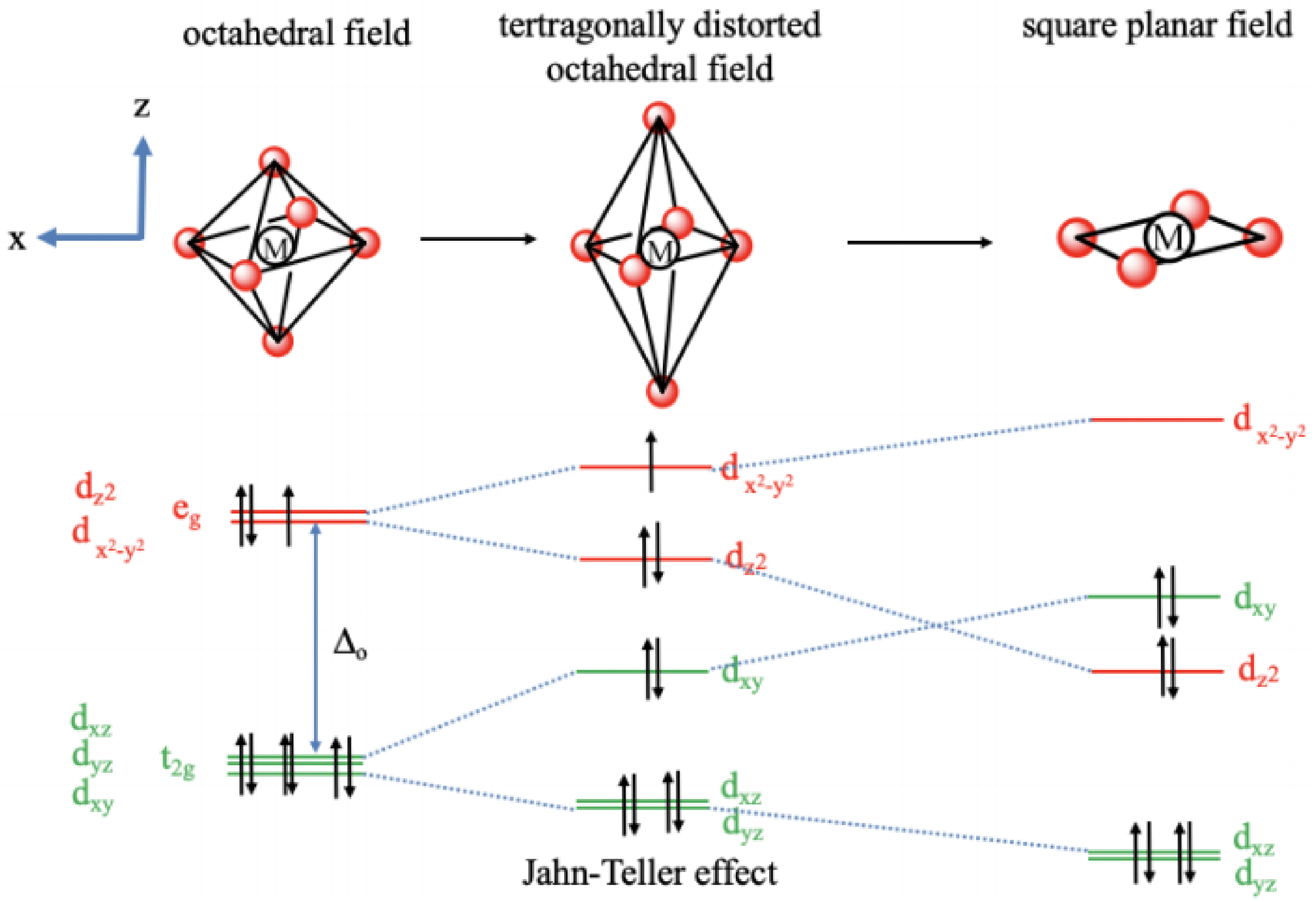
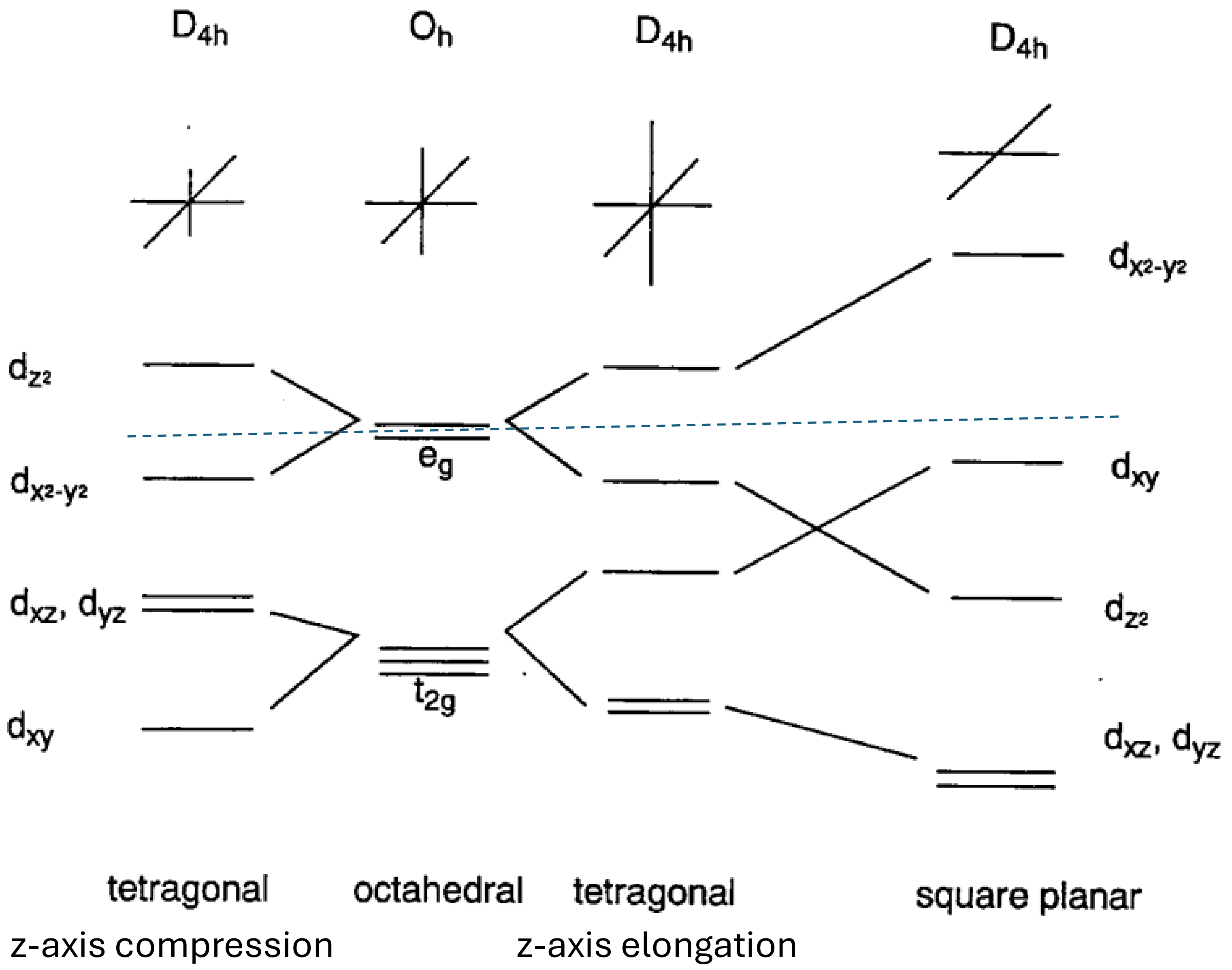


Figure 7.1.11 Tetragonal distortions in the octahedral crystal field leading to a square planar field.

Figure 6.2.7: Change of the orbital energy from octahedral to square planar complexes.

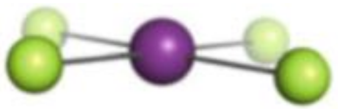


- In a ***tetragonally distorted complex there is a tetragonally distorted crystal field***.
- In an ***elongated octahedron two ligands*** are further away from the metal than the four others. Let us assume these two ligands are on the z-axis. Then, the crystal field is weaker on the z-axis. To keep the overall crystal field constant we must bring the other four ligands closer to the metal center. That means that we compress along the x and the y-axis. What will happen to the energy of the orbitals as the octahedron distorts? Because we elongate in the z-direction, the d_{z^2} orbital, that has most electron density on the z-axis goes down in energy. Because we compress on the x and the y-axis, the energy of the $d_{x^2-y^2}$ orbital increases.
- What about the t_{2g} orbitals? The d_{xy} orbital goes up in energy because it has electron density in the xy plane, but not along the z-axis. The d_{yz} and d_{zx} decrease in energy because they have a significant electron density in z-direction, and the electron density in z-direction is the same for both orbitals.
- We can now also think about, if the crystal field theory can explain why tetragonal distortion occurs preferentially for certain electron configurations of the metal. For example, it is known that metal ions with d^9 electron configuration often make octahedral complexes with tetragonal distortions. For instance, the hexaaqua copper (2+) complex is an example of a tetragonally distorted complex with a Cu^{2+} ion that has a d^9 electron configuration.
- We can understand that the **tetragonal distortion occurs when comparing the energy of the electrons in the undistorted vs the distorted octahedron**. *In the undistorted octahedron we have three electrons in the degenerated e_g orbitals*. **As we distort, we can move two electrons in the energetically lower d_{z^2}-orbital and fill the third one into the energetically higher $d_{x^2-y^2}$-orbital**. Thus, **overall, the electrons have a lower energy explaining the distortion**. This is an example of the **Jahn-Teller effect**.
- In general, ***the Jahn-Teller effect can occur when there are partially occupied degenerate orbitals***. In this case a molecule can lower its energy through distortion. Note that not only partial occupation of the e_g-orbitals, but also ***partial occupation of the t_{2g}-orbitals can cause the Jahn-Teller effect, although the effect is typically smaller***. For example in complexes with metal ions the d^4-electron configuration, all four electrons can be stabilized through Jahn-Teller distortion. It should also be noted that in addition to an elongation, there is also the possibility of compressing the octahedron along the z-axis. In this case the order of energy of the d_{z^2} and the $d_{x^2-y^2}$ orbital reverses, and the order of

- Finally, let us look at the ***square planar crystal field in square planar complexes***.
- To understand the square planar crystal field it helps to understand the square planar shape as an infinitely elongated octahedron. If we move the two ligands along the z-axis infinitely far away from the metal ion, then we have created a square planar structure.
- How will the orbital energies change compared to an elongated octahedron?
- The $d_{x^2-y^2}$ orbital will have an even higher energy due to the necessity to compensate for the decreased field associated with the ligands on the z-axis by further compressing along the x and the y-axis. The d_{z^2}-orbital is even further decreased in energy because the ligands along the z-axis are now completely gone. The d_{xy} orbital is increased in energy because of the enhanced field in the xy-plane. It is now higher than the d_{z^2}-orbital. The d_{yz} and the d_{zx} orbitals are further decreased in energy because they have significant electron density in z-direction.
- Viewing the square planar shape as an extreme case of a tetragonally elongated octahedron also lets us understand why the square planar shape is so often adopted by d^8-metal complexes.
- We can see that the stabilization energy, and thus the tendency to distort is the greatest when two electrons are in the metal e_g orbitals. In this case two electrons lower their energy through distortion and no electron has an increased energy. Thus, we would understand that the distortion becomes so great, so that the octahedral complex eventually loses two ligands and adopts the square planar shape. This is a nice example how crystal field theory can explain shapes and the number of bonds in a complex without actually being a bonding theory.

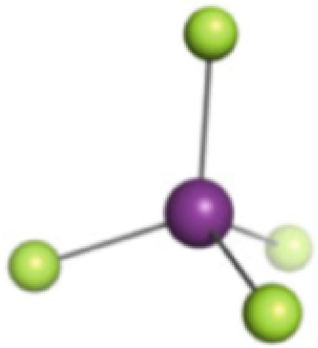
❑ Coordination number four: Square Planar Complex

➢ *Which of the following geometries is the* **less** *sterically demanding arrangement?*

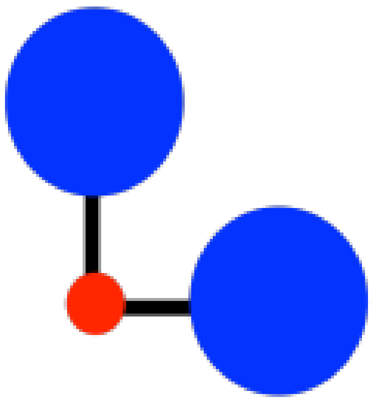


Square-planar

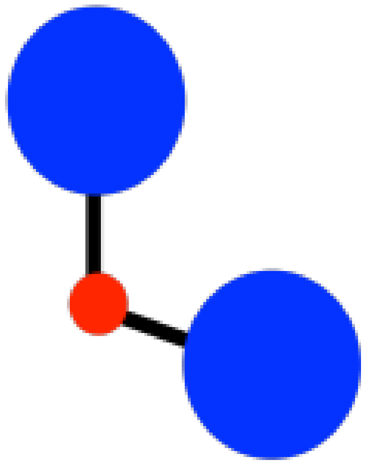
vs.



Tetrahedral



square planar
90^o



tetrahedral
109^o

- ➢ In a tetrahedron, all the ligands are 109.5^o from each other.
- ➢ In a square planar geometry, the ligands are only 90^o from each other.
- ➢ Tetrahedral geometry is always less crowded than square planar, so that factor always provides a bias toward tetrahedral geometry.
- ➢ As a result, we might expect *square planar geometry to occur only when steric is heavily outweighed by ligand field stabilization energy*.

- Complexes with four ligands in a plane containing the central metal are termed square planar complexes.
- It is easier to understand the electronic energy levels of the d orbitals in square planar complexes by starting from those for hexacoordinate octahedral complexes.
- Placing the six ligands along the Cartesian axes, the two ligands on the z-axis are gradually removed from the central metal, and finally, only four ligands are left on the xy plane.
- The interaction of the two z-coordinate ligands with the d_{z^2}, d_{xz}, and d_{yz} orbitals becomes smaller, and the energy levels of these ligands lower.
- On the other hand, the remaining four ligands approach the metal, and the $d_{x^2-y^2}$ and d_{xy} energy levels rise as a result of the removal of the two ligands.
- This results in the order of the energy levels of five d orbitals being d_{xz}, $d_{yz} < d_{z^2} < d_{xy} << d_{x^2-y^2}$ (Figure 6.2.7).
- Rh^+, Ir^+, Pd^{2+}, Pt^{2+}, and Au^{3+} complexes with a d^8 configuration tend to form square planar structures because eight electrons occupy the lower orbitals leaving the highest $d_{x^2-y^2}$ orbital empty.

- The energy levels therefore split as indicated in Fig. 13.8a with the d_{z^2} level and the d_{xz} and d_{yz} levels falling below the others.
- The *$d_{x^2-y^2}$ and d_{xy} levels rise slightly as the metal-ligand distances in the xy plane shorten a little because of the decrease in repulsion from the z ligands*.
- If the z ligands are *removed completely* to give the ***square-planar configuration***, the energy level diagram of Fig. 13.8b results. Here, the d_{z^2} and d_{xy} levels have crossed over. Notice that, ***as the configuration in the xy plane is similar to that in the octahedron***, the energy separation between $d_{x^2-y^2}$ and d_{xy} remains practically the same as the **octahedral Δ_o**.

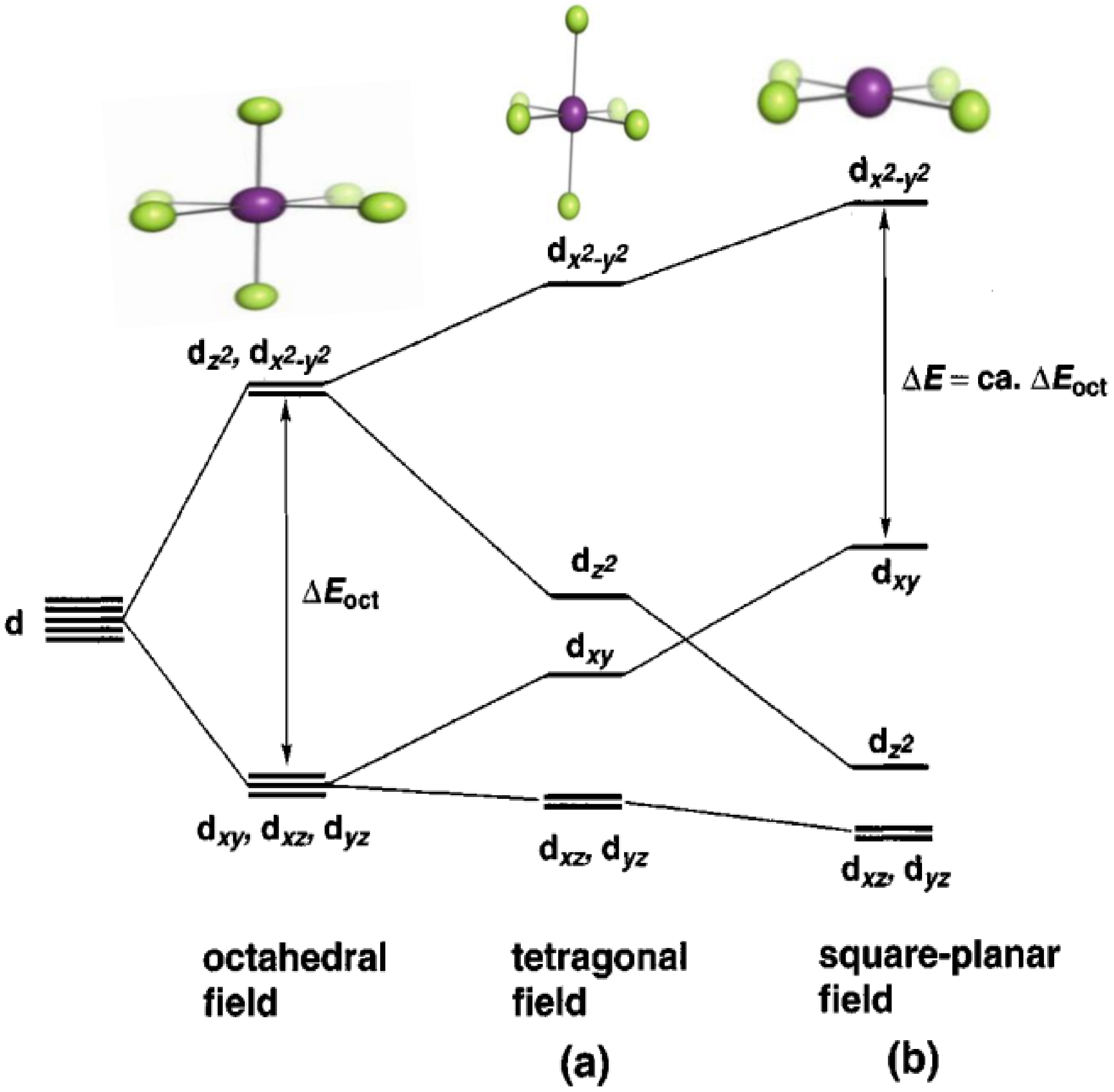


FIG. 13.8 Effect on orbital energies of removing ligands on the z axis, starting from an octahedron: (a) the tetragonal field with nonequivalent z ligands, (b) the square-planar field with no z ligands

Jahn-Teller Distortions

- The Jahn-Teller effect is a geometric distortion of a non-linear molecular system that reduces its symmetry and energy. This distortion is typically observed among octahedral complexes where the two axial bonds can be shorter or longer than those of the equatorial bonds. This effect can also be observed in tetrahedral compounds. This effect is dependent on the electronic state of the system.
- **Introduction**
- In 1937, Hermann Jahn and Edward Teller postulated a theorem stating that "stability and degeneracy are not possible simultaneously unless the molecule is a linear one," in regard to its electronic state. This leads to a break in degeneracy which stabilizes the molecule and by consequence, reduces its symmetry. Since 1937, the theorem has been revised which Housecroft and Sharpe have eloquently phrased as "**any non-linear molecular system in a degenerate electronic state will be unstable and will undergo distortion to form a system of lower symmetry and lower energy, thereby removing the degeneracy**."
- This is most commonly observed with transition metal octahedral complexes, however, it can be observed in tetrahedral compounds as well.
- For a given octahedral complex, the five d atomic orbitals are split into two degenerate sets when constructing a molecular orbital diagram. These are represented by the sets' symmetry labels: t_{2g} (d_{xz}, d_{yz}, d_{xy}) and eg (d_{z^2} and $d_{x^2-y^2}$). When a molecule possesses a degenerate electronic ground state, it will distort to remove the degeneracy and form a lower energy (and by consequence, lower symmetry) system. The octahedral complex will either elongate or compress the z ligand bonds as shown in Figure 9.3.1 below:

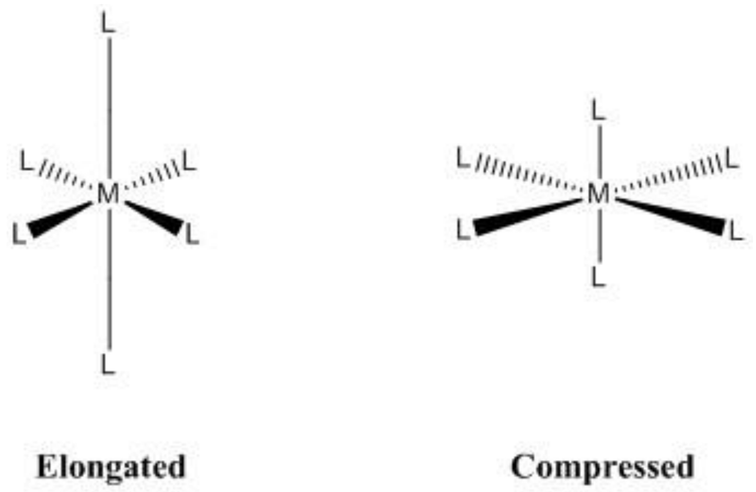


Figure 9.3.1: Jahn-Teller distortions for an octahedral complex.

- When an octahedral complex exhibits elongation, the axial bonds are longer than the equatorial bonds. For a compression, it is the reverse; the equatorial bonds are longer than the axial bonds. Elongation and compression effects are dictated by the amount of overlap between the metal and ligand orbitals. Thus, this distortion varies greatly depending on the type of metal and ligands. In general, the stronger the metal-ligand orbital interactions are, the greater the chance for a Jahn-Teller effect to be observed.

Elongation

- Elongation Jahn-Teller distortions occur when the degeneracy is broken by the stabilization (lowering in energy) of the d orbitals with a z component, while the orbitals without a z component are destabilized (higher in energy) as shown in Figure 9.3.2 below:

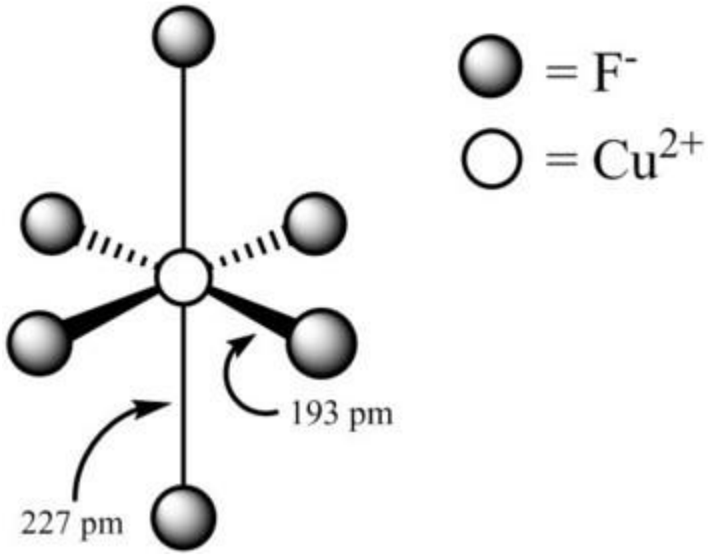


Figure 9.3.3: Structure of octahedral copper(II) fluoride.

Notice that the two axial bonds are both elongated and the four shorter equatorial bonds are the same length as each other. According the theorem, the orbital degeneracy is eliminated by distortion, making the molecule more stable based on the model presented in Figure 9.3.2.

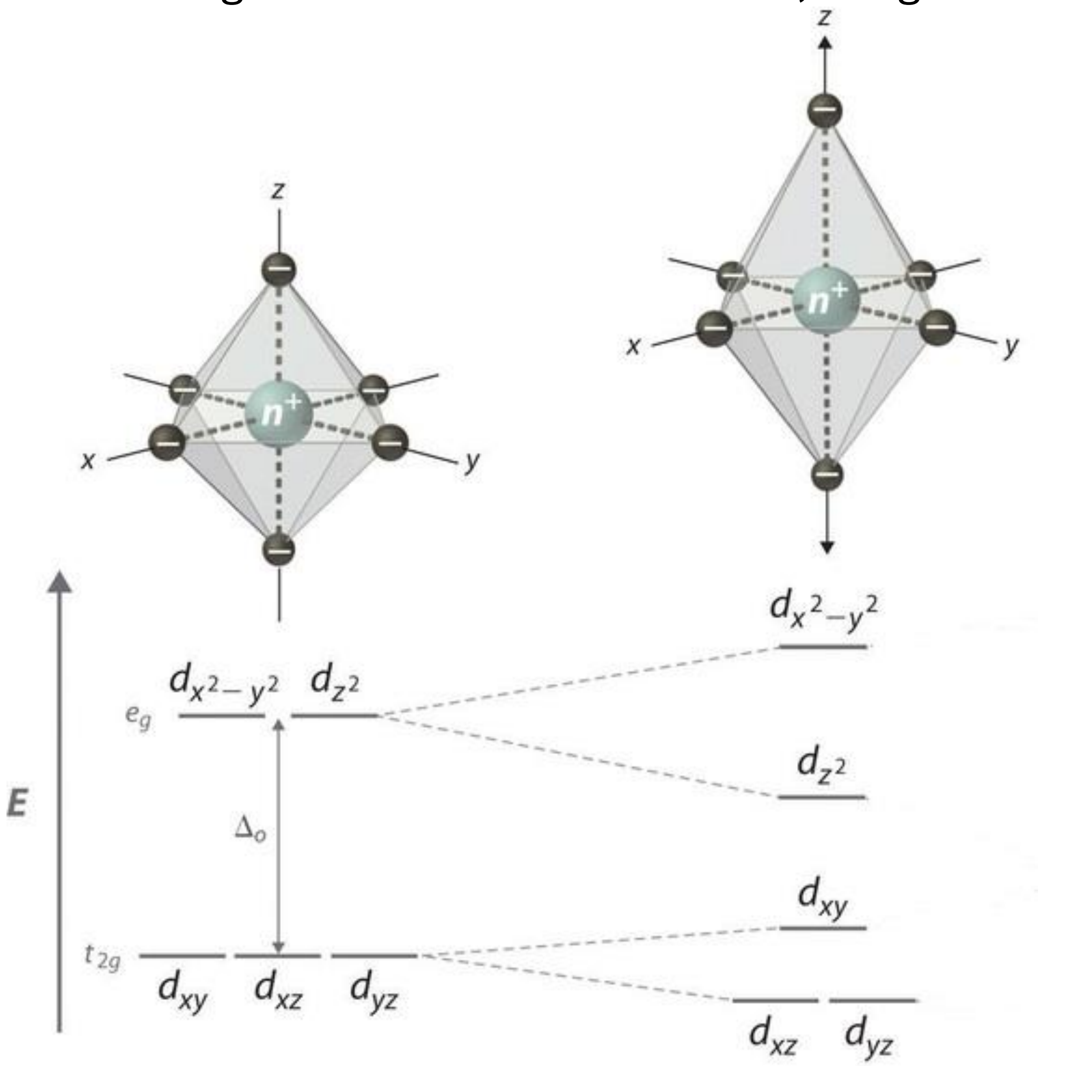
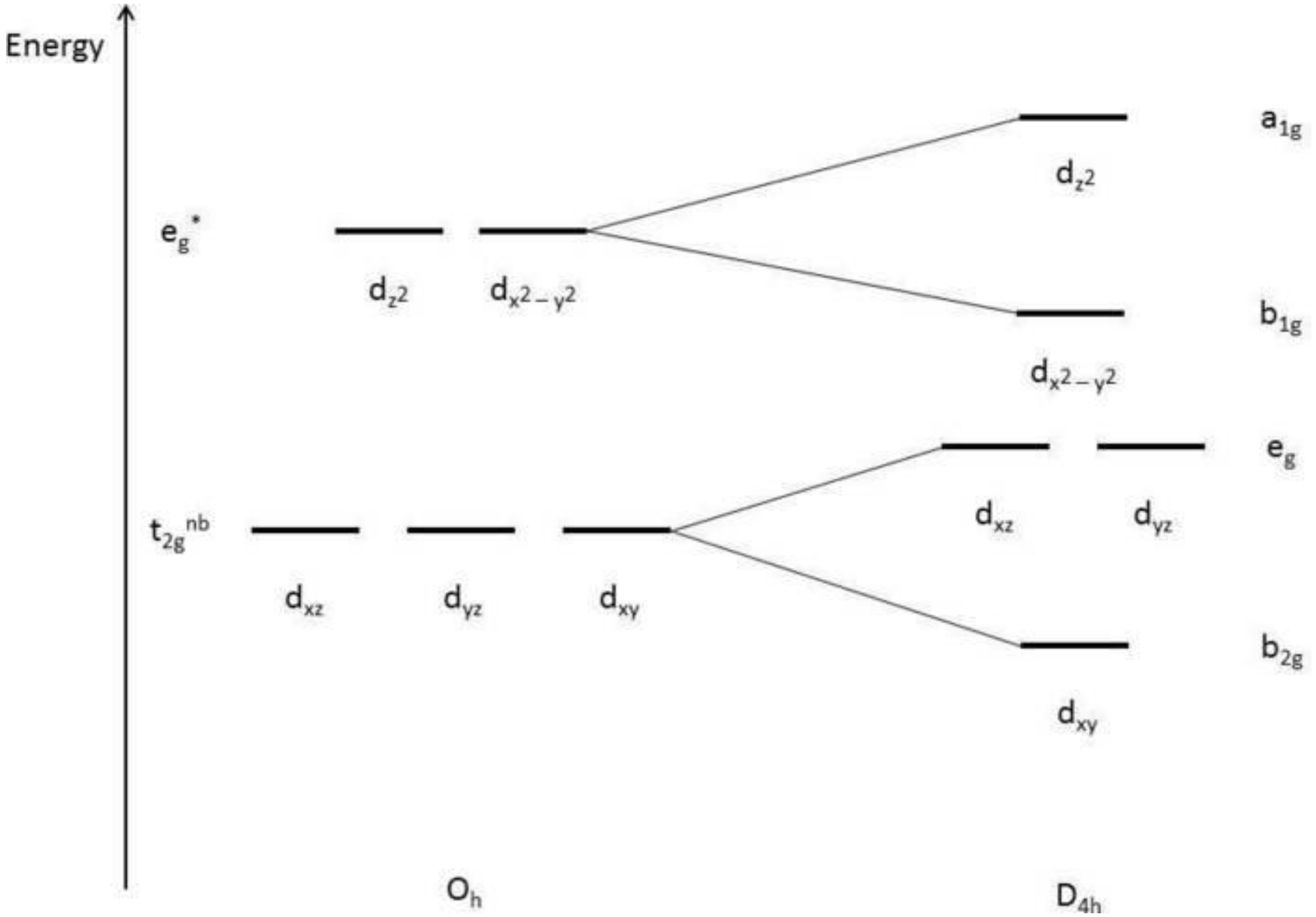


Figure 9.3.2: Illustration of tetragonal distortion (elongation) for an octahedral complex.

Compression

- Compression Jahn-Teller distortions occur when the degeneracy is broken by the stabilization (lowering in energy) of the d orbitals without a z component, while the orbitals with a z component are destabilized (higher in energy) as shown in Figure 9.3.4 below:



- This is due to the z-component d orbitals having greater overlap with the ligand orbitals, resulting in the orbitals being higher in energy. Since the dz^2 orbital is antibonding, it is expected to increase in energy due to compression. The d_{xz} and d_{yz} orbitals are still nonbonding but are destabilized due to the interactions.

Figure 9.3.4: Illustration of tetragonal distortion (compression) for an octahedral complex.

Electronic Configurations and Jahn-Teller Effects

- For Jahn-Teller effects to occur in transition metals there must be degeneracy in either the t2g or eg orbitals. The electronic states of octahedral complexes depend on the number of d-electrons and the splitting energy, Δ. When Δ is large and is greater than the energy required to pair electrons, electrons pair in t2g before occupying eg. On the other hand, when Δ small and is less than the pairing energy, electrons will occupy eg before pairing in t2g. The Δ of an octahedral complex is dictated by the chemical environment (ligand identity), and the identity and charge of the metal ion. If there electron configurations for any d-electron count is different depending on Δ, the configuration with more paired electrons is called low spin while the one with more unpaired electrons is called high spin.

- The electron configurations diagrams for d^1 through d^{10} with large and small Δ are illustrated in the figures below. Notice that the electron configurations for d^1, d^2, d^3, d^8, d^9, and d^{10} are the same no matter what the magnitude of Δ. Low spin and high spin configurations exist only for the electron counts d^4, d^5, d^6, and d^7.

- The figure below illustrates the electron configurations in the case of large Δ. The electron configurations highlighted in red (d^3, low spin d^6, d^8, and d^{10}) **do not** exhibit Jahn-Teller distortions. On the other hand d^1, d^2, low spin d^4, low spin d^5, low spin d^7, and d^9, would be expected to exhibit Jhan-Teller distortion. These electronic configurations correspond to a variety of transition metals. Some common examples include Cr^{3+}, Co^{3+}, and Ni^{2+}.

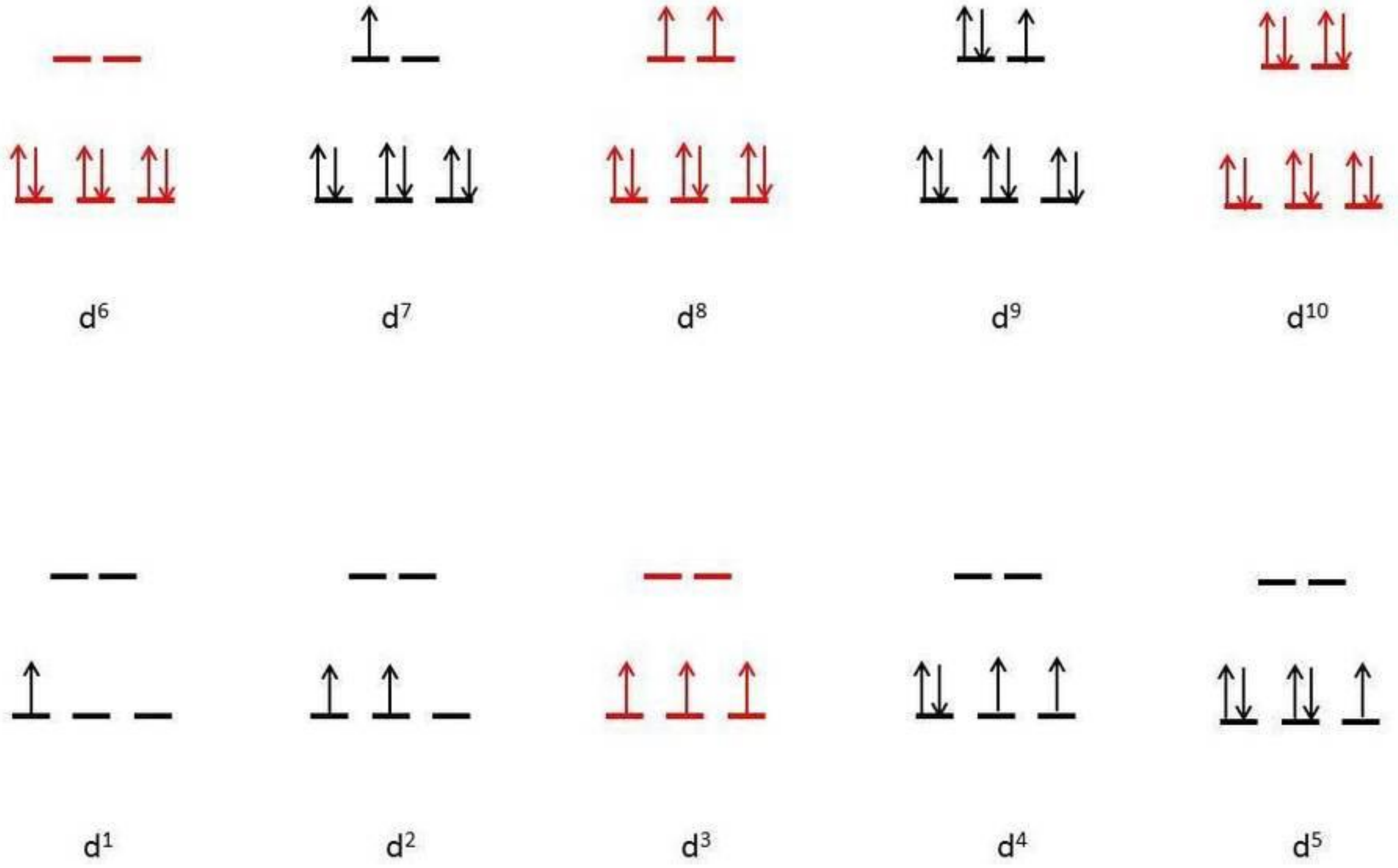
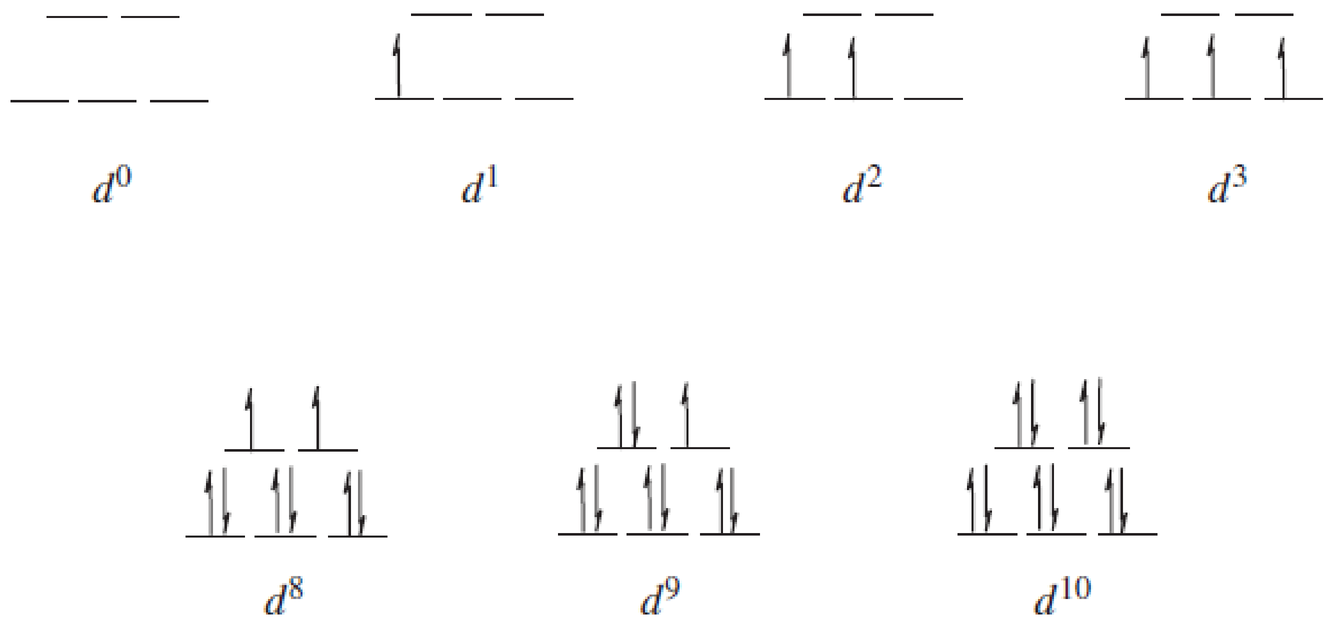
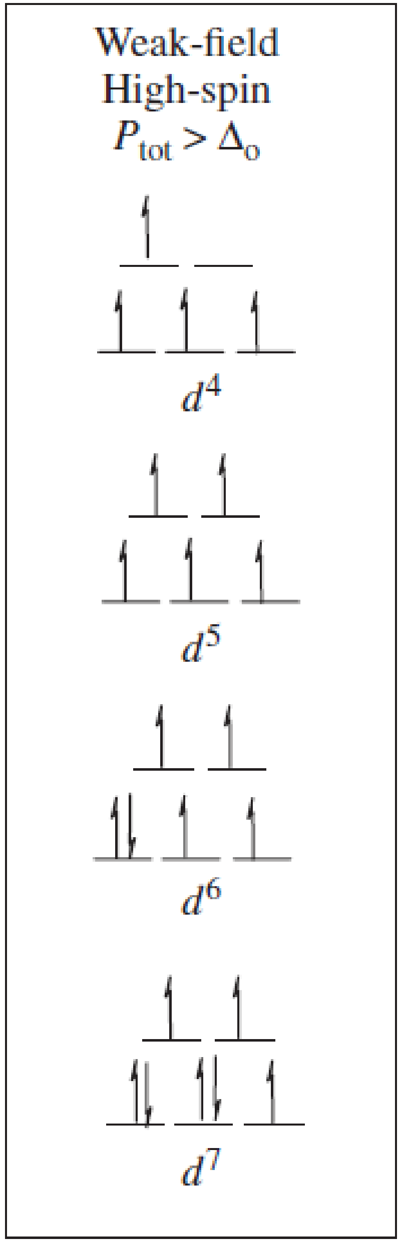
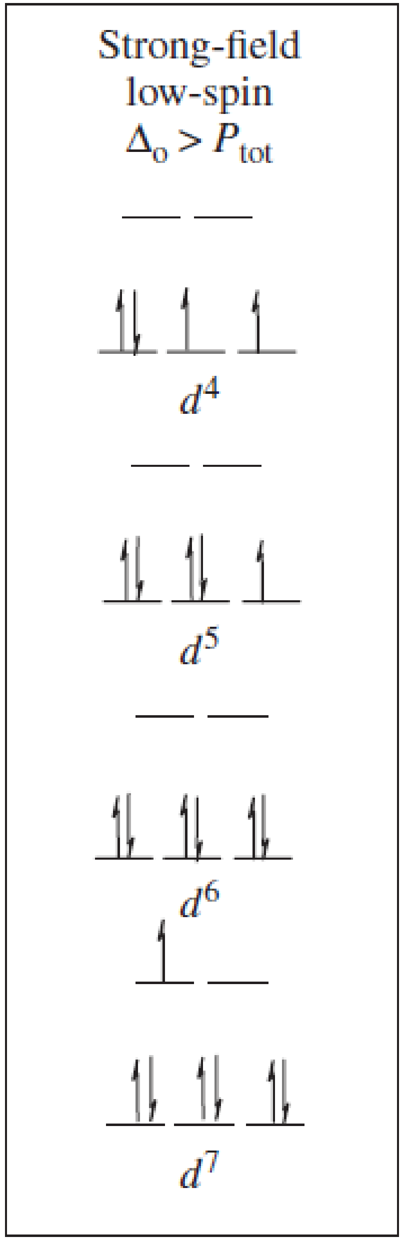


Figure 9.3.5: Electron configuration diagram of octahedral complexes (red indicates no degeneracies possible, thus no Jahn-Teller effects).

❑ Which d^n electron configurations would show a Jahn–Teller effect?



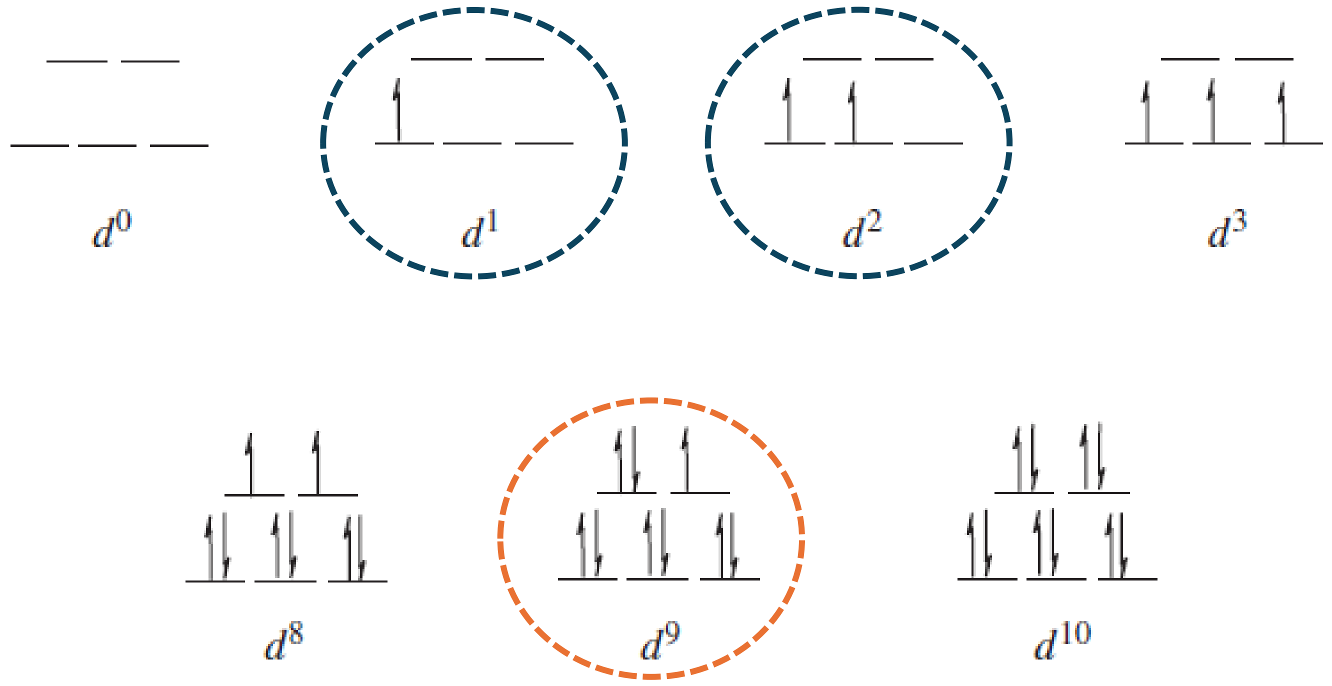


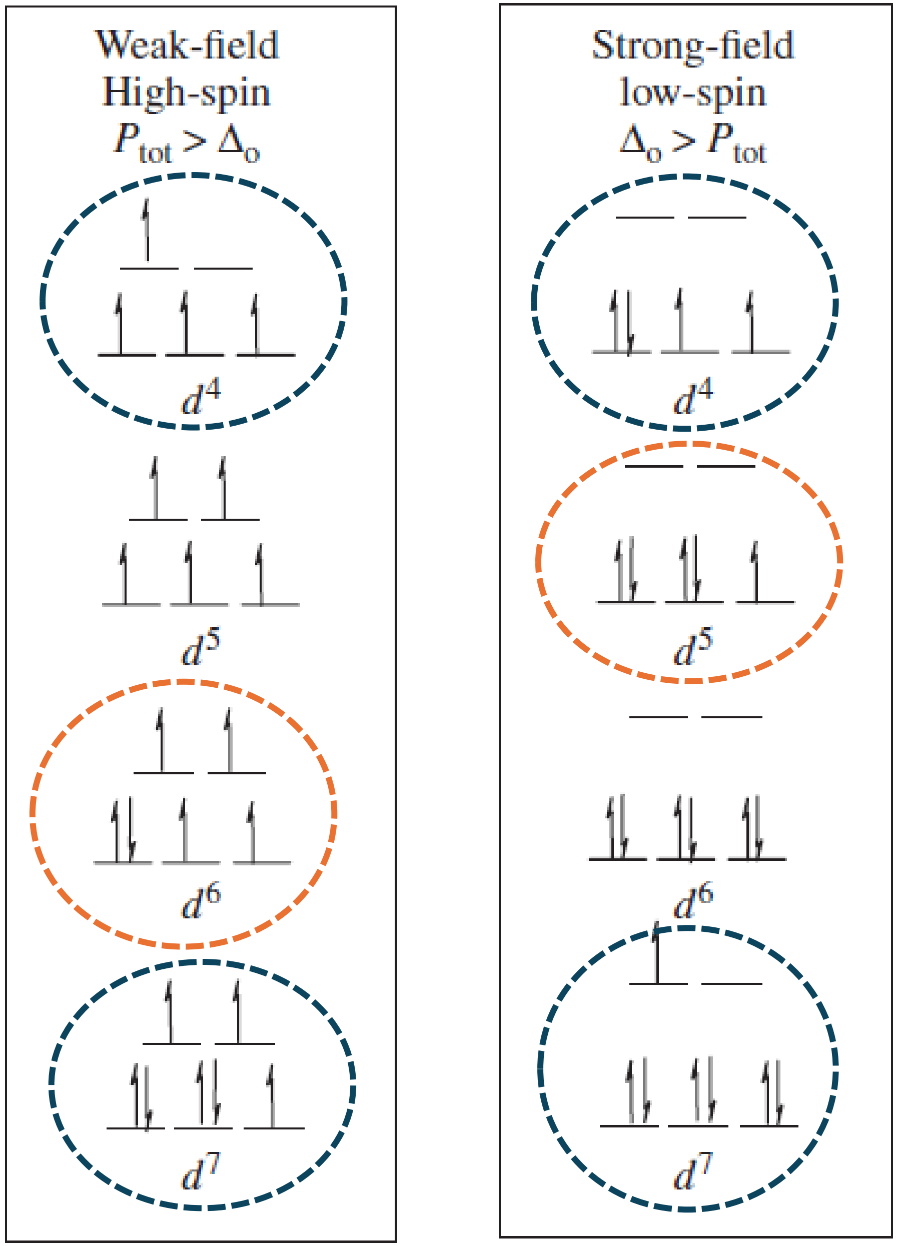


FIGURE…………

Crystal field splitting diagrams for the different d^n configurations in an **octahedral** CF

❑ Which d^n electron configurations would show a Jahn–Teller effect?





FIGURE…………

Crystal field splitting diagram for the different d^n configurations in an **octahedral** CF. Encircled d^n systems show JT effects.

❑ Which d^n electron configurations would show a Jahn–Teller effect?

Number of *d* electrons	*Electronic configuration*				
	e		t_2		
1	↑				
2	↑	↑			
3	↑	↑	↑		
4	↑	↑	↑	↑	
5	↑	↑	↑	↑	↑
6	↑↓	↑	↑	↑	↑
7	↑↓	↑↓	↑	↑	↑
8	↑↓	↑↓	↑↓	↑	↑
9	↑↓	↑↓	↑↓	↑↓	↑
10	↑↓	↑↓	↑↓	↑↓	↑↓

FIGURE...........
Crystal field splitting diagram for the different d^n configurations in a **tetrahedral** CF.

❑ Which d^n electron configurations would show a Jahn–Teller effect?

FIGURE…………

Crystal field splitting diagram for the different d^n configurations in a **tetrahedral** CF.

Number of d electrons	*Electronic configuration*				
	e		t_2		
1	↑				
2	↑	↑			
3	↑	↑	↑		
4	↑	↑	↑	↑	
5	↑	↑	↑	↑	↑
6	↑↓	↑	↑	↑	↑
7	↑↓	↑↓	↑	↑	↑
8	↑↓	↑↓	↑↓	↑	↑
9	↑↓	↑↓	↑↓	↑↓	↑
10	↑↓	↑↓	↑↓	↑↓	↑↓

❖ Two Types of Distortions:

- The Jahn–Teller effect identifies an unstable geometry (***a nonlinear complex with an orbitally degenerate ground state***); however, *it does not predict the preferred distortion*.

- For instance, with an octahedral complex, *instead of axial elongation and equatorial compression,* the degeneracy can also be removed by axial compression and equatorial elongation.

- Which distortion (elongation or compression) occurs in practice is **a matter of energetics**, not symmetry.

- However, *because axial elongation weakens two bonds, but equatorial elongation weakens four, axial elongation is more common than axial compression*.

❖ The Strength of Jahn-Teller Distortion

General Strength Trend of Jahn-Teller Distortion

For octahedral complexes:

- **weak** Jahn-Teller effect if $\mathbf{t_{2g}}$ is unevenly occupied
- **strong** Jahn-Teller effect if $\mathbf{e_g}$ **is unevenly** occupied

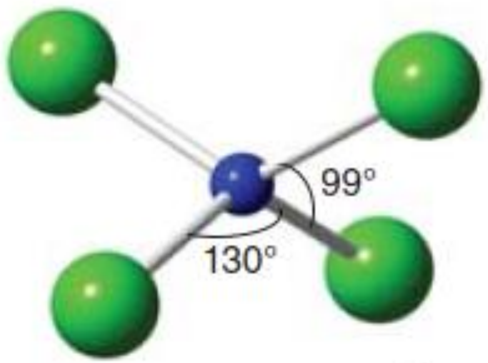
Table of Jahn-Teller Distortion Strength

Number of d-electrons	1	2	3	4	5	6	7	8	9	10
High-spin	w	w	-	s	-	w	w	-	s	-
Low-spin	w	w	-	w	w	-	s	-	s	-

The table of d-orbitals Jahn-Teller distortion strength w = weak Jahn-Teller effect expected; s = strong Jahn-Teller effect expected; - = no Jahn-Teller distortion

- ***A Jahn–Teller effect is possible for other electron configurations of octahedral complexes (the d^1, d^2, low-spin d^4 and d^5, high-spin d^6, and d^7 configurations)*** and for tetrahedral complexes (the d^1, d^3, d^4, d^6, d^8, and d^9 configurations).

- However, as *neither the t_{2g} orbitals in an octahedral complex* nor *any of the d orbitals in a tetrahedral complex point directly at the ligands*, ***the effect is normally much smaller***, as is any measurable distortion. Tetrahedral Cu^{2+} compounds often show a slightly 'flattened' tetrahedral geometry as seen for the $[CuCl_4]^{2-}$ anion in Cs_2CuCl_4.



The flattened $[CuCl_4]^{2-}$ anion

- Various forms of ***experimental verification*** of the Jahn–Teller effect include the ***electronic spectra***, ***ESR spectra*** of coordination compounds, ***X-ray crystallographic data*** showing different M–L bond lengths, and ***thermodynamic*** data.

- As an example (of thermodynamic data), consider ***the stepwise formation constants*** in Table 16.28 for $[Cu(NH_3)_6]^{2+}$, which has the d^9 electron configuration. As expected, the ***stability constants decrease with increasing n because there is less positive charge on the metal*** as it *becomes more saturated with ligands*. However, ***there is an abrupt drop in the stability of the Cu(II) compounds between n= 4 and n=5***. The reason that it is ***so difficult to add the fifth and sixth NH_3 ligands*** to Cu^{2+} is that the ***octahedral complex undergoes a "z-out" tetragonal distortion, which significantly weakens the two bonds along the z-axis***.

TABLE 16.28 Stepwise formation constants for $[Cu(NH_3)_n]^{2+}$.

K_1	2×10^4	K_4	2×10^2
K_2	4×10^3	K_5	3×10^{-1}
K_3	1×10^3	K_6	Very small

- A Jahn–Teller distortion can hop from one orientation to another and give rise to the dynamic Jahn–Teller effect. For example, below 20 K the EPR spectrum of $[Cu(OH_2)_6]^{2+}$ shows a static distortion (more precisely, one that is effectively stationary on the timescale of the resonance experiment). However, above 20 K the distortion disappears because it hops more rapidly than the timescale of the EPR observation.

❖Jahn-Teller Distortion and Spectra: Spectra of the First-order Jahn-Teller Distortion

➢ In the first-order Jahn-Teller distortion, two electron transitions can be observed. This phenomenon can lead to a "double-hump" result on the absorption spectra as shown in Figure 1.17.5 and 1.17.6.

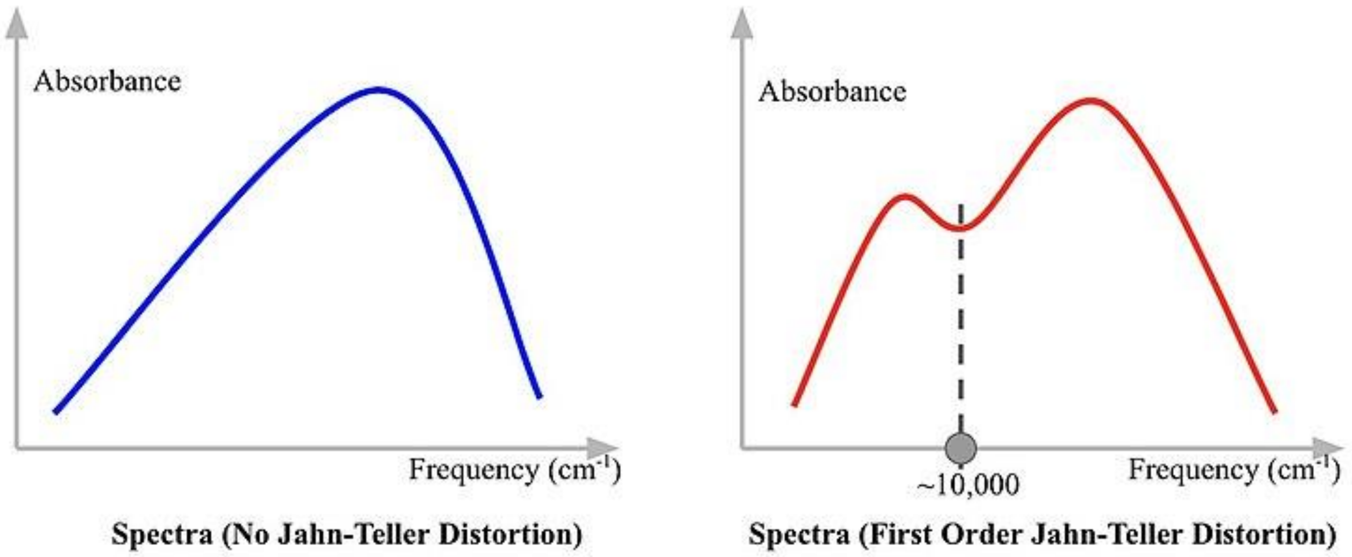


Figure 1.17.5: The absorption spectra of a first-order Jahn-Teller distortion

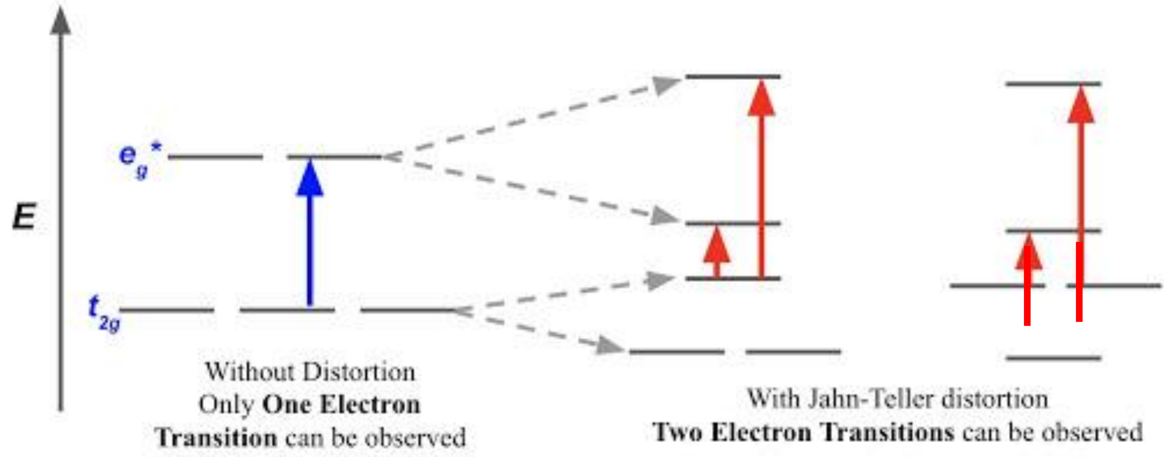
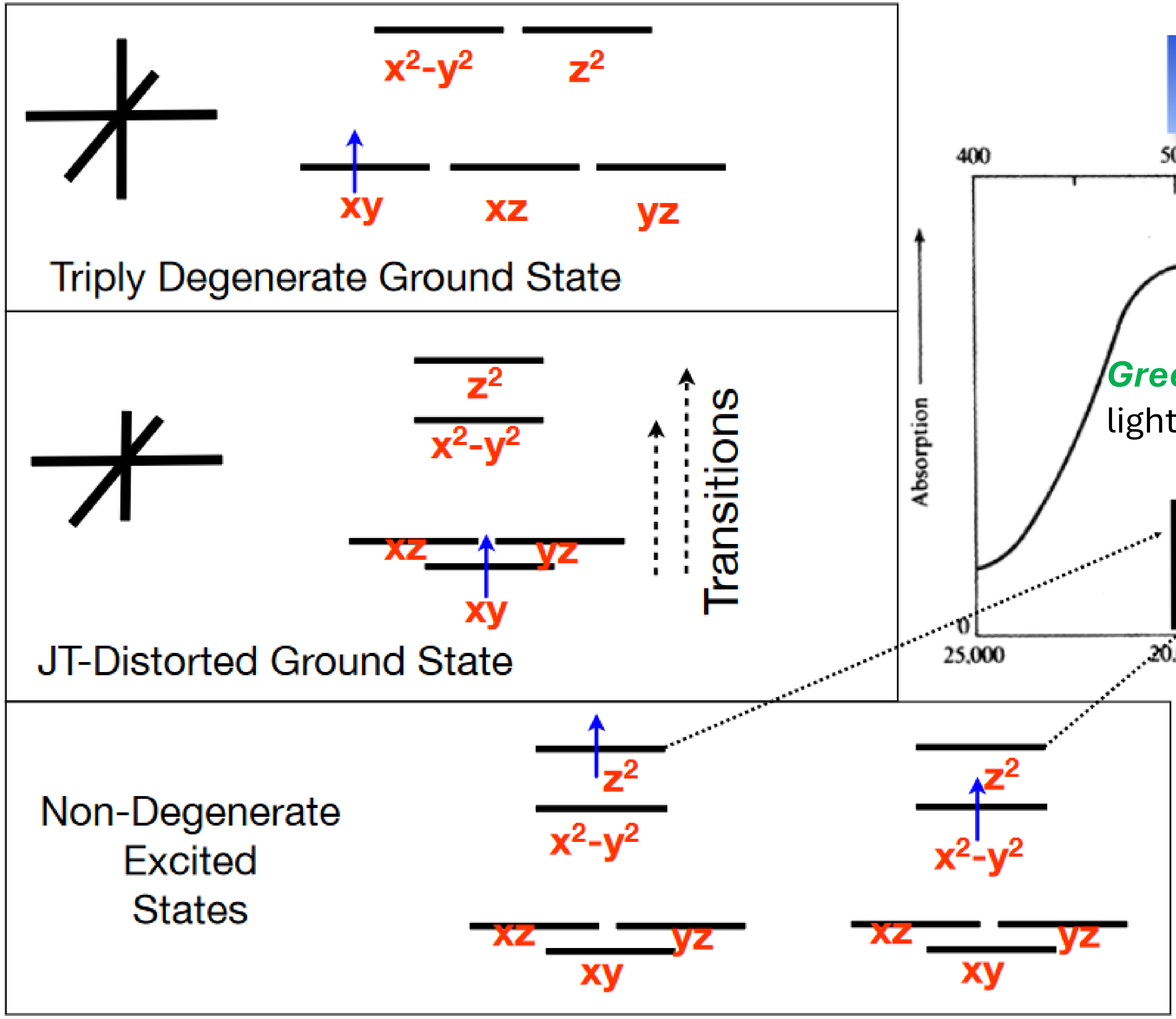


Figure 1.17.6: The electron transition in non-distorted (left) and distorted (right) orbitals

- Splitting of terms is observed in the ***electronic transitions*** of coordination compounds due to JT distortions. For instance, splitting of the terms for a d^1 $[TiCl_6]^{3-}$ ion as it undergoes a tetragonal distortion from octahedral symmetry, illustrating the ***two observed spin-allowed transitions*** for the complex. The electronic spectrum of this d^1 coordination compound is ***expected*** to have ***a single LF transition*** in the absence of Jahn–Teller splitting. In fact, the spectrum of the $[TiCl_6]^{3-}$ ion consists of ***two closely spaced peaks*** separated by approximately 1400cm^{-1}.

The Jahn-Teller Effect: An Example



$[Ti(H_2O)_6]^{3+}$

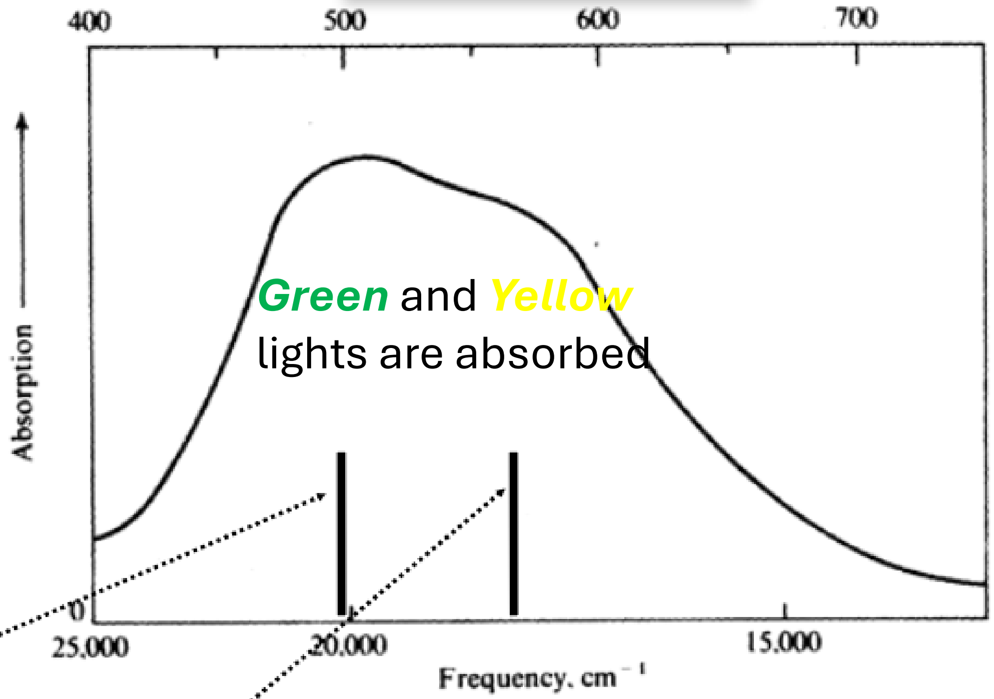


FIGURE 16.51. Splitting of the terms for a d^1 $[Ti(H_2O)_6]^{3+}$ ion as it undergoes a tetragonal distortion from octahedral symmetry, illustrating the two observed spin-allowed transitions for the complex. (Note: not drawn to scale.)

- These distortions are manifestations of the **Jahn–Teller effect**: *if the ground electronic configuration of a* ***nonlinear*** *complex is orbitally degenerate and asymmetrically filled, then the complex distorts so as to remove the degeneracy and achieve a lower energy.*

- The physical origin of the effect is quite easy to identify. Thus, *a tetragonal distortion of a regular octahedron*, *corresponding* to extension along the z-axis and compression on the x- and y-axes, *lowers the energy* of the $e_g(d_{z^2})$ orbital and increases the energy of the $e_g(d_{x^2-y^2})$ orbital (Fig. 20.12).

- For example, in a d^9 complex (with a configuration that would be $t_{2g}^6 e_g^3$ in O_h), such a distortion leaves two electrons in the d_{z^2} orbital with a lower energy and one in the $d_{x^2-y^2}$ orbital with a higher energy.

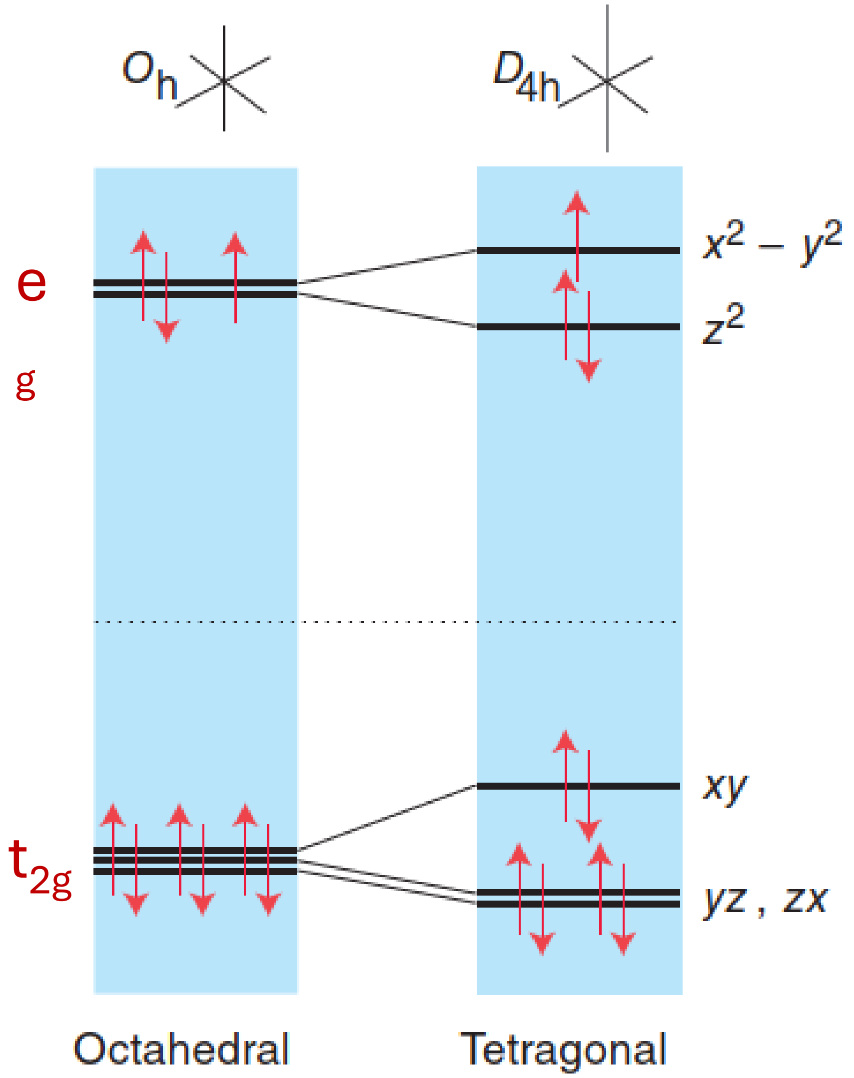


Figure 20.12 The effect of tetragonal distortions (compression along x and y and
extension along z) on the energies of d orbitals. The orbital occupation is for a d^9

❑ Square-planar complexes : Coordination number four

- Although a *tetrahedral arrangement of four ligands is the least sterically demanding arrangement, some complexes exist with* four ligands in *an apparently **higher-energy** square-planar arrangement*.
- If only electrostatic interactions are considered, a square-planar arrangement of ligands gives the d-orbital splitting shown in Fig. 20.10 , with $d_{x^2-y^2}$ raised above all the others.
- This arrangement may become energetically favorable ***when there are eight d electrons, and the crystal field is strong enough to favor the low-spin*** $\mathbf{d_{yz}^2\, d_{zx}^2\, d_{z^2}^2\, d_{xy}^2}$ ***configuration***.
- *In this configuration the **electronic stabilization energy** can **more than compensate** for any **unfavorable steric** interactions.*

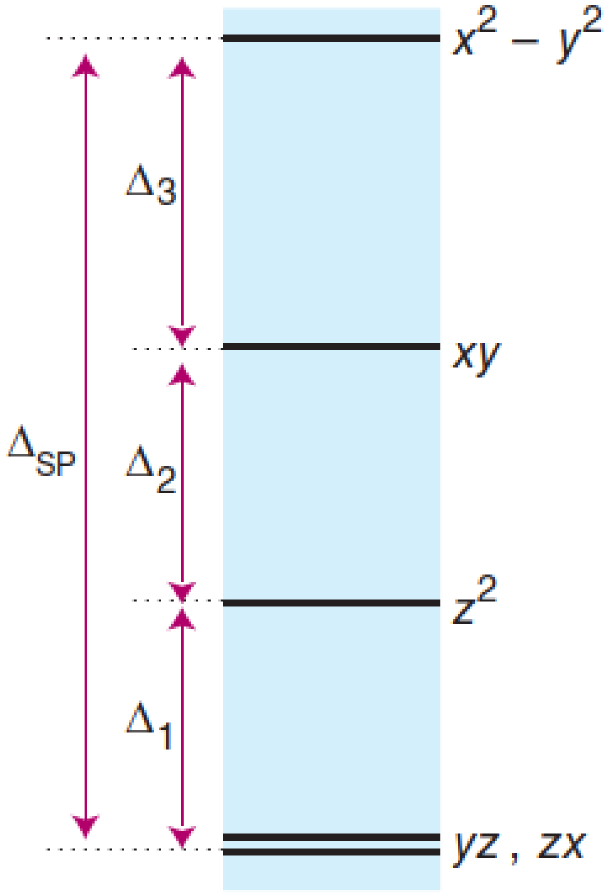


Figure 20.10 The orbital splitting parameters for a square-planar complex.

❑ Square-planar complexes : Coordination number four

Key points: A d^8 configuration, *coupled with a strong crystal field*, favors the formation of square-planar complexes.

- ***This tendency is enhanced with the 4d and 5d metals because of their larger size and the greater ease of electron pairing.***
- Thus, many square-planar complexes are found for complexes of the **large** $4d^8$ and $5d^8$ Rh(I), Ir(I), Pt(II), Pd(II), and Au(III) ions, in which *unfavorable steric constraints have less effect, and there is a large ligand-field splitting associated with the 4d- and 5d-series metals*.
- One should note that ***pairing energies*** for the 4*d*- and 5*d*-series metals *tend to be lower* than for the 3*d*-series metals, and *this difference provides a further factor* that favors the formation of low-spin square-planar complexes with these metals.

- The d^8 metal ions, such as $[PdCl_4]^{2-}$, form square planar complex.
- By contrast, small 3*d*-series metal complexes such as $[NiX_4]^{2-}$, with X a halide, are *generally* tetrahedral because the ligand-field splitting parameter is generally quite small and *will not compensate sufficiently for the unfavorable steric interactions*.
- However, only *when the ligand is high in the spectrochemical series is the CFSE large enough to result in the formation of a square-planar complex*, as, for example, with $[Ni(CN)_4]^{2-}$.
- In the ***square planar case*, only *low-spin complexes*** have been found.
- *Square-planar complexes are* ***low spin*** and *usually* ***diamagnetic*** because the four pairs of *d* electrons fill the four lowest-energy orbitals.

Factors That Affect Crystal Field Splitting

Magnitude of Crystal Field Splitting

- The magnitude of the crystal field splitting (Δ) dictates whether a complex with four, five, six, or seven d electrons (in an octahedral complex) is high spin or low spin, which affects its magnetic properties, structure, and reactivity. Large values of Δ (i.e., $\Delta > P$) yield a low-spin complex, whereas small values of Δ (i.e., $\Delta < P$) produce a high-spin complex. The magnitude of Δ depends on four factors: the valence of the metal, the principal quantum number of the metal (and thus its location in the periodic table), the geometry, and the nature of the ligand(s). Values of Δ for some representative transition metal complexes are given in Table below.

Valence of the metal

- Increasing the valence of a metal ion has two effects: the radius of the metal decreases and ligands are more strongly attracted to it due to Coulombic attraction. Both factors decrease the metal–ligand distance, which in turn causes the ligands to interact more strongly with the d-orbitals. Consequently, the magnitude of Δ_o increases as the valence of the metal increases. Typically, Δ_o for a M(III) is about 50% greater than for the M(II) of the same metal; for example, for $[V(H_2O)_6]^{2+}$, Δ_o = 11,800 cm^{-1}; for $[V(H_2O)_6]^{3+}$, Δ_o = 17,850 cm^{-1}.

Principal quantum number of the metal

- For a series of complexes of metals from the same group in the periodic table with the same charge and the same ligands, the magnitude of Δ_o increases with increasing principal quantum number: Δ (3d) < Δ (4d) < Δ (5d). The data for hexaammine complexes of the trivalent Group 9 metals illustrate this point:

$[Co(NH_3)_6]^{3+}$: Δ_o = 22,900 cm^{-1}
$[Rh(NH_3)_6]^{3+}$: Δ_o = 34,100 cm^{-1}
$[Ir(NH_3)_6]^{3+}$: Δ_o = 40,000 cm^{-1}

- The increase in Δ with increasing principal quantum number is due to the larger radial extension of the d orbitals as n increases. In addition, repulsive ligand–ligand interactions are most important for smaller metal ions. Relatively speaking, this results in shorter M–L distances and stronger d orbital–ligand interactions. The increase in Δ going from 3d to 4d and 5d is so large that all 4d and 5d metals will form low spin complexes.

Geometry of the complex

- The number of ligands in a complex as well as how well the ligand geometry overlaps with the d orbitals is also a factor in the magnitude of the crystal field splitting. For example, comparing octahedral and tetrahedral geometries the octahedral geometry has 6 ligands, and those 6 ligands overlap directly with the two d orbitals that lie along the axes, d_{z^2} and $d_{x^2-y^2}$. **The tetrahedral geometry has fewer ligands than octahedral**, and those ligands overlap less ideally with the ligands that lie between the axes, [d_{xy}, d_{yz}, and d_{zx}]. Because of this the crystal field splitting of a tetrahedral complex is generally less than half than an octahedral complex with the same metal and ligands. For example, $[FeCl_4]^-$ has a Δ_t = 5,200 cm^{-1} and $[FeCl_6]^{3-}$ has a Δ_o = 11,600 cm^{-1}.

Nature of the ligands

- In crystal field theory ligands are all modeled as negative point charges, which means that all ligands should behave identically. Experimentally, it is found that the Δ_o observed for a series of complexes of the same metal ion depends strongly on the nature of the ligands. For a series of chemically similar ligands, the magnitude of Δ_o decreases as the size of the donor atom increases. For example, Δ_o values for halide complexes generally decrease in the order F > Cl > Br > I because smaller, more localized charges, such as we see for F, interact more strongly with the d-orbitals of the metal. In addition, a small neutral ligand with a highly localized lone pair, such as NH_3, results in significantly larger Δ_o values than might be expected. Because the lone pair points directly at the metal ion, the electron density along the M–L axis is greater than for a spherical anion such as F. The experimentally observed order of the crystal field splitting energies produced by different ligands is called the spectrochemical series.

- Ligands are classified as strong field or weak field based on the spectrochemical series:

weak field $I^- < Br^- < Cl^- < \underline{S}CN^- < F^- < OH^- < ox^{-2} < \underline{O}NO < H_2O < SC\underline{N}^- < NH_3 < en < \underline{N}O_2 < CN^-$, CO **strong field**

- Note that SCN and NO_2 ligands are represented twice in the above spectrochemical series since there are two different Lewis base sites (e.g., free electron pairs to share) on each ligand (e.g., for the SCN ligand, the electron pair on the sulfur or the nitrogen can form the bond to a metal). The specific atom that binds in such ligands is underlined.

- Ligands on the weak field end of the series (halogens, OH, H_2O) will tend to form high spin complexes and ligands on the strong field end of the series (CN, CO, $\underline{N}O_2$) will tend to form low spin complexes.
- Intermediate ligands in the middle of the series could form high or low spin complexes depending on other factors.

Table: Crystal Field Splitting Energies for Some Octahedral (Δo)* and Tetrahedral (Δt) Transition-Metal Complexes

Octahedral Complexes	Δ_o (cm^{-1})	Octahedral Complexes	Δ_o (cm^{-1})	Tetrahedral Complexes	Δ_t (cm^{-1})
$[Ti(H_2O)_6]^{3+}$	20,300	$[Fe(CN)_6]^{4-}$	32,800	VCl_4	9010
$[V(H_2O)_6]^{2+}$	12,600	$[Fe(CN)_6]^{3-}$	35,000	$[CoCl_4]^{2-}$	3300
$[V(H_2O)_6]^{3+}$	18,900	$[CoF_6]^{3-}$	13,000	$[CoBr_4]^{2-}$	2900
$[CrCl_6]^{3-}$	13,000	$[Co(H_2O)_6]^{2+}$	9300	$[CoI_4]^{2-}$	2700
$[Cr(H_2O)_6]^{2+}$	13,900	$[Co(H_2O)_6]^{3+}$	27,000		
$[Cr(H_2O)_6]^{3+}$	17,400	$[Co(NH_3)_6]^{3+}$	22,900		
$[Cr(NH_3)_6]^{3+}$	21,500	$[Co(CN)_6]^{3-}$	34,800		
$[Cr(CN)_6]^{3-}$	26,600	$[Ni(H_2O)_6]^{2+}$	8500		
$Cr(CO)_6$	34,150	$[Ni(NH_3)_6]^{2+}$	10,800		
$[MnCl_6]^{4-}$	7500	$[RhCl_6]^{3-}$	20,400		
$[Mn(H_2O)_6]^{2+}$	8500	$[Rh(H_2O)_6]^{3+}$	27,000		
$[MnCl_6]^{3-}$	20,000	$[Rh(NH_3)_6]^{3+}$	34,000		
$[Mn(H_2O)_6]^{3+}$	21,000	$[Rh(CN)_6]^{3-}$	45,500		
$[Fe(H_2O)_6]^{2+}$	10,400	$[IrCl_6]^{3-}$	25,000		
$[Fe(H_2O)_6]^{3+}$	14,300	$[Ir(NH_3)_6]^{3+}$	41,000		

*Energies obtained by spectroscopic measurements are often given in units of wave numbers (cm^{-1}); the wave number is the reciprocal of the wavelength of the corresponding electromagnetic radiation expressed in centimeters: 1 cm^{-1} = 11.96 J/mol.

❑ Optical Properties of Coordination Complexes

❖ *A solution of $CuSO_4$* (in fact, $[Cu(H_2O)_6]^{2+}$) *appears* ***blue***. *Why?*

❑ Origin of Color?

White light consists of all wavelengths (λ) in the visible range and can be dispersed into colors of a narrower wavelength range.

Objects appear colored in white light because they absorb only certain wavelengths: an opaque object reflects the other wavelengths, and a clear one transmits them. If an object absorbs all visible wavelengths, it appears black; if it reflects all, it appears white.

Each color has a complementary color; for example, green and red are complementary colors. Figure 22.15 shows these relationships on an artist's color wheel in which complementary colors are wedges opposite each other. *A mixture of complementary colors absorbs all visible wavelengths and appears black*.

An object has a particular color for one of two reasons:

- It **reflects** (or **transmits**) light of that color. Thus, if an object absorbs all wavelengths except green, the reflected (or transmitted) light is seen as green.
- It **absorbs** light of the *complementary color*. Thus, if the object absorbs only red, the complement of green, the remaining mixture of reflected (or transmitted) wavelengths is also seen as green.

❖ *A solution of $CuSO_4$ appears blue to us. Why?*

❖ *Consider the hydrated cupric ion*, $[Cu(H_2O)_6]^{2+}$, *which absorbs light in the orange region of the spectrum so that a solution of $CuSO_4$ appears blue to us.*

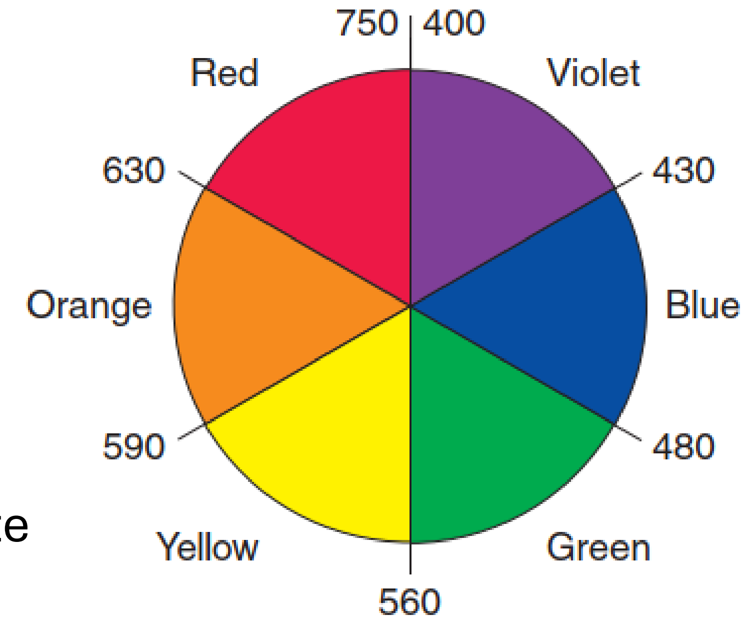


Figure 22.15 An artist's wheel. Colors, with approximate wavelength ranges (in nm), are shown as wedges.

Table 22.9 lists the color absorbed and the resulting color observed.

Table 22.9 Relation Between Absorbed and Observed Colors

Absorbed Color	λ (nm)	Observed Color	λ (nm)
Violet	400	Green-yellow	560
Blue	450	Yellow	600
Blue-green	490	Red	620
Yellow-green	570	Violet	410
Yellow	580	Dark blue	430
Orange	600	Blue	450
Red	650	Green	520

❑ Optical Properties of Coordination Complexes

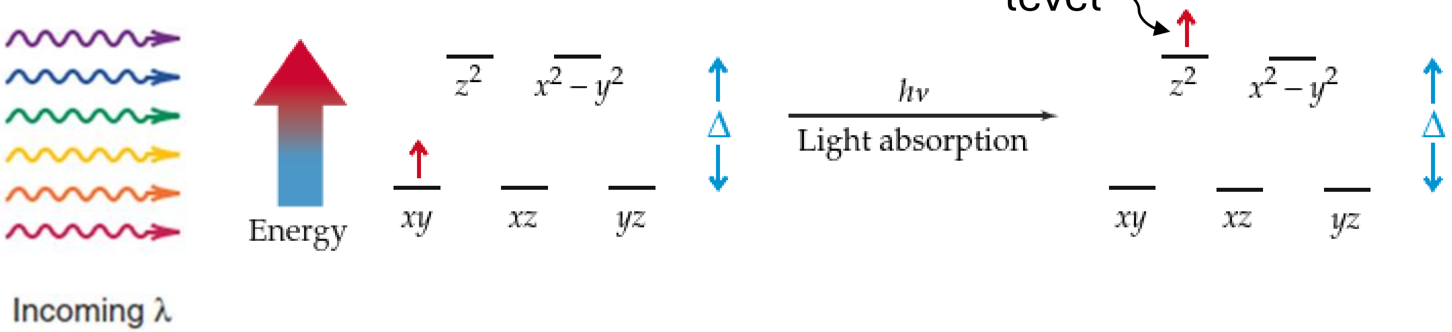
❖ THE ELECTRONIC SPECTROSCOPY OF TRANSITION-METAL COMPLEXES

- No account of the theory of the transition-metal ions would be complete without some reference to their electronic spectroscopy. But this is a vast and complicated subject which is well treated in a number of specialist volumes and here we can only draw attention to a few significant points.

- When a coordination complex absorbs light, an electron is excited from a d orbital of lower energy to a d orbital of higher energy. The larger the Δ, the smaller the wavelength of the absorbed light. In the case of an octahedral complex, absorption of light would excite an electron from a t_{2g} to an e_g orbital.

Fig… When a light wave hits a d-orbital electron, the electron may absorb the energy and jump to a higher energy orbital. The rest of the light wave transmits out of the sample.

e- excited to higher energy level



- A d-to-d (**d → d) transition must occur for a transition metal complex to show color.** Therefore, *ions with d^0 or d^{10} electron configurations are usually colorless*.

✓ Compare the colors of aq. $CuSO_4$ and aq. $ZnSO_4$. **Explain the origin of the difference.**

- UV-Vis spectroscopy is an analytical chemistry technique used to study the optical properties of various transition metal compounds. UV-Vis spectroscopy works by exciting a metal's d-electron from the ground state configuration to an excited state using UV-visible light. In short, when energy in the form of light is directed at a transition metal complex, a d-electron will gain energy and a UV-Vis spectrophotometer measures the absorbance light by the transition metal complexes with excited electrons at a specific wavelength of light, from the visible light region to the UV region.

- When using a UV-Vis Spectrophotometer, the solution to be analyzed is prepared by placing the sample in a cuvette then placing the cuvette inside the spectrophotometer. The machine then shines light waves from the visible and ultraviolet wavelengths and measures how much light of each wavelength the sample absorbs and then emits.

- Absorbance of the sample can be calculated via Beer's Law: $\boldsymbol{A = \varepsilon l c}$, where $\boldsymbol{A}$ is the absorbance, $\boldsymbol{\varepsilon}$ is the molar absorptivity of the sample, $\boldsymbol{l}$ is the length of the cuvette used, and $\boldsymbol{c}$ is the concentration of the sample. When the spectrophotometer produces the absorption graph, the molar absorptivity can then be calculated. To illustrate how this absorption graph looks like, you will find a sample absorbance spectrum below. As can be seen, the y-axis represents absorbance ($\boldsymbol{A}$) and the x-axis represents the wavelengths of light being scanned. This specific transition metal complex, $[CrCl(NH_3)_5]^{2+}$, has the highest absorbance in the UV region of light, right around 250-275 nm, and two slight absorbance peaks near 400 nm and 575 nm respectively. The two latter peaks are much less pronounced than the former peak due to the ***electron's transition being partially forbidden***—a concept that will be discussed later in this section. *If a transition is forbidden, not many transition metal electrons will undergo the excitation*.

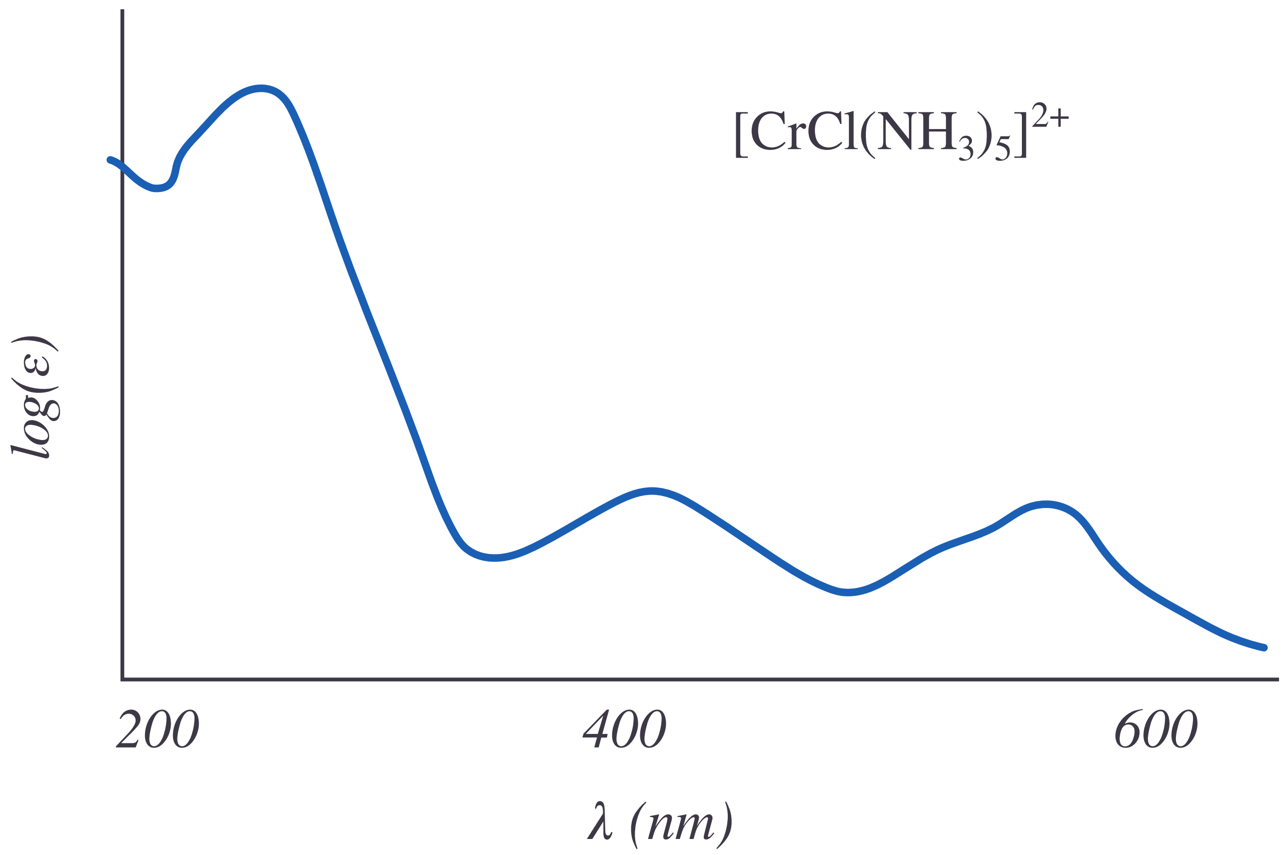


Figure 1.16.1: UV-Vis Spectrum of a Chromium(III) complex.

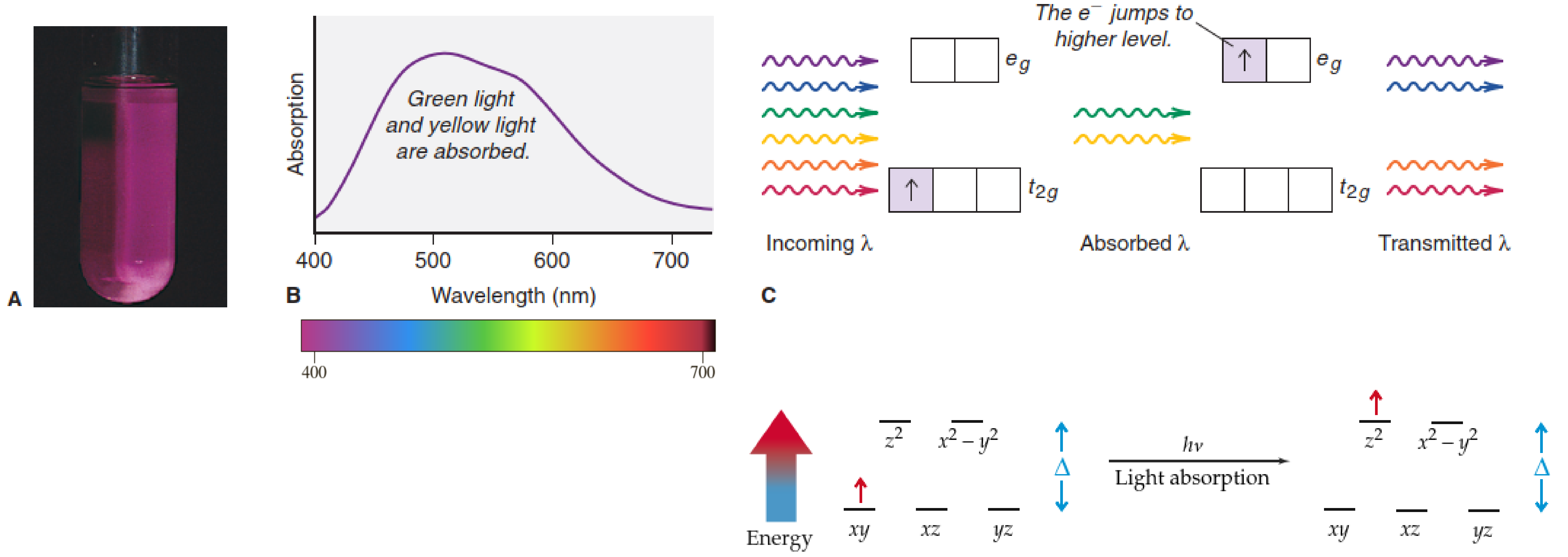


Figure 22.19 The color of $[Ti(H_2O)_6]^{3+}$.

A: The hydrated Ti^{3+} ion shows purple color.

B: An absorption spectrum shows that ***green light*** and ***yellow light*** are absorbed and other wavelengths are transmitted.

C: The process of photon absorption. An orbital diagram depicts the light colors absorbed when the d electron jumps to the higher level when the energy of the incoming photon is equal to the crystal field splitting. The ***maximum absorption peak*** in the visible region occurs at ~498 nm.

- We have been drawn into consideration of electronic spectra because *spectral data provide the most direct way of determining the crystal field splitting, Δ, (d-orbital splitting) energy*, the essential parameters of the CF model.

- For example, an octahedral $[Cr(H_2O)_6]^{2+}$ complex has a smaller Δ compared to an octahedral $[Cr(CN)_6]^{4-}$ complex (Fig. 7.1.16), and thus the aqua complex absorbs light of longer wavelength compared to the cyano complex. Vice versa, measuring the absorption spectra of the complexes, allows us to make statements about the relative crystal field strength of the ligands.
- By measuring the absorption spectrum of many complexes with a variety of ligands we can develop a so-called ***spectrochemical series*** that orders ligands according to their field strength.

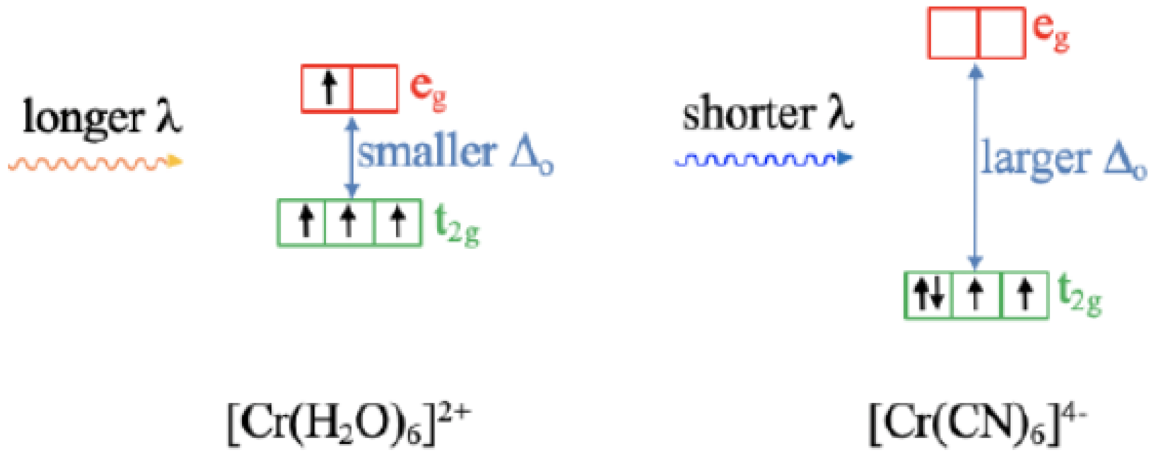


Figure 7.1.16 Octahedral $[Cr(H_2O)_6]^{2+}$ complex compared to an octahedral $[Cr(CN)_6]^{4-}$ complex

❖ Explaining the Colors of Transition Metal Complexes:

- The color of a coordination compound is determined by Δ of its complex ion. When the ion absorbs radiant energy, electrons can move from the lower energy level to the higher energy level (say, t_{2g} to e_g). Recall that the ***difference between two energy levels is equal to the energy (and inversely related to the wavelength) of the absorbed photon***:

$$\Delta E_{\text{electron}} = E_{\text{photon}} = h\nu = hc/\lambda$$

- Consider the $[Ti(H_2O)_6]^{3+}$ ion, which appears **purple** in aqueous solution (Figure 22.19, see below).

- Hydrated Ti^{3+} has its *one d electron* in one of the three lower energy t_{2g} orbitals.

- The energy difference (Δ) between the t_{2g} and e_g orbitals in this ion corresponds to photons between the green and yellow range. When white light shines on the solution, these colors of light are absorbed, and the electron jumps to one of the e_g orbitals.

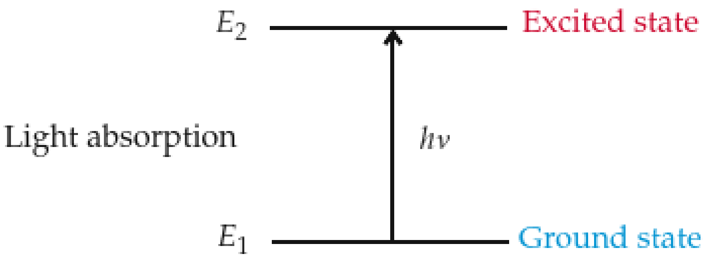
- Red, blue, and violet light are transmitted, so the solution appears purple.

- The best way to measure crystal field splitting is to use spectroscopy to determine the wavelength at which light is absorbed. The $[Ti(H_2O)_6]^{3+}$ ion provides a straightforward example, because Ti^{3+} has only **one** 3d electron (Figure 20.14). The $[Ti(H_2O)_6]^{3+}$ ion absorbs light in the visible region of the spectrum (Figure 20.15). The wavelength corresponding to maximum absorption is 498 nm [Figure 20.14(b)]. This information enables us to calculate the crystal field splitting as follows. We start by writing:

$$\Delta = h\nu \tag{20.1}$$

Also

$$\nu = \frac{c}{\lambda}$$



where c is the speed of light and λ is the wavelength. Therefore,

$$\Delta = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\ \mathrm{J \cdot s})(3.00 \times 10^{8}\ \mathrm{m/s})}{(498\ \mathrm{nm})(1 \times 10^{-9}\ \mathrm{m/1\ nm})}$$
$$= 3.99 \times 10^{-19}\ \mathrm{J}$$

Equation (7.3) shows that $E = hc/\lambda$.

This is the energy required to excite *one* $[Ti(H_2O)_6]^{3+}$ ion. To express this energy difference in the more convenient units of kilojoules per mole, we write

$$\Delta = (3.99 \times 10^{-19}\ \mathrm{J/ion})(6.02 \times 10^{23}\ \mathrm{ions/mol})$$
$$= 240{,}000\ \mathrm{J/mol}$$
$$= 240\ \mathrm{kJ/mol}$$

❑ The Spectrochemical Series:

- ➢ The fact that *color depends on the ligand allows us to create a spectrochemical series*, which ranks the ability of a ligand to split d-orbital energies (Figure 22.21). Using this series, we can predict the relative magnitude of Δ for a series of octahedral complexes of a given metal ion.
- ✓ Although we cannot predict the actual color of a given complex, we can determine whether a complex will absorb longer or shorter wavelengths than other complexes in the series.

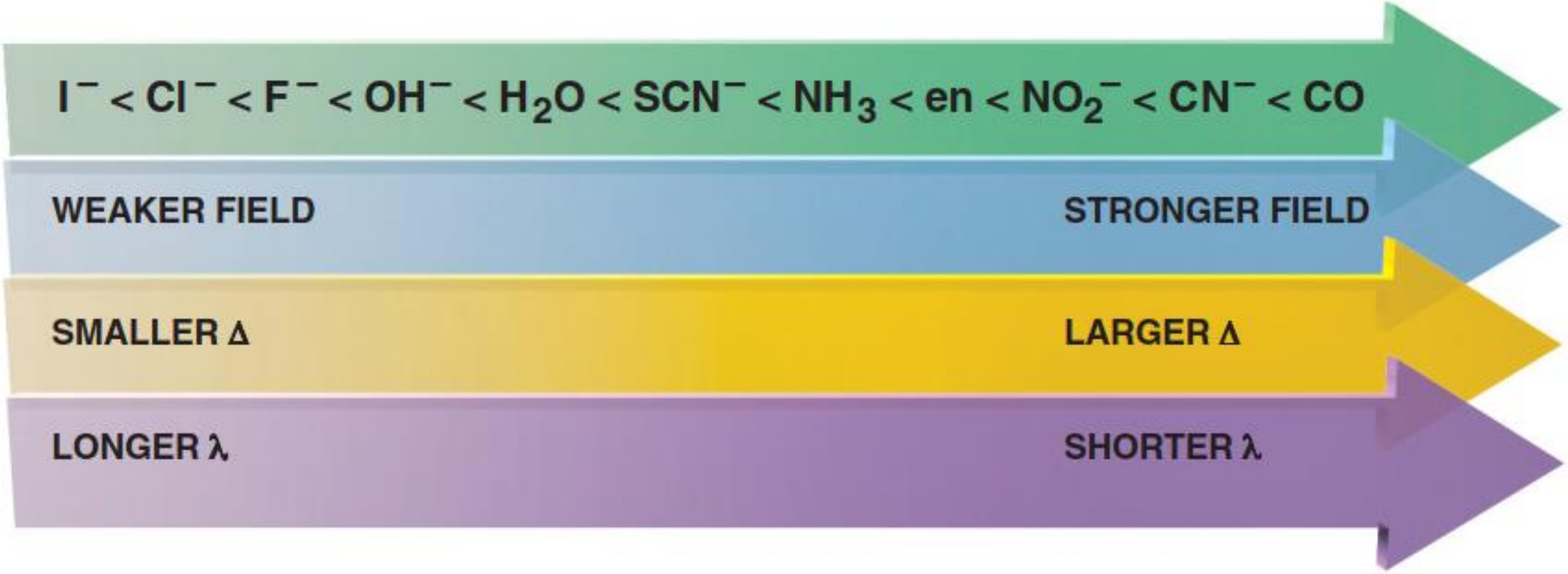


Figure 22.21 The spectrochemical series. As *Δ increases, shorter wavelengths (higher energies) of light must be absorbed to excite electrons*. For reference, water is a weak-field ligand.

- We can see for example, that an aqua ligand is a stronger field ligand compared to a fluoro ligand, but a weaker ligand than an ammine ligand.
- Generally, ligands at the left end of the series produce weaker fields, smaller Δ, and absorb light with longer wavelengths. Ligands at the right end of the series produce stronger fields, create larger Δ, and absorb light of shorter wavelengths.
- However, ***crystal filed theory cannot explain different ligand strength***. Why does one ligand produce a stronger field than another?
- To **answer this question, we need the ligand field theory**.

- Absorption spectra can show the wavelengths absorbed by: (1) a metal ion with different ligands and (2) different metal ions with the same ligand.
- Such data allow us to relate the energy of the absorbed light to Δ and make two key observations:

- ***For a given ligand,* color depends** on the **oxidation state** of the **metal ion**. In Figure 22.20A, aqueous $[V(H_2O)_6]^{2+}$ (left) is violet and $[V(H_2O)_6]^{3+}$ (right) is yellow.
- ***For a given metal ion,* color depends** on the **ligand**. A single ligand substitution can affect the wavelengths absorbed and, thus, the color (Figure 22.20B).

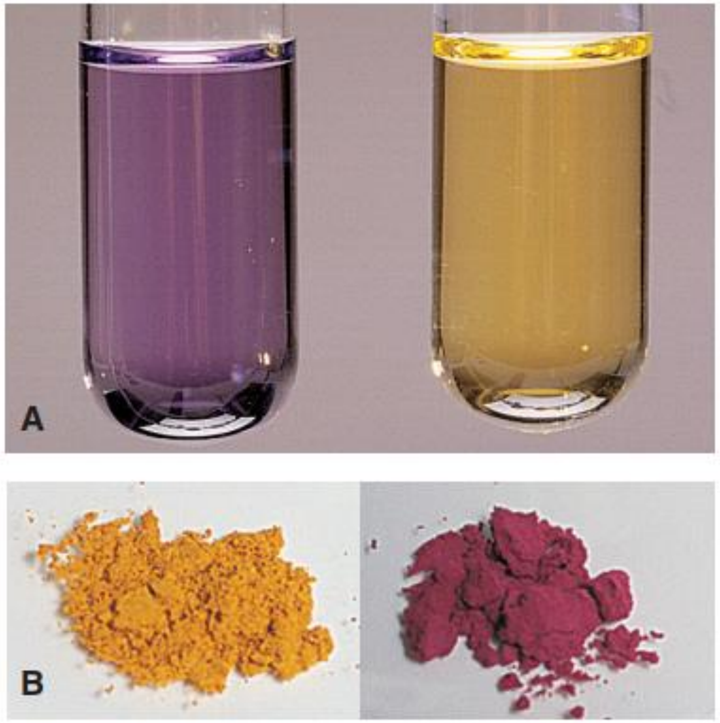
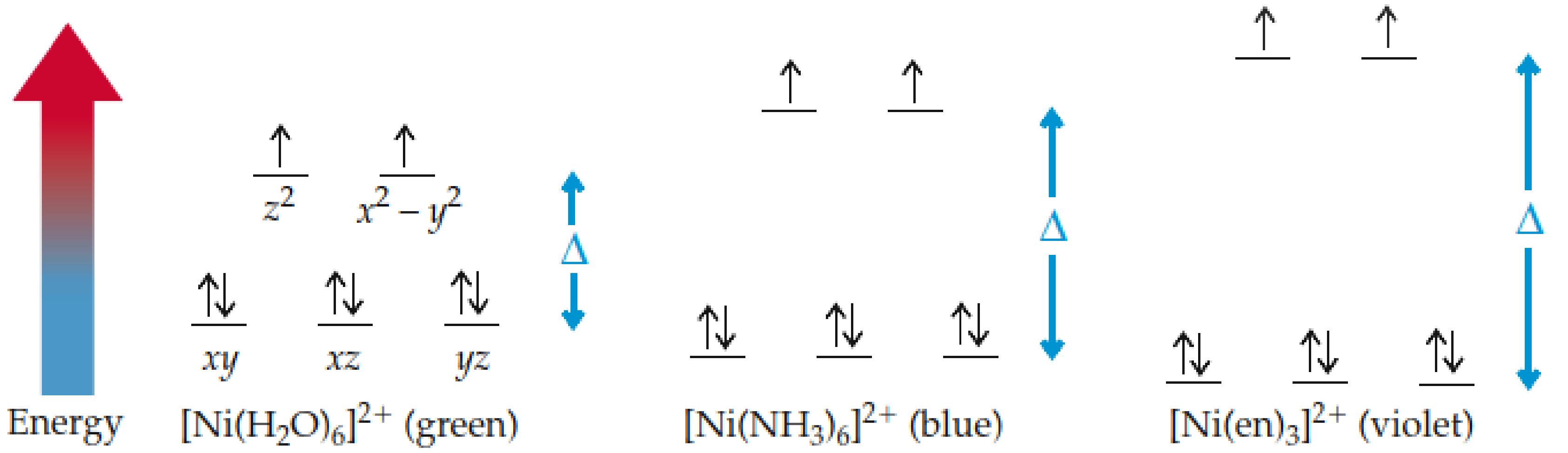


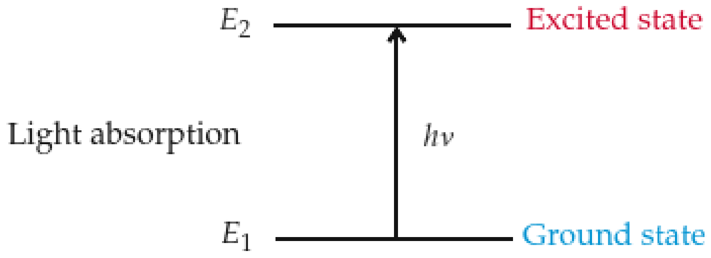
Figure 22.20 Effects of *oxidation state* and *ligand* on color.

A: Solutions of two hydrated vanadium ions: O.N. of V is +2 (left); O.N. of V is +3 (right).

B: A change in one *ligand can influence the color*. $[Cr(NH_3)_6]^{3+}$ is yellow (left), and $[Cr(NH_3)_5Cl]^{2+}$ is purple (right).

➢ The absorption spectra of different complexes indicate that the *size of the crystal field splitting depends on the nature of the ligands*. For example, Δ for Ni^{2+} ([Ar] $3d^8$) complexes increases as the ligand varies from H_2O to NH_3 to ethylenediamine (en). Accordingly, the electronic transitions shift to higher energy (shorter wavelength) as the ligand varies from H_2O to NH_3 to en, thus accounting for the observed variation in color (Figure 20.24):





$$\Delta E_{\text{electron}} = E_{\text{photon}} = h\nu = hc/\lambda$$

Δ increases, higher energies (shorter wavelengths)

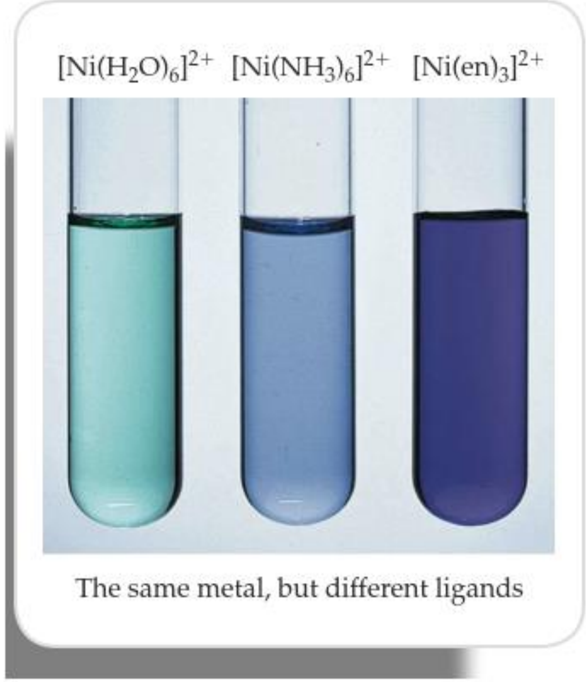


FIGURE 20.24

Aqueous solutions that contain $[Ni(H_2O)_6]^{2+}$, $[Ni(NH_3)_6]^{2+}$, and $[Ni(en)_3]^{2+}$.

Ranking Crystal Field Splitting Energies (Δ) for Complex Ions of a Metal

Problem Rank $[Ti(H_2O)_6]^{3+}$, $[Ti(CN)_6]^{3-}$, and $[Ti(NH_3)_6]^{3+}$ in terms of Δ and of the energy of visible light absorbed.

Plan The formulas show that Ti has an oxidation state of +3 in the three ions. From Figure 22.21, we rank the ligands by crystal field strength: the stronger the ligand, the greater the splitting, and the higher the energy of light absorbed.

Solution The ligand field strength is in the order $CN^- > NH_3 > H_2O$, so the relative size of Δ and energy of light absorbed is

$$Ti(CN)_6{}^{3-} > Ti(NH_3)_6{}^{3+} > Ti(H_2O)_6{}^{3+}$$

FOLLOW-UP PROBLEM 22.4 Which complex ion absorbs visible light of higher energy, $[V(H_2O)_6]^{3+}$ or $[V(NH_3)_6]^{3+}$?

❖ REVIEW OF CONCEPTS

The Cr^{3+} ion forms octahedral complexes with two neutral ligands X and Y. The color of $[CrX_6]^{3+}$ is blue while that of $[CrY_6]^{3+}$ is yellow. Which is a stronger field ligand?

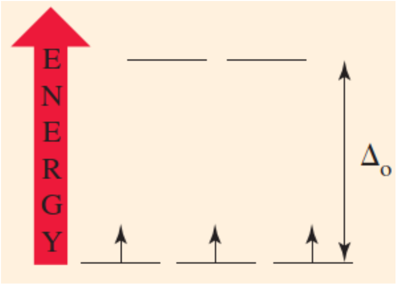
Q. **The color of $[Cr(H_2O)_6]Cl_3$ is violet whereas that of $[Cr(NH_3)_6]Cl_3$ is yellow. Explain this difference in color.**

....Both chromium(III) complexes are octahedral, and the electron configuration of Cr^{3+} is $[Ar]d^3$. From these facts, we can construct the energy-level diagram shown here. The three unpaired electrons go into the three lower-energy d orbitals. When a photon of light is absorbed, an electron is promoted from a d orbital of lower energy to a d orbital of higher energy. The quantity of energy required for this promotion depends on the energy level separation, Δ_O.

According to the spectrochemical series, NH_3 produces a greater splitting of the d energy level than does H_2O.

We should expect $[Cr(NH_3)_6]^{3+}$ to absorb light of a shorter wavelength (higher energy) than does $[Cr(H_2O)_6]^{3+}$.

Thus, $Cr(NH_3)_6]^{3+}$ absorbs in the violet region of the spectrum, and the transmitted light is yellow.



$[Cr(H_2O)_6]^{3+}$ absorbs in the yellow region of the spectrum, and the transmitted light is violet.

If we measure the wavelength, λ, of the photon absorbed, we can then calculate the splitting energy, $\Delta_O = hc/\lambda$.

Stronger field ligands increase the magnitude of Δ_O therefore, shorter wavelengths of light are absorbed and longer wavelengths are transmitted.

PRACTICE Example: The color of $[Co(H_2O)_6]^{2+}$ is light pink, whereas that of tetrahedral $[CoCl_4]^{2-}$ is deep blue. Explain this difference in color.

❖ Rules of Color Intensity and Forbidden Transitions

- The **intensity of the transmitted color** is based on two rules:

(1) If there is **a center of symmetry** in the molecule (i.e. center of inversion) then a **g** to **g** or ***u*** to ***u*** electron excitation is **not allowed**. **Spectroscopic *selection rules*** say that *transitions between states of the* ***same parity*** *are* ***strictly forbidden****; the Laporte rule*.

(2) Spin multiplicity: the **spin multiplicity of a complex cannot change** when an electron is excited. Multiplicity can be calculated via the equation **2S+1** where S is calculated by (1/2)(number of unpaired d-electrons). *Allowed transitions occur between states of the same multiplicity, say, singlet to singlet or triplet to triplet.*

If a complex breaks one of these rules, we say ***it is a forbidden transition***. If one rule is broken, it is *singly forbidden*. If two rules are broken, we call it *doubly forbidden* and so on. Even though the convention is to call it forbidden, this does not mean it will not happen; rather, the more rules the complex breaks, the more faded its color will be because the more unlikely the chances the transition will happen. Let's again look at the example (Figure 22.19)

If we apply the intensity rules to it:

(1) If we assume this molecule is octahedral in symmetry, this means it has an inversion center and thus the transition of e_g* to t_{2g} is forbidden under rule 2 due to both orbitals being gerade (g).

(2) Multiplicity before transition = 2[0.5(1 unpaired electron)] + 1 = 3, Multiplicity after transition = 2[0.5(1 unpaired electron)] + 1 = 3. Both multiplicities are the same, so this transition is allowed under rule 2.

Based on these rules, we can see that this transition is only singly forbidden, and thus it will appear only slightly faded and light rather than a deep, rich green.

- The **Laporte selection rule** for centrosymmetric molecules (those with a center of inversion) and atoms states that: The only ***allowed transitions are transitions that are accompanied by a change of parity***. That is, u → g and g → u transitions are allowed, but g → g and u → u transitions are forbidden.

- ✓ According to the Laporte rule, d–d transitions are ***parity-forbidden in octahedral complexes*** because they are g → g transitions (e_g–t_{2g} transitions). *If the ligand environment is not strictly octahedral* (e.g., because of a slight static distortion), then the ***center of symmetry of the complex as a whole may be lost***, and with it the definitive even parity of the d orbitals and states. The fact that these **d → d bands**, ***though rather weak, are indeed observed*** is due to the loss of parity which can arise as vibronic transitions as a result of coupling to asymmetrical vibrations such as that shown in Fig. 11.7 (in either a static or a dynamic manner).
- ✓ A d → d transition may then become weakly allowed, frequently because the lowered symmetry allows p and d orbitals to mix. This is particularly true of *tetrahedral complexes*. ***The tetrahedron has no center of symmetry*** *and the* ***d → d transitions of tetrahedral complexes are invariably more intense*** *than those of their octahedral counterparts.*
- ✓ Further, allowed transitions occur between states of the same multiplicity, say, singlet to singlet or triplet to triplet. *d → d transitions which involve a change of multiplicity, say, triplet to singlet, are doubly forbidden* and are very weak indeed. But as *spin-orbit coupling grows* in the 4d and 5d series of metals, and especially in the rare earths, nominally spin-forbidden transitions gain in intensity because of the loss of a clearly defined multiplicity for each state.

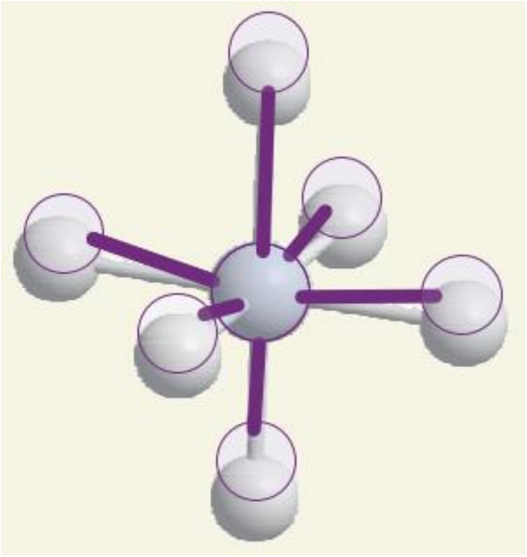


Fig. 11.7 A d–d transition is parity-forbidden because it corresponds to a g–g transition. However, a vibration of the molecule can destroy the inversion symmetry of the molecule and the g,u classification no longer applies. The removal of the centre of symmetry gives rise to a vibronically allowed transition.

- One can thus only expect absorption or emission of light by molecules that possess a dipole moment. Typically, these will be permanent dipoles, but in some applications dipoles may also be induced in molecules that have no permanent dipole momentum.
- *Chromophores* are the light absorbing moieties within a molecule. Due to differences in electronegativity between individual atoms, they possess a spatial distribution of electric charge. This results in a dipole moment $\bar{\mu}_0$ (ground state), such as for example the permanent dipole momentum of a carboxylic acid or an amide group.
- When light is absorbed by the chromophore, the distribution of electric charge is altered and the dipole moment changes accordingly ($\vec{\mu}_1$; excited state). The transition dipole momentum $\vec{\mu}_{01}$ is the vector difference between the dipole moment of the chromophore in the ground and the excited state. ***This transition dipole momentum is a measure for transition probability***, and its **dipole strength, *D*$_{01}$**, is defined is **the squared length of the transition dipole momentum vector**:
- $\boldsymbol{D_{01} = |\vec{\mu}_{01}|^2}$
- The ***strength of this transition dipole momentum is directly related to the probability*** with which a transition occurs, i.e. the strength of an absorption band.
- In **charge-transfer transitions** the ***electron moves through a considerable distance*** **(**the transfer of an electron from the ligands into the d orbitals of the central atom, or vice versa***)****, which means that the* ***transition dipole moment may be large*** *and the* ***absorption*** *is correspondingly* ***intense***.
- **The intensities of charge-transfer transitions are proportional to the square of the transition dipole moment**, in the usual way. We can think of the transition moment as a measure of the distance moved by the electron as it migrates from metal to ligand or vice versa, with a large distance of migration corresponding to a large transition dipole moment and therefore a high intensity of absorption.

Extension of Crystal Field Theory

Experimentally it is known that the nature of the ligand plays an important role in the magnitude of the crystal field splitting in metal complexes and can determine whether a complex is high or low spin. Empirically ligands can be ordered in the spectrochemical series but using crystal field theory we cannot rationalize where ligands fall in the series. For example, why is the neutral CO one of the strongest field ligands in the series while the anionic halogens are weak field ligands? To understand why ligands behave differently we have to consider how ligand orbitals interact with the metal d orbitals. Ligand field theory is essentially a combination between crystal field theory and molecular orbital theory where we focus only on the orbital overlap between the metal d orbitals and select ligand orbitals (rather than considering all of the valence orbitals on both the metal and ligand).

Metal-Ligand σ Bonding:

The bond between a metal and a ligand is primarily a Lewis acid-base interaction where the ligand is a Lewis base and donates a pair of electron to the metal, the Lewis acid. This interaction forms a coordinate covalent bond. As with molecular orbital theory we have to decide how (or if) the ligand lone pairs will overlap with each of the d orbitals, and what type of bonding will be formed In an octahedral complex the ligands and their lone pairs lie along the x, y, and z axes. As shown in Figure 9.5.1, the two on axis **d orbitals (dx2−y2 and dz2)** have to correct orientation to **form sigma bonds** with the ligands. With the three off axis d orbitals (dxy, dxz, and dyz) the ligand lone pair overlaps with a node in these orbitals (Figure 9.5.1c). This means there is no net overlap and no bonding (or antibonding) molecular orbital can be formed.

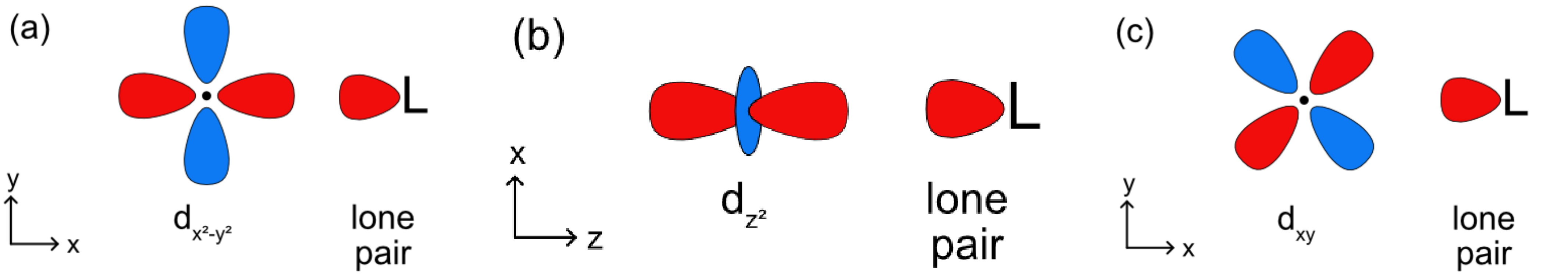


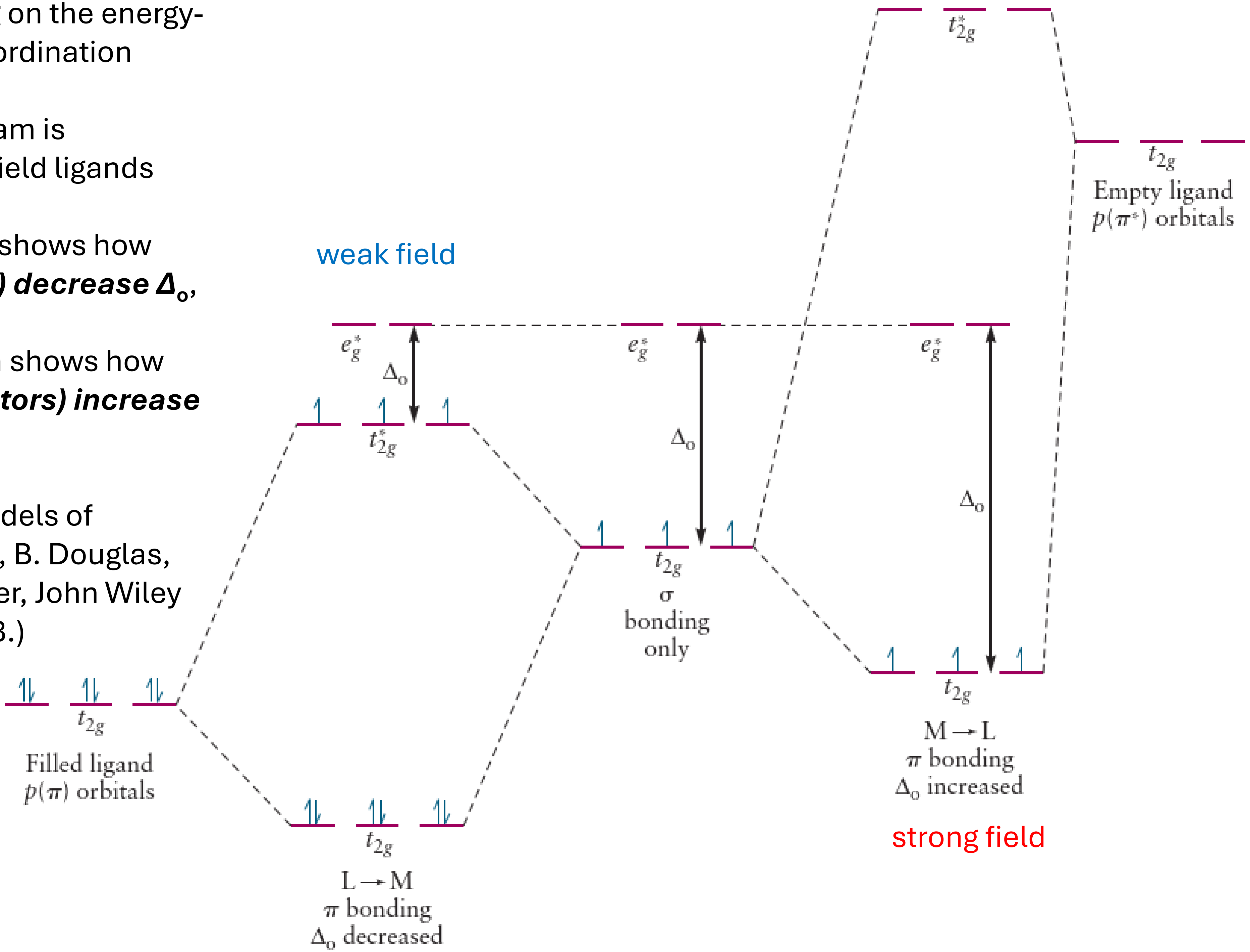
Figure 9.5.1. Overlap between a ligand lone pair orbital and the metal d orbitals.

Knowing this, we can construct a ligand field splitting diagram (a simplified molecular orbital diagram) (Figure 9.5.2). In the free metal ion all 5 of the d orbitals are degenerate and before they interact with the metal all 6 of the ligand lone pair orbitals are at the same energy as well. Generally***, based on electronegativity the ligand lone pair orbitals will be lower in energy than the metal d orbitals***. When the ligand lone pairs interact with metal d orbitals the on axis set of d orbitals (dx2−y2 and dz2) will form sigma bonding (eg) and sigma antibonding (eg*) molecular orbitals. The off axis set of d orbitals are nonbonding and are not stabilized or destabilized by the ligand lone pairs. Unlike a complete molecular orbital diagrams, a ligand field diagram won't necessarily have the same number of molecular orbitals as atomic orbitals because not every possible bonding interaction is included (for example, any overlap between the ligand orbitals and the metal s and p orbitals are not included). As in crystal field theory we can define an octahedral ligand field splitting energy (Δo), which will be the difference between the two molecular orbitals with the most metal d orbital character. In Figure 9.5.2 ***the t2g set of d orbitals are nonbonding*** and therefore 100% metal character. The eg bonding orbital is closer in energy to the ligand lone pairs and will have a higher % contribution from the ligands, and the **eg* antibonding orbital is closer in energy to the metal d orbitals** and will have ***a higher % contribution from the metal***. The relevant energy difference is between the t2g and eg* orbitals, as highlighted in green in Figure 9.5.2. The splitting energy will increase as there is a stronger interaction between the metal and ligand orbitals (based on the criteria discussed previously) and the eg* is more destabilized.

FIGURE 8.32 Effect of **π bonding** on the energy-level structure for octahedral coordination complexes.

- ✓ The **center** energy-level diagram is appropriate for intermediate field ligands that are ***σ donors only***.
- ✓ The **left** energy-level diagram shows how ***weak field ligands (π donors) decrease Δ_o***, and
- ✓ the **right** energy-level diagram shows how ***strong field ligands (π acceptors) increase Δ_o***.

(Adapted from Concepts and Models of Inorganic Chemistry, 2nd edition, B. Douglas, D. H. McDaniel, and J. J. Alexander, John Wiley and Sons, New York, 1983, p. 293.)



Metal-Ligand π Bonding:

- In addition to sigma bonding, some ligands have orbitals available which can form π bonds with metals as well. These could include p orbitals on a single donor atom such as a halogen ion or π bonding or antibonding molecular orbitals on a molecular ligand. To determine how π bonding will change the ligand field splitting picture in Figure 9.5.2, we will visualize how a ligand p orbital will interact with the metal d orbitals. The $d_{x^2-y^2}$ and d_{z^2} orbitals do not have the correct symmetry to form any type of bond with a ligand p orbital as shown in Figures 9.5.3a−b The d_{xy}, d_{xz}, and d_{yz} orbitals all have the correct symmetry to form π bonds with a ligand p orbital (Figure 9.5.3c).

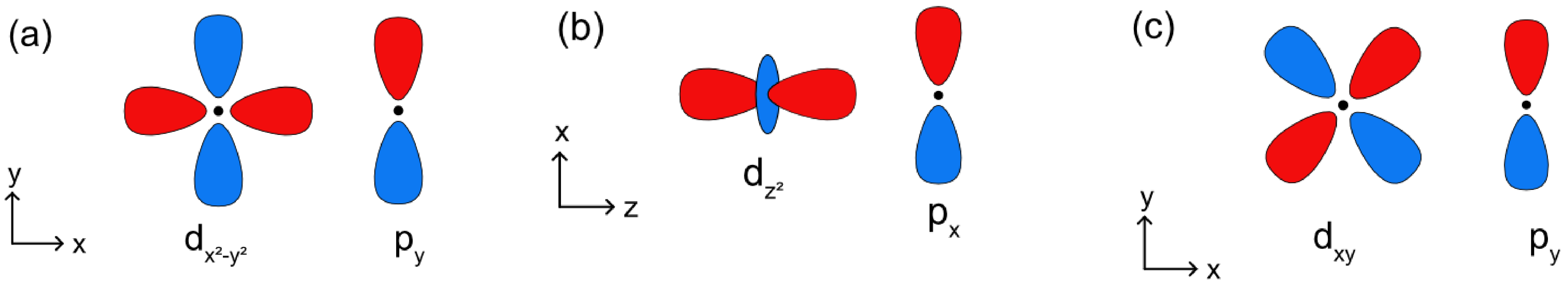


Figure 9.5.3a: There is no net overlap between the ligand p orbital and the $d_{x^2-y^2}$ or d_{z^2} metal orbitals, so this is a nonbonding interaction. Overlap between a ligand p orbital and the metal d_{xy} orbital to form a π bond.

- Unlike metal-ligand σ bonding, where electrons are always donated from the ligand to the metal, ***electrons in metal-ligand π bonding can be donated*** from either the **metal** or the **ligand** depending on the nature of the orbitals involved. In cases where ligands have empty p or π symmetry orbitals that are close in energy to the metal d orbitals, they can accept electrons from a filled metal orbital. This is also known as π backbonding because the direction of electron donation is opposite of that found in sigma metal ligand bonding. These ligands are called π acceptor ligands. In cases where ligands have filled p or π symmetry orbitals that are close in energy to the metal d orbitals, they can donate electrons to an empty metal orbital. These ligands are called π donor ligands.

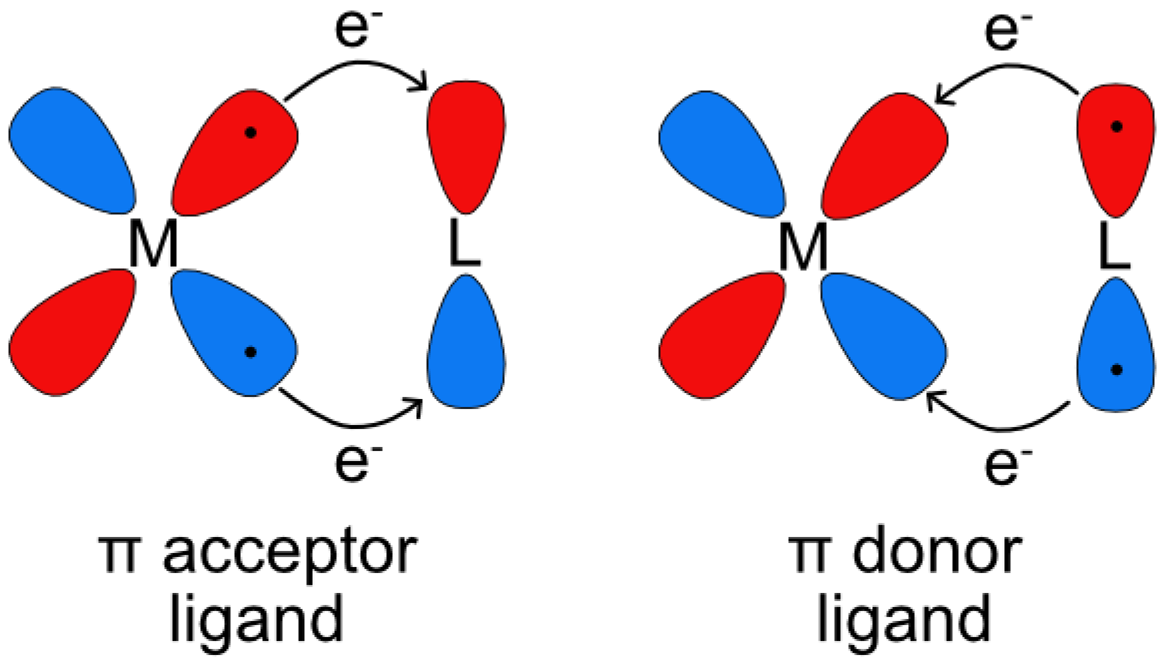


Figure 9.5.4: Illustration of metal-ligand π bonding with π acceptor and donor ligands. Filled orbitals are represented with black dots.

1) Π Acceptors

- Ligands that are π acceptors have empty, low energy π symmetry orbitals available to accept electrons from the metal center. Most often this is a π* antibonding molecular orbital but it could also be an empty p orbital.
- Due to the aufbau principle, these empty orbitals will be higher in energy than any filled lone pairs on the ligands and will also generally be higher energy than the metal d orbitals. Similar to multiple bonding in main group compounds π bonding always occurs in combination with sigma bonding so in the ligand field splitting diagram of metal-ligand bonding with a π acceptor ligand the lone pairs for sigma bonding must be included as well as the empty orbitals for π bonding. As seen in Figure 9.5.3, the $d_{x^2-y^2}$ and d_{z^2} orbitals do not overlap with the ligand p orbitals. The on-axis d orbitals will form sigma bonding (e_g) and sigma antibonding (e_g*) molecular orbitals. Unlike the sigma bonding only diagram, the off-axis d orbitals (d_{xy}, d_{xz}, and d_{yz}) overlap with the ligand p orbitals to form π bonding (t_{2g}) and π antibonding (t_{2g}*) molecular orbitals. Rather than being nonbonding, like in the sigma bonding only picture, the metal based t_{2g} molecular orbital is bonding. This increases the energy difference between the t_{2g} and e_g* and therefore increases Δ_o.

2) Π Donor Ligands

- Ligands that are π donors have filled π symmetry orbitals that are high enough in energy to donate electrons to the metal center. Most often this is a filled p orbital, but it could also be a π bonding molecular orbital. As lone pairs are generally the highest occupied orbitals is a molecule, the filled π orbitals will be lower in energy. The sigma interaction between the ligand lone pairs and the metal $d_{x^2-y^2}$ and d_{z^2} remain the same as previously described, forming the e_g bonding and e_g* antibonding molecular orbitals. The d_{xy}, d_{xz}, and d_{yz} overlap with the filled π orbitals to form π bonding (t_{2g}) and π antibonding (t_{2g}*) molecular orbitals. The difference between the pi acceptor and donor ligands is the energy and character of the t_{2g} and t_{2g}* molecular orbitals. In the π acceptor case above, the t_{2g} orbital is above the e_g and has a higher contribution from the metal and the t_{2g}* is above the e_g* and has a higher contribution from the ligand. In the π donor case the opposite is true. The t_{2g} orbital is below the e_g and has a higher contribution from the ligand and the t_{2g}* is below the e_g* and has a higher contribution from the metal. In this case Δ_o is smaller than in the sigma only picture because the t_{2g} symmetry molecular orbital has gone from nonbonding to antibonding and the energy difference between t_{2g}* and e_g* is smaller.

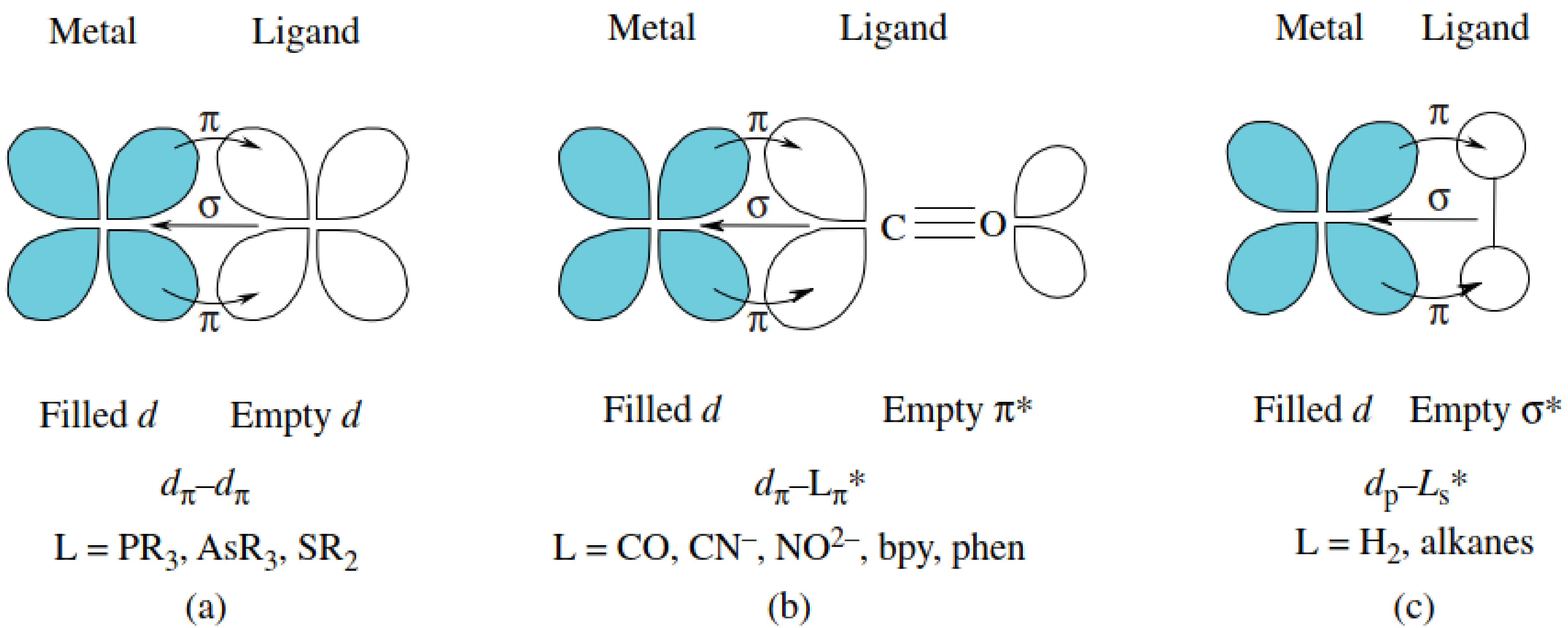


FIGURE 16.21 (a–c) Three different types of pi backbonding interactions.

The Spectrochemical Series

As discussed above, crystal field theory alone can't distinguish the behaviors of different ligands or explain why some ligands inherently produce larger splittings than others. Armed with ligand field theory we can now revisit the spectrochemical series and explain why ligands fall where they do within the series.

Strong field ligands

Strong field ligands are ones that produce large splittings between the d orbitals and form low spin complexes. Examples of strong field ligands include CO, CN^-, and NO_2. From a ligand field theory perspective all three of these ligands have strong π bonds (either double or triple bonds) and have empty π* molecular orbitals available for backbonding. Strong field ligands are both sigma donors (all ligands are sigma donors) and π acceptors. π backbonding between a metal and a ligand stabilizes the metal based t2g molecular orbital and increases Δ_o.

Weak field ligands

Weak field ligands are ones that produce small splittings between the d orbitals and form high spin complexes. Examples of weak field ligands include the halogens, OH^- and H_2O. From a ligand field perspective these ions or molecules all have filled orbitals which can π bond with the metal d orbitals. In the case of the halogens these are filled p orbitals and in the case of OH^- and H_2O these are lone pairs which are perpendicular to the metal-ligand bond. Weak field ligands are both sigma donors and π donors. The π donation from the ligand to the metal destabilizes the metal based t_{2g} molecular orbital and decreases Δ_o.

Intermediate field ligands

Intermediate field ligands produce intermediate splittings between the d orbitals and could form high or low spin complexes. Examples of intermediate field ligands include NH_3 or ethylenediamine (en). From a ligand field perspective these ligands only have a single lone pair available for sigma bonding. All of the other orbitals on nitrogen are used for sigma bonding to the H or C atoms within the molecule. Intermediate ligands are those which are only sigma donors and are not π donors or π acceptors. In this case the t_{2g} symmetry d orbitals are nonbonding and the Δ_o is determined by the strength of the metal ligand sigma interaction (and amount of destabilization of the e_g^* molecular orbital).